

Peace or Push? Understanding Global Patterns of Refugee Return

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Abstract

When do refugees return? Existing research and the design of the refugee regime imply that refugees remain abroad while home is unsafe and return as conditions improve. Yet using a novel global dataset, I show that most returns occur during ongoing conflict, not after peace. I argue that many refugees return not because they want to go home, but because coercive host-government policies or conflict in asylum make remaining infeasible. Quantitative research has largely missed this driver because it focuses on stated intentions in individual cases and because standard UNHCR data undercount forced returns. I introduce the Global Refugee Return Dataset, covering 293 origin–asylum dyads from 1980 to 2023, with new measures of realized return and forced-return policies drawn from archival country reports. Severe forced-return policies are associated with nearly three times the annual return rate observed in their absence, including while conflict continues at home. In aggregate, asylum-side shocks coincide with substantially more return than post-war periods because more refugees are exposed to them. These findings recast refugee return as a constrained choice, shaped not only by where refugees want to live, but by whether they can remain in asylum.

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1 Introduction

When do refugees return? The expectation in both the international refugee regime and existing research is that refugees return once the conflict or persecution that caused their displacement has ended. The regime is designed to protect refugees in asylum for as long as they would face persecution or serious harm upon return. Its legal cornerstone is the principle of *non-refoulement*, which prohibits states from returning refugees under those conditions.¹ UNHCR treats voluntary repatriation as its preferred “durable solution.”² Studies of return intentions likewise find that refugees become more willing to return as security in their country of origin improves (e.g., Alrababah et al., 2023; Beaman, Onder and Onder, 2022; Adema et al., 2025; Al Husein and Wagner, 2023).

When we examine global, historical return patterns however, we see that realized refugee return does not align with these expectations. Using the new Global Refugee Return dataset, described below, I document two striking patterns. First, more than three-quarters of recorded returns from 1980 to 2023 occurred while conflict was ongoing in the country of origin. Second, refugees often do not return when there is peace: of the 41 wars in the data that ended and stayed ended for at least five years, only 12 saw even half of the refugee population return within those five years. Refugees therefore frequently return when prevailing accounts predict that they should remain abroad, while many remain displaced when those accounts predict that they should return.

Why do refugees return to war? I argue that many return not because conditions in their country of origin have improved, but because they are forced out of asylum. Most refugees live in low- and middle-income countries, and 67% are hosted in a country neighboring their country of origin (UNHCR, 2025). This proximity leaves refugees vulnerable if violence spreads across the border or a separate conflict begins in their asylum country. Host governments may also induce return by revoking refugees’ legal status, restricting access to employment and services, withdrawing assistance and closing camps, or threatening detention and removal. Conflict and coercive policies in asylum can therefore compel refugees to return even when they would prefer to remain abroad and conflict continues at home.

Consider South Sudanese refugees in Sudan. When war broke out in Sudan in 2023, nearly three-quarters of the roughly two million South Sudanese refugees there returned to South Sudan, despite ongoing conflict in their home country. By contrast, most

¹The principle is codified in Article 33(1) of the 1951 Convention Relating to the Status of Refugees and its 1967 Protocol. It is also recognized as a rule of customary international law and therefore binds states regardless of whether they are parties to the Convention.

²The other two durable solutions are local integration and third-country resettlement. UNHCR has treated voluntary repatriation as its preferred durable solution since the mid-1980s (Barnett and Finnemore, 2004; Loescher, 2001).

South Sudanese refugees in Uganda, Ethiopia, and Kenya remained abroad. Afghan returns in the 1990s followed a similar pattern. Despite renewed civil war and the Taliban’s rise during 1993 and 2001, roughly three million Afghans returned. Many did so unwillingly under coercive measures imposed by Iran and, to a lesser extent, Pakistan, despite international calls to halt their forced return to war.

Although coercive repatriation has long been documented in historical and legal scholarship, it remains largely absent from quantitative research on refugee return. There are two main reasons for this omission. First, most studies use cross-sectional surveys within individual cases to examine refugees’ *intentions* to return. Drawing on models of migration choice, this research examines how refugees’ relative welfare in origin and asylum countries shapes where they prefer to live. These studies have taught us a great deal about refugees’ preferences, particularly the importance of security at home relative to economic conditions in asylum. Yet preferences shape realized return only when refugees can meaningfully choose between remaining in asylum and returning home.

Second, UNHCR data systematically undercount forced returns. Its return statistics are designed to measure voluntary repatriation, so refoulement generally falls outside the data. Forced returns may also be missed when governments revoke refugees’ status before removing them or when conflict and restricted access prevent UNHCR from observing returns. Finally, governments may inflate or suppress refugee and return figures for political reasons, and UNHCR may be unable to verify them independently or challenge them without jeopardizing access.

To address these challenges, I constructed the Global Refugee Return (GRR) dataset, covering annual refugee stocks and returns for 293 origin–asylum dyads from 1980 to 2023. Its core series draws on the *World Refugee Survey* (WRS), an annual NGO publication covering major asylum countries from 1980–2008. The WRS combined site visits and interviews conducted by local research partners with input from host governments, UNHCR field offices, and media reporting. The extended GRR series continues through 2023 using U.S. Department of State Human Rights Reports, which likewise document refugee conditions, forced-return policies, and return movements. Together, these sources allow GRR to measure both the asylum-country forces that may induce return and realized return itself, including unregistered and forced movements omitted from official data.

I evaluate the argument in three ways. First, I document when refugees return. Return is highly concentrated: the largest 5% of origin–asylum dyad-years account for approximately 94% of all return, and the median dyad experiencing any return records half of its lifetime return in a single year. This concentration points to common shocks that move many refugees at once, rather than gradual shifts in individual preferences. I then classify dyad-years according to whether they experience an asylum-side shock, a post-war window, both, or neither. Nearly four-fifths of all return occurs in years

with an asylum-side shock, whereas only 6.6% occurs after war in the absence of such a shock. This difference largely reflects exposure, as far more refugees are exposed to asylum-side shocks than to post-war windows alone.

Second, I estimate within-dyad models that compare the same refugee populations over time. Severe forced-return policies are associated with an increase in the annual return rate from roughly 7% to 18%. This association is similar whether or not war continues at home, and most severe-policy episodes occur during ongoing origin-country conflict. A stacked event study shows no preexisting increase in return before a policy begins, followed by a sharp increase at its onset. The association also persists when comparing refugees from the same origin in the same year across different asylum countries. Conflict in asylum is also associated with greater return, although less precisely. The conventional account still explains an important part of refugee return. War endings also predict higher return, but less strongly than severe forced-return policies and among far fewer refugees, making post-war return a smaller share of realized return globally.

Third, I trace these mechanisms through comparisons of Afghan refugees in Iran and Pakistan and Rwandan refugees in Burundi, the Democratic Republic of the Congo, and Tanzania. By comparing refugees from the same origin across asylum countries, these cases hold conditions at home relatively constant and show how return responds to the timing of coercion and conflict in asylum.

This research makes three contributions. First, the paper reconceptualizes refugee return as a constrained choice. Most quantitative research asks whether refugees *want* to return, but realized return also depends on whether they can remain abroad. By examining observed movements across countries and over time rather than stated intentions, I connect preference-based research with qualitative and legal scholarship on coercive repatriation. This approach also broadens the empirical scope beyond high-income asylum countries to the low- and middle-income countries where most refugees live, often near the conflicts from which they fled and under very different legal and security conditions. This broader perspective shows how conditions in asylum can eliminate the option to stay and force refugees to return even when they would prefer to remain abroad.

Second, the paper provides new global measures of realized refugee return, including coerced movements, and of the de facto conditions refugees face in asylum. Although *World Refugee Survey* reports have long informed qualitative research, they have not previously been digitized and systematically coded at this scale. With 36 research assistants, I digitized and coded the full corpus, producing 7,526 dyad-year observations spanning four decades. The value of this broader measurement is evident in the results: when I substitute UNHCR's voluntary-return series for GRR, the asylum-side associations largely disappear, consistent with official statistics omitting these returns. GRR also complements de jure policy datasets such as the Dataset of World Refugee

and Asylum Policies (DWRAP) (Blair, Grossman and Weinstein, 2022) by measuring conditions in practice, including several dimensions not analyzed here, and by making the underlying reports searchable as a reusable resource for studying the refugee regime.

Third, the findings speak to research on international cooperation, law, and institutions. *Non-refoulement* is the core legal principle of the international refugee regime, yet states use coercive policies to induce return, including to countries experiencing ongoing conflict. UNHCR sometimes facilitates such repatriations because refusing may leave refugees without assistance or jeopardize the organization’s access to them. These patterns illustrate the limits of international law and organizations in constraining state behavior. They also raise a broader puzzle: if host governments can push refugees out, and many refugees remain long after wars end, why do states continue to provide asylum? The paper therefore documents a consequential limit of international cooperation while raising the broader question of why states sustain the refugee regime despite weak enforcement and potentially enduring costs.

The article proceeds by first discussing existing literature and documenting the surprising pattern of return during conflict. I then develop the argument about forced return, explain the limitations of UNHCR data, and introduce the GRR dataset. I test the argument using descriptive evidence, within-dyad models, and a model-based global decomposition, before tracing the mechanisms through comparisons of Afghan and Rwandan refugee populations.

2 Puzzling Patterns of Refugee Return

Existing research consistently identifies safety and security in the country of origin as central to refugees’ return preferences (Alrababah et al., 2023; Kaya and Orchard, 2020; Al Husein and Wagner, 2023; Adema et al., 2025; Beaman, Onder and Onder, 2022; Kayaoglu, Şahin-Mencütek and Erdoğan, 2022). Alrababah et al. (2023) argue that refugees consider other conditions only once a basic threshold of safety at home has been met. Syrian refugees in Lebanon were unlikely to consider returning while Syria remained unsafe, even when they faced hostility and poor living conditions in asylum. Similarly, lower conflict intensity predicts the spontaneous return of Syrian refugees from Lebanon, Jordan, and Iraq (Beaman, Onder and Onder, 2022), while territorial integrity and credible security guarantees are especially important for Ukrainian refugees’ willingness to return after the war (Adema et al., 2025).

Security is not the only factor that matters. Political conditions and regime change, economic prospects, military conscription, and attachment to the country of origin also shape return preferences (Kaya and Orchard, 2020; Kayaoglu, Şahin-Mencütek and Erdoğan, 2022; Ghosn et al., 2021; Müller-Funk and Fransen, 2023; Alrababah et al., 2023; Beaman, Onder and Onder, 2022; Adema et al., 2025). Conditions in

asylum may matter as well, but the evidence is more mixed. Some studies find that hardship or discrimination increases willingness to return, while others find little relationship; Beaman, Onder and Onder (2022), in contrast, find that better conditions in exile are associated with greater realized return (Kayaoglu, Şahin-Mencütek and Erdoğan, 2022; Ghosn et al., 2021; Müller-Funk and Fransen, 2023; Alrababah et al., 2023; Beaman, Onder and Onder, 2022). Most of this research operationalizes conditions in asylum through economic circumstances, legal status, discrimination, social integration, and general living conditions, rather than systematically examining severe asylum-side shocks, such as forced-return policies or armed conflict, that can make remaining unsafe or infeasible.

This literature yields two expectations for realized return. First, return should be more common when the origin country is at peace. Second, large-scale return should be rare while intense conflict continues. To examine whether realized return follows these expectations globally, I use the Global Refugee Return (GRR) Dataset, which records annual refugee stocks and return flows for 293 origin–asylum dyads from 1980 to 2023 and is described in Section 4.2. Figure 1 shows a striking pattern: most refugees return to war, not peace.

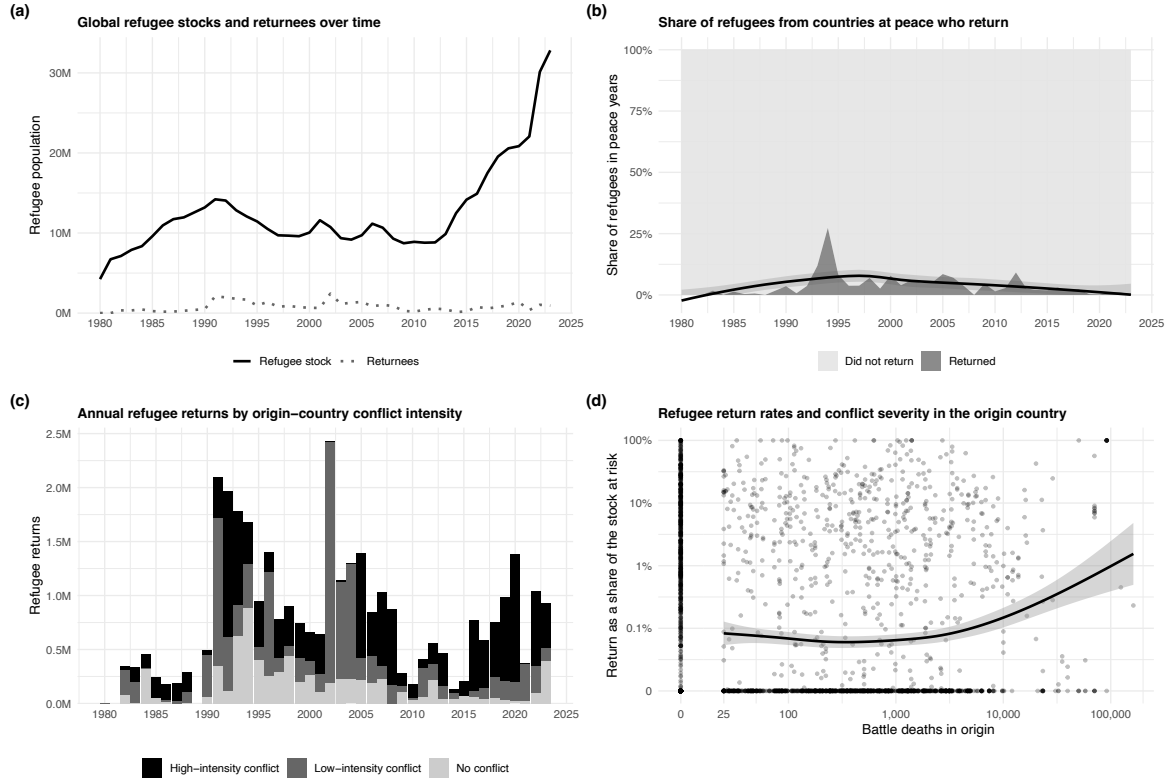
Panel (a) first shows that return flows are generally small relative to the size of displaced populations. Across dyad-years with more than 1,000 refugees, the median return rate is zero and the mean is 5.34%; in 89% of dyad-years, fewer than 5% of refugees return. Panel (b) shows that peace rarely produces return. Among refugee populations displaced by war, the average annual return rate is only 3.42% when there is no conflict in the country of origin.³ Of the 41 wars in the data that ended and remained ended for at least five years, only 12 were followed by the return of even half the refugee population within those five years.

Panel (c) shows that most return occurs during war, not after it. Between 1980 and 2023, more than three-quarters of all recorded returns took place while conflict was ongoing in the country of origin. Roughly two-fifths occurred during high-intensity conflict, more than one-third during low-intensity conflict, and only one-fifth during peace.

Some of this pattern reflects exposure. As Table 1 shows, dyad-years with conflict in the origin constitute 34.1% of dyad-years but contain 72.5% of refugee stock-years at risk, because protracted conflicts produce the largest displaced populations. Their 79.0% share of recorded return therefore exceeds their share of the population at risk by only 6.5 percentage points. But exposure does not resolve the puzzle. It explains why most refugees at risk come from countries at war, not why so many return while war continues. Both the refugee regime and preference-based accounts imply that

³This calculation is restricted to origin–asylum dyads in which refugees generally fled war rather than broader non-conflict persecution.

Figure 1: Return during conflict and non-return during peace



Notes: (a) Global refugee stock (solid) and recorded returns (dotted). (b) Share returning, among refugees who fled war, in dyad-years when the origin is at peace. (c) Annual returns by origin-country conflict intensity (UCDP: ≥ 25 and $\geq 1,000$ battle deaths). (d) Return rate against log battle deaths (dyad-year). Lines are LOESS smoothers; 1989 is omitted. *Sources:* GRR; UCDP/PRIO.

refugees should be reluctant to return while home remains unsafe. Moreover, return rates are actually higher during war than peace, at 5.45% compared with 3.82%, a ratio of 1.42. Panel (d), which scales returns by the refugee stock at risk, reinforces this pattern: return rates do not decline consistently as battle deaths increase and rise at the highest levels of violence.

3 Refugee Return as a Constrained Choice

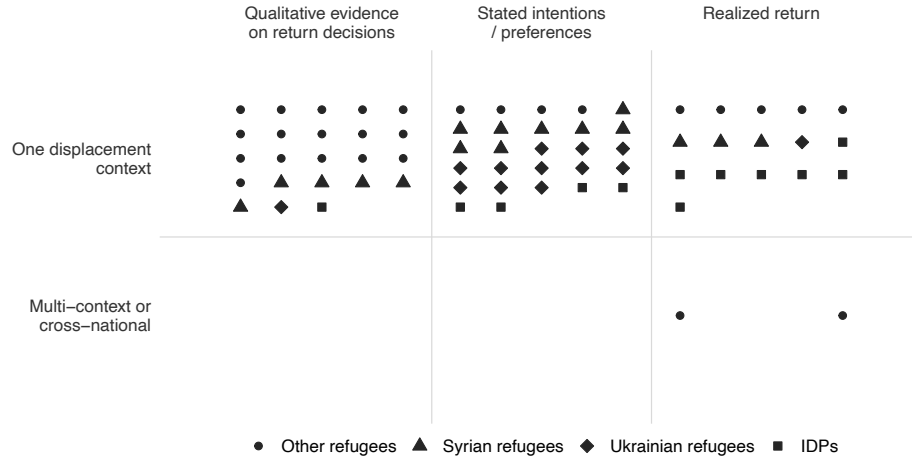
Existing research primarily explains when refugees want to return. Drawing on models of migration choice, it treats refugees as comparing expected welfare in origin and asylum and preferring the location that offers greater welfare (Borjas, 1987). Because preferences are not directly observable, empirical studies generally examine stated return intentions within individual displacement contexts. As Figure 2 shows, most focus on Syrian and, more recently, Ukrainian refugees. These studies tell us a great deal about refugees' preferences, but less about when those preferences translate into

Table 1: Exposure, return, and return rates by conditions in the origin country

Conditions in the origin	Dyad-years	Share of dyad-years	Share of stock at risk	Share of all return	Return rate
No conflict	3,366	65.9%	27.5%	21.0%	3.82%
Low-intensity conflict	1,134	22.2%	32.6%	36.8%	5.65%
High-intensity conflict	609	11.9%	39.9%	42.1%	5.28%
<i>All dyad-years</i>	5,109	100.0%	100.0%	100.0%	5.00%

Notes: Dyad-years with more than 1,000 refugees at risk in the previous year, excluding 1989, for which the *World Refugee Survey* was not published: 5,109 dyad-years holding 665.6 million refugee stock-years and 33.3 million recorded returns. The stock at risk is the lagged refugee stock, raised to recorded return in the few dyad-years where return exceeds it. Conditions in the origin are the maximum UCDP intensity recorded for the origin country that year. The return rate is the final column and equals the share of return divided by the share of the stock at risk.

Figure 2: Scope and outcomes in empirical research on refugee return decisions



Notes: Each point represents an empirical study of refugee return identified through a systematic search of nineteen political science, migration, and development journals from 2006–2026, supplemented by citation tracing, the 2024 World Bank–UNHCR Joint Data Center review, and targeted searches. A displacement context may include multiple origin populations in one host country or one origin population across multiple host countries; only studies spanning multiple contexts are classified as cross-national. See the Appendix for the full list of studies.

realized return or whether the same relationships hold across contexts.

I instead conceptualize realized return as a constrained choice. Refugees may prefer to remain in asylum but be compelled to leave, so observed return need not indicate that home has become safer or more attractive or reveal a preference for return. Preference-based explanations therefore apply when refugees can meaningfully choose between remaining in asylum and returning home. I argue that many cannot, and that constraints on remaining in asylum account for a substantial share of realized return globally.

3.1 Constraints on Remaining in Asylum

Two types of shocks can make remaining in asylum infeasible: host-government coercion and armed conflict in the country of asylum. First, host governments may pressure refugees to leave before conditions at home have improved. Despite the prohibition on forced return under the 1951 Refugee Convention,⁴ states may circumvent or ignore *non-refoulement* (Schwartz, 2025; Chimni, 2004; Harrell-Bond, 1989; Long, 2013; Stein and Cuny, 1994). Deportation is only one means of inducing return. Governments may revoke residence permits, restrict access to work and services, withdraw assistance, close camps, or threaten detention and expulsion. These measures can make remaining increasingly difficult while the resulting return is still recorded as “voluntary.”⁵ Consider the treatment of Afghan refugees in Iran in 2005:

“UNHCR [complained] that Iran was pressuring [Afghan refugees] to return by suspending education and medical services and revoking their residence permits. UNHCR also accused Iran of levying a punitive tax on refugees to force them to repatriate. Afghan media reports accused Iranian police of beating refugees and demanding bribes to extend residence permits. ” (*WRS, Iran, 2006*)

Second, armed conflict in the country of asylum can make remaining there more dangerous than returning home. Most refugees remain near their countries of origin, often in neighboring states that share similar structural risk factors for conflict and are vulnerable to the cross-border spread of war (UNHCR, 2025; Fearon and Laitin, 2003; Collier and Hoeffler, 2004; Gleditsch, 2007; Lischer, 2005). Refugees may therefore escape one conflict only to encounter another in asylum. Violence can directly harm refugees or disrupt the humanitarian assistance on which they depend. Under these conditions, refugees may return even while their country of origin remains at war. The 1996 return of Rwandan refugees from the DRC⁶ illustrates this mechanism:

“In October, an outbreak of civil war in Zaire [DRC], which engulfed Rwandan refugee camps there, produced what UNHCR described as the ‘largest and swiftest’ repatriation in memory. About a half-million Rwandan refugees returned home during a four-day period in mid-November. Most returnees repatriated on foot, jamming the main Rwandan highway from the border. The sea of humanity stretched nearly 100 miles during the height of the return.” (*WRS,*

⁴*Non-refoulement* is generally recognized as binding all states as customary international law, while states parties to the 1951 Convention or its 1967 Protocol are additionally bound by their treaty obligations.

⁵UNHCR may facilitate repatriation under these conditions, both to reduce the harm refugees would face if expelled without assistance and because it depends on host-government cooperation to maintain access to refugee populations (Barnett, 2001*a,b*; Barnett and Finnemore, 2004; Whitaker, 2003). Its participation does not necessarily establish that refugees would have returned in the absence of government pressure.

⁶The country was officially known as Zaire in 1996 and was renamed the Democratic Republic of the Congo in May 1997. For consistency with current usage, I refer to it as the DRC in the text; quotations and source labels retain the contemporaneous name.

Table 2: Preferences, external shocks, and realized return

Case	Conditions	Implication
Preference-based return	$P_{iat} = 1$	The refugee prefers the origin country and would return even in the absence of an external shock.
Forced return	$P_{iat} = 0, S_{at} \geq \kappa_{iat}$	The refugee prefers asylum, but the external shock crosses their individual threshold and induces return.
Non-return	<i>No external shock:</i> $P_{iat} = 0, S_{at} = 0$	<i>Preference-based non-return:</i> The refugee prefers asylum and remains there in the absence of an external shock.
	<i>Shock below threshold:</i> $P_{iat} = 0, 0 < S_{at} < \kappa_{iat}$	<i>Non-return despite shock:</i> The refugee experiences an external shock, but it is not sufficiently severe to induce return.

Notes: P_{iat} indicates whether refugee i prefers returning to remaining in asylum. S_{at} denotes the severity of the external shock, with $S_{at} = 0$ indicating that no shock occurs. The individual threshold $\kappa_{iat} > 0$ captures how much pressure is required to induce refugee i to return. Because these thresholds differ across refugees, the same shock can induce some refugees to return while others remain.

Zaire [DRC], 1997)

When refugees are forced to leave asylum, onward movement to a third country is possible in principle but often infeasible. Onward migration may seem like a realistic alternative in high-income asylum countries, where refugees have often already traveled considerable distances. But most refugees live in low- and middle-income countries, often near the border they crossed and without the resources or legal permission to travel elsewhere. Many remain in peripheral regions because of community ties or government efforts to contain them there (Camarena, 2024); 79% of UNHCR camps in Africa lie within 50 kilometers of an international border (Fein and Van Den Hoek, 2022). Reaching a third country may therefore require a long, costly, and illegal journey that authorities can block, particularly when host governments are seeking to return refugees home (Whitaker, 2003). When conflict drives refugees from asylum, traveling farther into the host country may also be dangerous. For many refugees unable to remain in asylum, their country of origin is therefore the only destination they can realistically reach.

3.2 Formalization

I use a simple formalization to show how the conventional preference-based account is nested within a broader constrained-choice framework and to clarify how different combinations of preferences and constraints map onto observed return and non-return. Table 2 summarizes these possible cases.

Let i index refugees, o their country of origin, a their country of asylum, and t time. Let $U_{oi,t}$ denote refugee i 's expected utility from returning to the origin country at time t , and let $U_{ai,t}$ denote their expected utility from continuing to live in the asylum country. These utilities incorporate security, economic and political conditions, social ties, and other characteristics of the refugee and the two locations. Let $\mathbb{1}\{\cdot\}$ denote an indicator function that equals one when the condition inside the braces is satisfied and zero otherwise. Preference-based return is represented by

$$P_{iat} = \mathbb{1}\{U_{oi,t} > U_{ai,t}\}, \quad (1)$$

where $P_{iat} = 1$ indicates that refugee i prefers returning to remaining in asylum. Because refugees differ in their circumstances, this inequality becomes satisfied at different times for different people. Changes in preferences should therefore generally produce relatively small and temporally dispersed return flows.

Return can also be induced by an external shock in the country of asylum. Let S_{at} denote the severity of host-government coercion or armed conflict in asylum country a at time t . Let κ_{iat} denote the level of pressure required to induce refugee i to return despite preferring to remain in asylum. Refugees who are more strongly attached to asylum or better able to withstand the shock have a higher value of κ_{iat} . Define

$$F_{iat} = \mathbb{1}\{S_{at} \geq \kappa_{iat}\}, \quad (2)$$

where $F_{iat} = 1$ indicates that the external shock is sufficiently severe to induce refugee i 's return. A shock therefore need not move everyone. It induces return among refugees whose individual thresholds it crosses. The severity term accommodates different forms of pressure: reducing food assistance may cross only lower thresholds, whereas revoking residence permits may make remaining infeasible for a much larger group. Because the same shock can cross many thresholds at approximately the same time, return induced in this way should be more temporally concentrated than return generated by changes in individual preferences.

Let $R_{iat} = 1$ indicate that refugee i actually returns at time t . Realized return is

$$R_{iat} = P_{iat} + (1 - P_{iat})F_{iat}. \quad (3)$$

Equation 3 identifies two routes to return. Refugees return because they prefer the origin country, $P_{iat} = 1$, or because an external shock induces return among those who would otherwise remain, $P_{iat} = 0$ and $F_{iat} = 1$. The conventional preference-based account is therefore the nested case in which no external shock crosses refugees' thresholds and realized return is determined by P_{iat} alone. The argument developed

here is that external shocks can instead move substantial numbers of refugees together, producing concentrated return even when conditions in the origin country have not improved.

3.3 Hypotheses

The model implies that more severe shocks in countries of asylum cross more refugees' individual thresholds, increasing return even among those who prefer to remain abroad and while conflict continues in their country of origin.

H1 (Forced Return During War). External shocks in countries of asylum increase realized refugee return even while armed conflict continues in the country of origin.

H1a (Government Coercion). Return rates increase as host-government measures restricting refugees' ability to remain become more severe.

H1b (Conflict in Asylum). Return rates increase with the onset and intensity of armed conflict in the country of asylum.

An implication is that among refugees from the same country of origin, return should occur disproportionately from asylum countries experiencing coercive policies or conflict, producing differences in the timing of return across asylum countries even when origin-country conditions are held constant.

The preceding hypotheses concern whether asylum-side shocks increase return rates. H2 instead asks which conditions account for more observed return in aggregate, which depends on how often each condition occurs, how many refugees it affects, and the resulting return rate. Because the dyad-year data cannot identify whether each return was preferred or induced, I compare observable contexts. I classify dyad-years as experiencing only an asylum-side shock, defined as a forced-return policy or conflict in asylum; only a post-war period, defined as the first three peaceful years after an origin-country war; both; or neither. This comparison shows where return is concentrated without attributing the cause of each individual movement.

H2 (Aggregate Prevalence). In aggregate, asylum-shock-only dyad-years account for a larger share of observed refugee return than post-war-only dyad-years.

4 Data

4.1 Limitations of UNHCR Data

Testing the theory requires cross-national data on realized return. The principal existing source is UNHCR’s population statistics (United Nations High Commissioner for Refugees, n.d.*b*). These data are subject to four sources of measurement error that may lead returns, especially forced returns, to be systematically under-reported. I describe each problem below, drawing on examples from the *World Refugee Survey*, introduced in Section 4.2.

Unregistered refugees. The term “refugee” encompasses several legal and statistical categories. UNHCR’s data include recognized refugees, asylum-seekers, people in refugee-like situations, and others in need of international protection (United Nations High Commissioner for Refugees, n.d.*a*). These data often miss unregistered refugees, including people who qualify as refugees but never apply for asylum and those whose refugee status is later revoked and who are subsequently treated as irregular migrants.

This gap is especially important for measuring forced return because withdrawing formal status can itself facilitate removal. Governments may revoke refugees’ status before deporting them, so those individuals may no longer be counted as refugees and their return may not appear in official refugee statistics. The Croatian government’s treatment of Bosnian refugees in 1995 illustrates this problem:

“Croatia revoked the refugee status of 100,000 Bosnian refugees ... saying it was now safe for them to return. ... The irony here is that the Croatian army, which is occupying many of those same towns, has refused to allow Muslims from Bosnia to come into those same places.”

1996, Croatia

Dependence on institutional presence. Accurate return estimates often depend on UNHCR’s access and operational presence in the country. UNHCR may lack access because governments restrict its activities, because it has limited resources or no office, or because war and insecurity force personnel to evacuate. This creates a systematic measurement problem, since some of the same conditions that increase forced return—including host-government reluctance to cooperate on refugee protection and conflict in asylum—also make UNHCR less able to observe and record those movements. For example, UNHCR data records no returns from South Africa to Mozambique despite reports of large-scale removals:

“With no UNHCR presence in South Africa ... [n]either country recognizes them as refugees ... During 1985, South Africa was returning some 1,500 a month to the danger they had fled.”

1986, South Africa

The Somali civil war created a similar problem when it reached areas hosting Ethiopian refugees. UNHCR evacuated its personnel, and the resulting returns do not appear in its data:

“Beginning in December 1990, as fighting in the Somali civil war intensified ... UNHCR personnel were evacuated from Somalia ... Large numbers of Ethiopian refugees streamed home to escape the fighting, many on foot. ... More than 34,000 returned in January alone. ... By the end of May, 64,000 returnees had reached Ethiopia.”

1992, Ethiopia

Measurement of forced and spontaneous returns. UNHCR’s return statistics are designed to measure voluntary repatriation, not forced return (UNHCR, 2024). Deportations and refoulement therefore generally fall outside the data, while coerced movements may still be recorded as voluntary when they occur through formal repatriation channels. The data can thus both omit forced returns and obscure coercion among returns they do record. Spontaneous returns are also difficult to observe. Refugees may go uncounted if they leave without deregistering or cross at unmonitored border points.

The politics of population figures. UNHCR figures can also become objects of political negotiation. Host governments may understate returns to maintain higher refugee counts and attract more foreign aid. UNHCR may accept these figures when it cannot verify them independently or when challenging them could jeopardize government cooperation or access to refugee populations. Somalia provides one example:

“UNHCR had no firm estimate of the number of refugees in Somalia because the government did not permit a census. ... During 1988, large numbers of Ethiopian refugees returned to Ethiopia ... [but] [t]he Somali government denied that a mass repatriation had occurred. ... The Somali government claimed that about 537,000 refugees remained in the country at year’s end, but some voluntary agencies put the total at less than 400,000.”

1989, Somalia

These sources of error are systematic rather than random. Forced returns are especially likely to go unrecorded when refugees lose formal status, move outside assisted channels, return after humanitarian access has collapsed, or are classified as deportees rather than returnees. Analyses based only on UNHCR’s return series may therefore understate forced return, particularly during ongoing conflict or following coercive policies in the country of asylum.

4.2 The Global Refugee Return Dataset

To address these problems, I created the Global Refugee Return (GRR) Dataset, which reconstructs refugee populations and movements from contemporaneous reports drawing on local expertise rather than treating administrative figures as definitive. The principal source is the *World Refugee Survey* (WRS), published annually by USCRI. Its country reports cover 1980–2008 and document refugee populations, conditions, government policies, and movements in each country of asylum. USCRI describes its reporting process as follows:

“[A] rigorous method including, when possible, site visits to the countries on which we report. In most countries, we have Research Partners—local human rights NGOs that interview officials, refugees, and representatives of local civil society and international agencies based on our standard questionnaire . . . We consult with and seek the input of representatives of each government . . . We receive extensive input from the field offices of the UN High Commissioner for Refugees (UNHCR) and the UN Relief and Works Administration for Palestine Refugees in the Near East. Finally, we crosscheck all of this with secondary sources in the media, on the internet, and in major governmental and nongovernmental human rights reports.” (USCRI, 2008)

By combining locally collected information with official and nonofficial sources, the WRS often captures refugee dynamics missing from administrative data. Its reports document unregistered populations, spontaneous and forced returns, and disputes over official figures. Their coverage is also less dependent on the presence of any single organization. Because USCRI did not administer humanitarian assistance, its estimates were less exposed to the political pressures surrounding UNHCR statistics, which could influence the amount of aid host governments received.

Constructing GRR required digitizing WRS reports, often available only as scans or physical volumes, and converting them into standardized dyad–year observations. I hand-coded the 25 largest origin populations, covering 83 dyads, and worked with 36 trained research assistants to code the remainder using a common protocol. I then reviewed and adjudicated every observation, returning to the original reports, adjacent years, and desk research when GRR and UNHCR figures differed substantially. To my knowledge, this is the first project to digitize and systematically code the full WRS corpus.

The WRS trades some numerical precision for this broader coverage. Reports sometimes provide rounded estimates, ranges, or conflicting figures, and coverage is less complete for some high-income asylum countries. GRR applies standardized rules for coding these cases, described in the Appendix. The WRS was not published in 1990, so GRR has no observations for 1989.

GRR includes origin populations with at least 25,000 refugees in exile in any year. For each origin population, I include the principal asylum countries that collectively host at least 75% of that population.⁷ I provide core and extended versions of the dataset. The *core* dataset covers Global South asylum countries from 1980 through 2008 and relies on the WRS evidence in which I have the greatest confidence. It contains 105 dyads across 58 origin populations and 61 asylum countries.

The *extended* series adds high-income asylum countries and continues through 2023 using the U.S. Department of State’s annual country reports on human rights practices. These reports include a dedicated section on refugees and migrants based on in-country reporting, which documents instances of forced return and forced-return policies, as well as unregistered population figures (U.S. Department of State, 2024). Because *State* reports do not always provide stock or voluntary return estimates, the extended series uses UNHCR figures as a baseline and supplements them with additional evidence on forced returns and unregistered populations. The extended series therefore provides broader geographic and temporal coverage, while the core series offers more consistent measurement.

4.3 Comparison of GRR and UNHCR Data

Table 3 compares GRR with UNHCR in the core sample. The comparison excludes populations absent from UNHCR’s data, such as Palestinians under UNRWA, leaving 1,415 dyad-years across 96 origin-asylum dyads. At the aggregate level, the two datasets agree closely on voluntary return. UNHCR records 19.1 million returns across the comparison sample, compared with 19.5 million voluntary returns in GRR.

Once GRR’s 3.6 million forced returns are included, however, UNHCR records only 83% of total realized return. Of the 157 dyad-years in which GRR identifies forced return, 49 (31%) are invisible in UNHCR: it records no return in that dyad-year, the preceding year, or the following year. Another seven appear only as token movements, where UNHCR’s count across those three years is positive but less than one-tenth of GRR’s forced-return estimate for that one dyad-year. Together, these episodes account for 47% of GRR’s forced-return volume.

Moreover, 62% of forced-return volume occurs in dyad-years where UNHCR’s figure closely matches GRR’s voluntary-return estimate, suggesting that the forced component is omitted. In other cases, coercive movements appear in UNHCR data as voluntary repatriation. For example, GRR classifies Rohingya returns from Bangladesh between 1994 and 1997 as forced, while UNHCR records approximately the same number as voluntary.

⁷The 75% threshold is calculated using UNHCR data, supplemented with data on Palestinian refugees under UNRWA and Mozambican refugees in South Africa, whom the UNHCR data list under an unknown country of asylum.

Table 3: Comparison of GRR and UNHCR return data

	1980s	1990s	2000s	All
Sample				
Dyad-years compared	319	563	533	1,415
Dyads	51	83	79	96
Aggregate calibration				
GRR voluntary return	1,770,932	10,050,798	7,643,847	19,465,577
GRR forced return	259,027	957,565	2,337,900	3,554,492
UNHCR recorded return	1,250,530	9,757,187	8,105,436	19,113,153
UNHCR / GRR voluntary	0.71	0.97	1.06	0.98
UNHCR / GRR total	0.62	0.89	0.81	0.83
Episode capture				
Forced-return episodes	21	61	75	157
invisible in UNHCR	14 (67%)	14 (23%)	21 (28%)	49 (31%)
token in UNHCR	0	2	5	7
Share of forced-return volume				
Forced-return volume in invisible episodes	71%	16%	34%	32%
in invisible or token episodes	71%	26%	53%	47%
where UNHCR $\approx$ GRR voluntary	60%	36%	72%	62%

Notes: The comparison covers the GRR core dataset, 1980–2008, with asylum countries outside Europe and North America. It excludes dyad-years in which GRR’s voluntary-return estimate is copied from or defined using UNHCR data. The 2000s column covers 2000–2008. An episode is a dyad-year in which GRR records forced return. *Invisible* means that UNHCR records no return in that year or either adjacent year; *token* means that UNHCR records a positive number smaller than one-tenth of the GRR forced-return estimate. $UNHCR \approx GRR \text{ voluntary}$ denotes forced-return volume in dyad-years where UNHCR’s figure is within 5% of GRR’s voluntary-return estimate.

As Table 3 shows, the discrepancies are greatest in the 1980s but persist into later decades. Forced-return episodes remain likely to be omitted through the 2000s. Neither dataset provides an independent ground truth for every movement, but the concentration of disagreement in forced-return episodes is consistent

5 Empirical Analysis

The empirical analysis proceeds from aggregate patterns to within-dyad associations. I first evaluate H2 by comparing how frequently asylum-side shocks and post-war periods occur, how many refugees are exposed to them, and how observed return is distributed. To assess H1, I then examine whether return rates increase after host-country shocks within origin-asylum contexts, including while conflict continues in the country of origin. The case studies subsequently assess whether the mechanisms connecting these shocks to return are consistent with the argument.

5.1 Explanatory Variables

Forced-Return Explanations. *Forced Return Policy* is an ordinal measure of coercive policy at the asylum–origin dyad level, hand-coded from the WRS and U.S. Department of State human rights reports. It equals 0 when there is no policy, 1 for

moderate coercion, and 2 for severe coercion. Moderate policies include official threats or encouragement to return. Severe policies include implemented measures intended to produce widespread return, such as workplace raids or the revocation of residence permits. The coding is based on the policy imposed, not the amount of return subsequently observed; the Appendix provides the complete coding rules.⁸

Conflict in COA records the maximum UCDP conflict intensity in the country of asylum in a given year: no conflict, low-intensity conflict with 25–999 battle deaths, or high-intensity conflict with at least 1,000 battle deaths (UCDP, 2024). This is based on conflict location; a country participating in a war abroad is not coded as experiencing conflict in asylum. Both forced-return variables enter as indicators, with no forced-return policy and no conflict as the respective reference categories.

Preference-Based Explanations. Individual return preferences are not observed in cross-national data. I therefore include macro-level conditions that existing accounts identify as shaping the relative attractiveness of origin and asylum as controls when estimating the forced-return relationships.

For the country of origin, *Conflict in COO* distinguishes no, low-intensity, and high-intensity conflict, analogous to the COA measure, but is hand-coded to match the UCDP conflict that produced the refugee population rather than unrelated fighting in the same country. *War Ending* identifies the first three peaceful years after a high-intensity conflict spell ends in the country of origin. A year is excluded from this window if the conflict restarts. *Civil Liberties Restrictions (COO)* is the Freedom House civil liberties index, which ranges from 1 for the greatest protection to 7 for the least, and *GDP (COO)* is logged GDP per capita from the World Bank.

For the country of asylum, *GDP (COA)* is logged GDP per capita, and *Refugee Rights* is the DWRAP index of de jure refugee and asylum policy, with higher values indicating more liberal policies (Blair, Grossman and Weinstein, 2022). *Conflict Duration* is the logged cumulative number of years of conflict in the origin country. Because there is no data on when individual refugees entered asylum, this measure serves as a proxy for time in asylum.

Conflict duration is measured as the number of years elapsed in the origin-country conflict spell from which a given refugee population fled. The counter runs while conflict is active at either intensity, tolerates a single quiet year, and resets after two consecutive years without conflict, so it measures the age of an ongoing war rather than a country’s cumulative history of violence. Because the underlying conflict variable is matched to the specific war driving each displacement, the measure varies across the hosts of one origin where their displacements trace to different conflicts. Two-thirds

⁸An additional external measure of forced-return policies from Amnesty International reports is underway.

Table 4: Exposure and observed return across post-war and asylum-shock contexts

Context	Dyad-years	Stock-years at risk (m)	Return (m)	Share of return	Return rate
Panel A. Any forced-return policy or asylum conflict					
Neither	3,355 (65.7%)	221.30 (33.3%)	4.94	14.8%	2.23%
Post-war window only	242 (4.7%)	27.92 (4.2%)	2.18	6.6%	7.81%
Asylum shock only	1,302 (25.5%)	342.50 (51.5%)	19.39	58.3%	5.66%
Both	210 (4.1%)	73.83 (11.1%)	6.77	20.3%	9.16%
Panel B. Severe policy or high-intensity asylum conflict only					
Neither	4,225 (82.7%)	402.19 (60.4%)	11.83	35.5%	2.94%
Post-war window only	381 (7.5%)	77.84 (11.7%)	5.68	17.1%	7.30%
Asylum shock only	432 (8.5%)	161.61 (24.3%)	12.51	37.6%	7.74%
Both	71 (1.4%)	23.91 (3.6%)	3.26	9.8%	13.65%
<i>All dyad-years</i>	5,109 (100.0%)	665.55 (100.0%)	33.28	100.0%	5.00%

Notes: Dyad-years with more than 1,000 refugees at risk in the previous year, excluding 1989. Panel A defines an asylum-side shock as any forced-return policy or armed conflict in the country of asylum. Panel B uses a narrower definition comprising severe forced-return policy or high-intensity asylum conflict. Contexts are mutually exclusive within each panel. The post-war window covers the first three peaceful years following the end of a high-intensity origin-country conflict. The return rate equals observed return divided by refugee stock-years at risk.

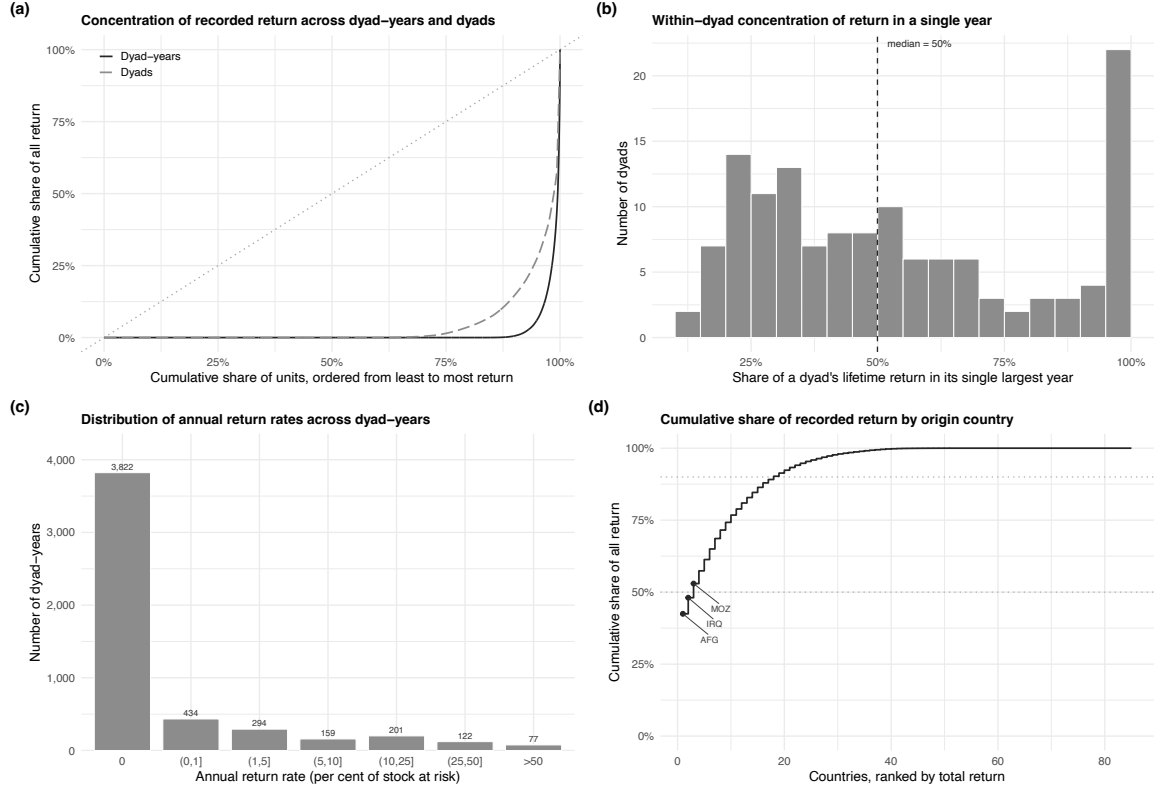
of dyad-years have no active spell, so the standardized measure separates ongoing war from its absence more sharply than it distinguishes long wars from short ones. It enters as a control, and none of the reported results depends on how it is coded.

The pooled model also includes six time-invariant measures absorbed by dyad fixed effects in the main specifications: whether the host is bound by the 1951 Convention or 1967 Protocol for the relevant origin population, accounting for states with geographic restrictions; distance and contiguity; common spoken language from CEPII GeoDist; religious similarity from the Correlates of War World Religion Project; and the share of the origin population with transnational ethnic kin in the asylum country from EPR-TEK (Vogt et al., 2015). The pooled model also includes log refugee stock as a covariate, in addition to its use as the offset, because it lacks dyad fixed effects to absorb persistent differences in population size across dyads.

5.2 Aggregate Distribution of Return

Figure 3 shows that refugee return is concentrated in a small number of displacement contexts and, within those contexts, in a handful of years. The top 5% of dyad-years account for 93.8% of all recorded return, and the top 1% for 63.5%. Among dyads that record any return, the median experiences 49.9% of its lifetime return in a single year, and 19.3% experience at least 90% in one year. Most dyad-years with more than 1,000 refugees record no return, while Afghanistan alone accounts for 42.5% of

Figure 3: Concentration of recorded refugee return



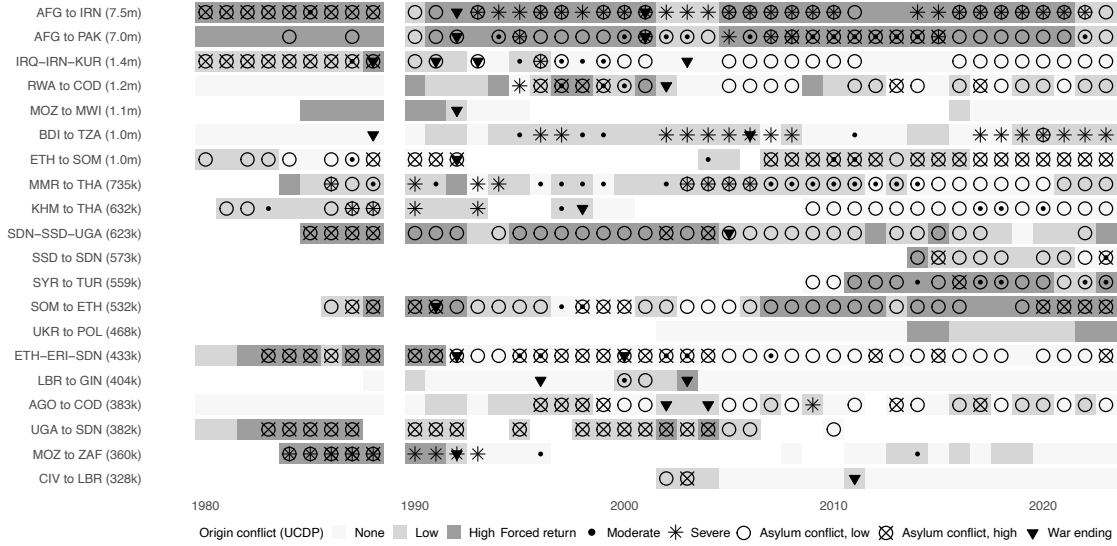
Notes: Panel (b) includes the 135 dyads observed for at least five years that record any return; panel (c) includes dyad-years with more than 1,000 refugees at risk. Panels (a), (b), and (d) use the full GRR panel of 7,498 dyad-years across 292 dyads. Dotted lines in panels (a) and (d) mark perfect equality and the 50th and 90th percentiles, respectively.

recorded return, the three largest origins for 53.0%, and the ten largest for 76.8%. Realized return is therefore not a steady annual flow. In most dyad-years, few or no refugees return; in a small number, large portions of a population move together. This concentration motivates examining shocks that can affect many refugees at once.

Table 4 examines whether these returns occur around war endings or asylum-side shocks. It divides dyad-years into four mutually exclusive contexts: neither condition, a post-war window only, an asylum-side shock only, or both. The broad definition includes any forced-return policy or conflict in asylum, while the narrow definition includes only severe policies or high-intensity conflict.

The comparison provides descriptive support for H2. Under the broad definition, asylum-shock-only dyad-years account for 58.3% of observed return, compared with 6.6% in post-war-only dyad-years. The same ordering holds under the narrow definition, at 37.6% and 17.1%, respectively. Most return also occurs before conflict at home

Figure 4: Conflict, war endings, and forced return in the 20 dyads with the most return



Notes: The twenty dyads with the most recorded return from 1980 to 2023 account for 78% of all return. Row labels report total return. Shading indicates origin-country conflict intensity; symbols indicate asylum-country conflict, forced-return policies, and war endings, as defined in the legend. Policy symbols overlay conflict symbols when both occur.

ends: 79% of all return, including 17 of the 20 largest episodes, takes place during ongoing origin-country conflict.

The difference between asylum-side shocks and post-war periods is driven primarily by exposure rather than higher return rates. Under the broad definition, asylum-shock-only dyad-years contain 51.5% of refugee stock-years at risk, compared with 4.2% in post-war-only dyad-years, even though their return rate is lower. Under the narrow definition, return rates in the two contexts are nearly identical. Asylum-side shocks therefore account for more aggregate return because far more refugees are exposed to them, not because they consistently produce higher return rates than war endings.

Figure 4 illustrates how common these asylum-side shocks are in the twenty dyads with the greatest total return. It plots each dyad over time, showing periods of origin-country conflict and war endings alongside forced-return policies and conflict in asylum. Asylum-side shocks are strikingly common across these major displacement contexts, including during periods when conflict in the origin country remains ongoing.

These descriptive patterns show where return is concentrated, but not whether asylum-side shocks are associated with higher return within the same refugee populations over time. I therefore turn to within-dyad models that compare each origin-asylum dyad to itself across years.

5.3 Regression Specification

I estimate Poisson pseudo-maximum likelihood (PPML) models with dyad and year fixed effects. The unit of analysis is the directed origin-asylum dyad-year, and the dependent variable is the number of refugees who return during the year. PPML is well suited to a non-negative, highly skewed outcome with many zeros and does not require transforming or dropping zero-valued observations. Dyad fixed effects absorb time-invariant differences across refugee populations and origin-asylum relationships, while year fixed effects absorb shocks common to all dyads. Because the analysis is observational, I interpret them as conditional associations rather than causal effects.

I include the log refugee population at risk as an offset, so the coefficients describe proportional changes in the return rate rather than differences driven by population size. The population at risk is the previous year’s refugee stock.⁹ I restrict the analysis to dyad-years with a refugee stock of more than 1,000 refugees in the dyad, leaving 5,131 observations across 285 dyads. The baseline model is

$$E[Y_{it} \mid \mathbf{X}_{it}] = \exp(\beta' \mathbf{X}_{it} + \alpha_i + \gamma_t + \ln S_{it-1}), \quad (4)$$

where Y_{it} is the number of returnees in dyad i and year t , $\mathbf{X}_{it}$ is the vector of time-varying covariates, α_i and γ_t are dyad and year fixed effects, and $\ln S_{it-1}$ is the offset. Standard errors are clustered by dyad.

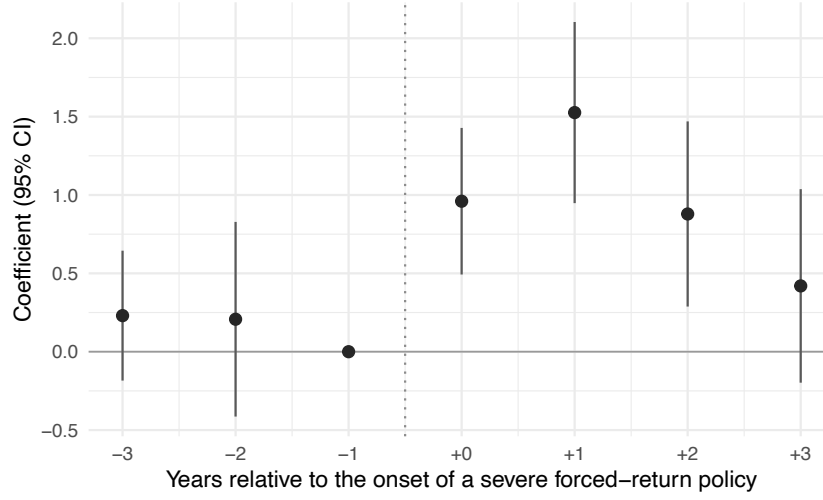
To examine the cross-host implication of H1, I estimate a specification with origin-by-year fixed effects, which absorbs conditions common to refugees from the same origin in a given year and identifies asylum-side coefficients from variation across the countries hosting them at the same time. I also estimate a complementary specification with asylum-by-year fixed effects, which absorbs conditions common to a host in a given year and identifies the forced-return policy coefficient from differences in how that government treats refugee groups from different origins. The latter absorbs asylum-country conflict by construction. Both specifications retain dyad fixed effects and are reported in the Appendix.

I estimate the fixed-effects specification using the full GRR panel (Model 1), the “core” GRR panel covering the years with the most detailed *WRS* reporting (Model 2), and UNHCR return counts over the same underlying core panel (Model 3).¹⁰ Model 4 is a pooled PPML model that drops the dyad fixed effects and retains zero-return dyads. It provides a descriptive analysis of correlates of return across contexts. Table 5 reports

⁹In 21 dyad-years, a large inflow arrives and returns within the same year, causing return to exceed the lagged stock; for these observations, I use the estimated within-year peak population.

¹⁰The final samples for Models 2 and 3 differ slightly because some UNHCR observations are missing and fixed-effects Poisson drops different dyads with no variation in each outcome. The appendix repeats the comparison using the 999 observations common to both models.

Figure 5: Return around the onset of a severe forced-return policy



Notes: Stacked event study around the 23 severe-policy onsets with a fully observed, policy-free three-year pre-period. Each onset forms a separate cohort. Controls are dyads observed throughout the cohort’s window whose host imposed no severe policy during that window. The model is PPML on the return count, with the log stock at risk as an offset, cohort-by-dyad and cohort-by-year fixed effects, standard errors clustered by dyad, and $t - 1$ omitted. Bars show 95% confidence intervals. The Appendix reports a two-year window and a linear specification of the return rate.

the estimates.

5.4 Regression Results

Forced-Return Policies (H1a). The strongest result in Table 5 concerns coercive host-government policy. Relative to years with no policy, a moderate forced-return policy is associated with a 64% higher return rate ($e^{0.50} = 1.64$, $SE = 0.21$). A severe policy is associated with 2.8 times the return rate ($e^{1.02} = 2.77$, $SE = 0.21$). Applied to the 6.6% mean rate in the Model 1 sample, this corresponds to an increase from roughly 7% to 18% of the population at risk returning in a year. The association remains positive in the core panel (0.62, $SE = 0.24$) and in the pooled model (1.03, $SE = 0.22$).

I find no evidence that this association depends on peace at home. Of the 129 severe-policy dyad-years in the estimation sample, only 30 fall within the post-war window, while 82 occur during ongoing conflict in the country of origin. When severe policy is interacted with ongoing origin-country conflict in a separate model, the severe-policy coefficient is 1.01 ($SE = 0.40$) and the interaction is 0.02 ($SE = 0.38$, $p = 0.95$). The estimated association between severe coercion and return is therefore nearly identical whether or not conflict continues in the origin country, as H1a anticipates.

Figure 5 examines the concern that severe policies merely coincide with return already

Table 5: Poisson pseudo-maximum likelihood estimates of refugee return

Term	(1) Extended GRR	(2) Core GRR	(3) UNHCR	(4) Pooled
Forced return				
Forced Return: severe	1.02*** (0.21)	0.62* (0.24)	-0.02 (0.39)	1.03*** (0.22)
Forced Return: moderate	0.50* (0.21)	0.40* (0.17)	0.03 (0.16)	0.39* (0.17)
Conflict in COA: high	0.54† (0.30)	0.92** (0.29)	0.16 (0.37)	0.36 (0.33)
Conflict in COA: low	0.82*** (0.21)	0.65** (0.23)	-0.09 (0.24)	0.53** (0.17)
Preference-based explanations				
War Ending	0.63** (0.20)	0.56* (0.26)	1.05*** (0.31)	0.72** (0.25)
Conflict in COO: high	0.13 (0.35)	0.11 (0.38)	0.92* (0.42)	0.15 (0.37)
Conflict in COO: low	0.02 (0.22)	0.08 (0.24)	0.14 (0.27)	0.05 (0.32)
GDP (COO)	0.38 (0.29)	0.36 (0.36)	0.05 (0.42)	-0.55*** (0.16)
GDP (COA)	-1.15† (0.64)	-0.15 (0.71)	0.69 (0.85)	-0.75*** (0.17)
Civil Liberties Restrictions (COO)	-0.55*** (0.16)	-0.49* (0.20)	0.08 (0.19)	-0.58*** (0.10)
Refugee Rights	0.04 (0.16)	0.02 (0.19)	-0.03 (0.29)	0.06 (0.08)
Conflict Duration	-0.09 (0.09)	-0.04 (0.13)	0.02 (0.13)	-0.00 (0.13)
<i>Factors in the pooled specification only</i>				
Refugee Convention	—	—	—	0.25 (0.21)
Distance (log)	—	—	—	0.01 (0.14)
Contiguous	—	—	—	0.65 (0.41)
Refugee Stock at Risk (log)	—	—	—	-0.06 (0.06)
Common spoken language	—	—	—	-0.51** (0.18)
Religious similarity (z)	—	—	—	0.26** (0.09)
Kin share of origin population (z)	—	—	—	-0.09 (0.07)
Specification and fit				
Outcome	GRR return	GRR return	UNHCR return	GRR return
Sample	Extended GRR	Core GRR	Core GRR	Extended GRR
Years	1980–2022	1980–2008	1980–2008	1980–2022
Dyad fixed effects	Yes	Yes	Yes	No
Year fixed effects	Yes	Yes	Yes	No
Dyad-years	2,327	1,056	1,075	3,810
Dyads	110	73	75	249
Pseudo R ²	0.794	0.809	0.788	0.712

Notes: COO = country of origin; COA = country of asylum. All models are PPML with the log refugee stock at risk as an offset and standard errors clustered by dyad. The core sample covers 1980–2008 and excludes asylum countries in Europe and North America; the extended sample is the full panel. Column 3 differs from column 2 only in the outcome, using UNHCR voluntary return rather than GRR total return. Column 4 has no fixed effects and instead includes dyad-level measures directly. Continuous variables are standardized separately.

underway. Governments might adopt such policies to accelerate an existing return movement, in which case policy would track rather than precede return. I estimate a stacked event study around each policy onset to address this concern, comparing the treated dyad with dyads whose hosts impose no severe policy during the same window. The estimates show no pre-trend two or three years before onset: the coefficients are 0.23 at $t - 3$ and 0.21 at $t - 2$, and the leads are jointly indistinguishable from zero ($p = 0.55$). Return rises in the onset year (0.96, $SE = 0.24$), peaks one year later (1.53, $SE = 0.29$), and declines thereafter. A linear model of the return rate produces the same pattern (see appendix). This timing is consistent with severe policy preceding increased return, although the evidence remains suggestive.¹¹

In the Appendix, I report a specification with origin-by-year fixed effects, comparing refugees from the same origin in the same year across asylum countries.¹² The severe forced-return coefficient remains large and significant at 1.34 ($SE = 0.34$); moderate coercion and low-intensity conflict in asylum are also positively associated with return, while the high-intensity conflict estimate remains positive but imprecise. A complementary asylum-by-year specification compares different refugee groups within the same asylum country and year, exploiting variation in whether the host targets some groups with severe forced-return policies but not others. Severe forced-return policy again remains positively associated with return (1.58, $SE = 0.80$).

Model 3 replaces GRR realized return with UNHCR voluntary return. On the same underlying panel, the four asylum-side coefficients shrink from 0.40–0.92 in Model 2 to -0.09 – 0.16 in Model 3. This implies that neither forced-return policies nor conflict in the asylum country is associated with return, a pattern difficult to reconcile with the movements documented in the archival record. The origin-side results are also unexpected. War endings are strongly associated with return (1.05, $SE = 0.31$), but so is high-intensity conflict in the country of origin (0.92, $SE = 0.42$), while no other explanatory variable is distinguishable from zero. Taken together, these results are difficult to reconcile with return driven primarily by improving conditions at home. The contrast between Models 2 and 3 is instead consistent with GRR recovering asylum-driven movements that the official series omits.

Conflict in Asylum (H1b). Conflict in the country of asylum is also associated with more return, although the evidence is less precise. In Model 1, low-intensity conflict is associated with 2.3 times the return rate observed during peace ($e^{0.82}$, $SE = 0.21$), while high-intensity conflict is associated with 1.7 times the return rate ($e^{0.54}$, $SE = 0.30$). Only the low-intensity estimate is statistically distinguishable from zero, the two coefficients do not differ significantly ($p = 0.34$), and their ordering reverses in the core panel. The evidence therefore supports an association between conflict

¹¹The event study contains only 23 onsets across 17 dyads, leaving the pre-policy estimates imprecise. Because $t - 1$ is omitted, it also cannot rule out anticipation immediately before implementation.

¹²The resulting sample is 1,397 dyad-years across 88 dyads.

in asylum and return, but not a clear difference by conflict intensity. Leave-one-out analyses in the Appendix further show that the weak high-intensity result is sensitive to the inclusion of Sub-Saharan Africa.

This association is not confined to periods when the origin country is at peace. For high-intensity asylum conflict, the interaction with ongoing origin-country conflict is small and statistically indistinguishable from zero (0.15, SE = 0.54, $p = 0.78$). For low-intensity conflict, the association is instead stronger during ongoing origin conflict (interaction = 0.64, SE = 0.29, $p = 0.03$). H1b therefore receives qualified support. The two asylum-side measures also capture largely distinct episodes: only 57 of 842 conflict-affected dyad-years have any forced-return policy.

Preference-Based Explanations. The results also show that refugees respond to conditions in origin and asylum. The coefficient on the post-war window is 0.63 (SE = 0.20) in Model 1, implying a return rate 1.9 times higher during the first three peaceful years after a high-intensity war ends. In the core panel, the corresponding coefficient is 0.56 (SE = 0.26), and on UNHCR’s voluntary-return count it is 1.05 (SE = 0.31). Peace thus remains an important predictor of return.

The broader level of violence at home is less predictive. Neither low- nor high-intensity conflict in the origin country differs significantly from peace (0.02, SE = 0.22; and 0.13, SE = 0.35). Political conditions matter more consistently: a one-standard-deviation increase in civil-liberties restrictions in the origin country is associated with less return (−0.55, SE = 0.16), with a similar estimate in the core panel (−0.49, SE = 0.20). Return is also lower as asylum-country income rises (−1.15, SE = 0.64), although this estimate is imprecise. Origin-country income, statutory refugee rights in asylum, and conflict duration are not distinguishable from zero in the fixed-effects models.

The pooled model broadly reproduces the main patterns in the fixed-effects results: forced-return policies and conflict in asylum are positively associated with return, war endings predict more return, and civil-liberties restrictions at home predict less. Income matters more in the pooled comparison, with higher GDP in both origin and asylum associated with lower return. Among the time-invariant correlates, return is lower where origin and asylum share a widely spoken language (−0.51, SE = 0.18), higher where their religious composition is more similar (0.26, SE = 0.09), and unrelated to transnational ethnic kin (−0.09, SE = 0.07). Refugee Convention status is also unrelated to return (0.25, SE = 0.21). Analysis in the Appendix finds that war endings are not predictive of return from wealthier asylum countries, but are associated with greater return from poorer ones. Interactions with origin-country income and civil-liberties restrictions are not distinguishable from zero.

Taken together, these results do not reject preference-based explanations. Refugees return at higher rates after wars end, remain abroad when repression at home is greater,

and appear less likely to leave wealthier asylum countries. The results instead show why these explanations are incomplete. Coercive policy and conflict in asylum are also associated with substantial return, including while conflict continues in the country of origin.

Model-Adjusted Aggregate Comparison. The mutually exclusive descriptive comparison shows where observed return is concentrated but does not account for compositional differences across dyad-years. As a model-adjusted check, a table in the appendix uses the pooled PPML in Model 4 to compare fitted return under the observed data with predictions obtained after setting each condition to its reference value, holding the remaining covariates fixed. The resulting ordering parallels the descriptive comparison: setting severe forced-return policy to zero reduces predicted return by 25.0%, compared with 16.9% for the post-war window and 4.4% for high-intensity conflict in asylum. The correspondence indicates that the descriptive ordering between severe policy and post-war periods is not solely an artifact of the observed composition of exposed dyads. These quantities are model-implied descriptive contrasts, not estimates of the number of refugees caused to return by each condition.

6 Alternative Explanations and Robustness

I address three broad concerns about the interpretation of the results. The Appendix reports the complete robustness analyses.

The first concern is the validity of the forced-return policy measure. Because coding archival reports requires judgment, GRR makes each classification auditable by providing the supporting text and coding rules. Policies are coded based on their intent and implementation, not on whether forced return actually occurs. Consistent with this, 36% of moderate-policy and 8% of severe-policy dyad-years record no return. Severe policy also remains significant when the outcome is restricted to returns coded as voluntary. As a consistency check, forced-return policies are less common where DWRAP records stronger protections against refoulement. Finally, I apply the same coding rules to Amnesty International reports to assess whether the classifications can be reproduced using an independent source corpus.¹³

A second concern is that other processes may explain the association. One alternative explanation is that return during conflict anticipates peace. Anticipation cannot account for the main result. There is some evidence of anticipatory return: return is modestly elevated before war endings (0.47, SE = 0.20), and coercion associated with an approaching ending predicts additional return (interaction = 1.45, SE = 0.51). However, this does not negate the main result on forced return. Only 10 of the 129 severe-policy dyad-years in the estimation sample fall within three years before a war

¹³The data collection and analysis is underway.

ending, and only three occur in the immediately preceding year. Excluding every dyad-year within three years on either side of a war ending leaves the severe-policy coefficient at 1.02 (SE = 0.30), indistinguishable from the baseline.

Another possibility is that the higher return associated with conflict reflects rejected asylum seekers returning rather than pressure on established refugee populations. This is unlikely for two reasons. First, asylum seekers constitute a small share of the relevant populations: in severe-policy dyad-years, their median share is 0.0% and exceeds 10% in only 9% of cases. Large asylum-seeker populations are concentrated in the Global North, partly because many countries in the Global South instead automatically recognize people fleeing particular conflicts as *prima facie* refugees (United Nations High Commissioner for Refugees, N.d.). Second, severe policies generally target large, long-established refugee populations. Ninety-four percent of severe-policy dyad-years involve at least 25,000 refugees, and the median occurs nine years after the population first exceeded that threshold. The association is therefore unlikely to be driven primarily by rejected or recent asylum applicants.

Beyond these checks, the severe-policy association is robust to alternative measures of conflict, war endings, repression, refugee stocks, and return; different estimators and clustering strategies; lagged covariates; and the exclusion of refugee-warrior cases, particular regions, and the largest origins, asylum countries, dyads, and return episodes. Influential-case checks are especially important because of the concentration of return. Removing the five largest dyads increases the severe-policy coefficient from 1.02 to 1.41, while removing the top 1% of dyad-years increases it to 1.50; both remain significant at $p < 0.001$. The preference-based estimates are less stable: restrictions on civil liberties and stronger ties to asylum generally predict less return, while the post-war estimate is sensitive to alternative samples and outcomes and origin-country conflict is consistently unrelated to return. Across these specifications, the severe-policy estimate remains substantively unchanged.

7 Case Studies

To complement the panel analysis, I compare Afghan refugees in Iran and Pakistan and Rwandan refugees in Burundi, the DRC, and Tanzania. These cases mirror the cross-host comparison above: refugees from the same origin faced broadly similar conditions at home but different coercive and security conditions across asylum countries. The case studies are illustrative rather than designed to identify causal effects or provide formal process tracing. They show how the mechanisms proposed above operated in major return episodes and how return varied across asylum contexts.

7.1 Afghan Refugees in Iran and Pakistan, 1993–2001

Afghanistan remained at war throughout the 1990s. After the Najibullah government fell in April 1992, rival mujahedin factions fought for control of Kabul, and the Taliban emerged in late 1994, capturing Herat in September 1995 and Kabul a year later. Large numbers of refugees returned from both Iran and Pakistan after Najibullah's fall amid expectations of peace in 1992–1993. But return continued even after renewed fighting made clear that the war was not over.

Iran began pressuring its roughly 2.8 million Afghan refugees to leave in 1993. Early measures included the “failure to renew residence cards, cutoffs of assistance, withdrawal of subsidies, and closures of schools and marketplaces” (1994, Iran). By 1995, Iran increasingly treated some Afghans as migrants rather than refugees, issuing “six-month permits in order to enable them to leave the country” and warning that those whose permits expired would be “apprehended and deported in the same way as any other undocumented aliens” (1995, Iran). In March 1995, the government announced that all Afghan refugees would be required to leave by March 1997 and subsequently refused to renew work permits, raided Afghan workplaces, and withdrew health subsidies. By June and July, “as many as 1,500 were repatriating daily” (1996, Iran).

Iran temporarily eased this pressure after the Taliban captured Herat in September 1995, through which many refugees returning from Iran had to travel. Authorities considered rerouting returnees through Turkmenistan but abandoned the plan. Large-scale deportations resumed later in the decade.

Pressure in Pakistan was generally weaker but intensified during two periods. In 1994–1995, UNHCR and the World Food Programme phased down assistance, including food rations and camp services, after the government expressed the expectation that most Afghans would repatriate following Najibullah's fall and rejected permanent settlement. Pressure intensified again in 2000–2001. In early 2001, authorities in Northwest Frontier Province authorized police to detain and deport Afghans without valid passports and visas, including long-term refugees, producing what a UN-commissioned study described as “mass harassment in cities and officially sanctioned forcible return to Afghanistan in a systematic manner” (2002, Pakistan).

Return patterns broadly track these differences, although not every policy episode produced an immediate increase. In 1992–1993, return was high from both Pakistan (22.8% and 17.3%) and Iran (9.9% and 16.5%). As Iran sustained severe pressure in 1994–1995, return remained high at 11.8% and 14.9%, while Pakistan's rate fell to 5.5% before rising to 12.8% during its own severe-policy episode in 1995. When Iran eased pressure after the Taliban captured Herat, its return rate fell to 3.9% in 1996 and 0.4% in 1997, compared with 14.0% and 6.8% from Pakistan. As large-scale deportations from Iran resumed, return there rose again to roughly 9–10% annually from 1998–2000, while Pakistan's rate declined from 6.8% to 3.8%. Conditions in Afghanistan were

common to refugees in both countries, but return varied substantially with changes in their ability to remain in each asylum country.

7.2 Rwandan Refugees in Burundi, the DRC, and Tanzania, 1996

After the 1994 genocide and the subsequent victory of the Tutsi-led Rwandan Patriotic Front, hundreds of thousands of Hutu fled to neighboring countries, including members of the former government and armed forces implicated in the genocide. Although the new Rwandan government encouraged return and UNHCR intensified repatriation efforts in 1995, most refugees remained in Burundi, the DRC, and Tanzania. Many feared violence or arrest in Rwanda, while former-regime leaders intimidated refugees and sometimes prevented them from leaving the camps. Efforts to promote voluntary repatriation therefore produced relatively little return on their own.

In Burundi, mass return followed direct government coercion. During 1996, officials told Rwandan refugees they were no longer welcome, soldiers and local militias attacked and burned camps, and authorities forcibly transported refugees across the border. In July, soldiers expelled 15,000 refugees and temporarily blocked UNHCR access to two camps. By year's end, "about 98 percent of Rwandan refugees had left Burundi, due to fear for their safety and coercive policies by Burundian authorities" (1997, Burundi).

In the DRC, repatriation efforts likewise produced little movement until civil war reached the refugee camps in October 1996. Rebel attacks dispersed some former-regime leaders who had obstructed departure, while the fighting itself forced refugees from the camps. More than half a million returned within four days in what UNHCR described as the "largest and swiftest repatriation in memory" (1997, Zaire). Some returned willingly once freed from camp leaders' control, while others returned to escape the fighting or because they had no other option. At least 200,000 instead moved deeper into the DRC, showing that displacement from the camps did not uniformly produce a preference for return.

Return from Tanzania also remained limited until government policy changed. Only 2% of Rwandan refugees returned during the first eleven months of 1996. On December 5, however, following the return from the DRC, the government proclaimed that "all Rwandese refugees can now return to their country in safety" and imposed a year-end deadline for departure (1997, Tanzania). Hundreds of thousands initially moved deeper into Tanzania, away from Rwanda, but security forces blocked their movement, closed the camps, and directed them toward the border. Return subsequently reached as many as 15,000 people per hour, while approximately 50,000 avoided repatriation by dispersing elsewhere in Tanzania or presenting themselves as Burundian.

The staggered timing is difficult to explain through conditions in Rwanda alone. Refugees

in all three countries faced the same Rwandan government and broadly similar conditions at home, yet mass return followed coercion in Burundi, conflict reaching the camps in the DRC, and an expulsion order in Tanzania. Refugees' attempts to move deeper into the DRC and Tanzania further show that many did not simply prefer return once it became possible. Across all three hosts, mass movement followed changes in refugees' ability to remain in asylum.

8 Conclusion

This paper revisits a foundational premise of the international refugee regime: that refugees receive protection while conditions at home remain unsafe and return once those conditions improve. Using the Global Refugee Return dataset, which covers 293 origin–asylum dyads from 1980 to 2023, I show that realized return often follows a different pattern. Most recorded returns occur while conflict is still ongoing in the country of origin, while many refugees remain abroad long after war has ended.

I argue that this pattern arises because realized return depends not only on whether refugees want to go home, but also on whether they can remain in asylum. Host-government coercion and armed conflict in the country of asylum can make staying infeasible, inducing refugees to return even when conditions at home remain unsafe. Two main findings support this argument.

First, conditions in asylum are strongly associated with when refugees return. Comparing the same refugee populations over time, severe forced-return policies are associated with 2.8 times the annual return rate observed in their absence, compared with 1.9 times the rate following the end of a war. The association persists when comparing refugees from the same origin in the same year across different asylum countries. Return also rises sharply when severe policies begin, with no evidence of an increase in the preceding two to three years, and the relationship is robust across alternative measures, specifications, and samples. Conflict in asylum is likewise associated with greater return, although the estimates are less precise.

Second, severe asylum-side shocks account for substantially more observed return globally than post-war periods. 37.6% of return occurs in dyad-years experiencing a severe forced-return policy or high-intensity conflict in asylum but no recent war ending, compared with 17.1% in post-war-only dyad-years. This difference is driven primarily by exposure rather than higher conditional return rates, as far more refugees are exposed to severe asylum-side shocks. The same ordering appears in the model-adjusted comparison, where removing severe forced-return policies produces a larger reduction in predicted return than removing post-war periods. War endings therefore remain an important predictor of return, but severe asylum-side shocks coincide with substantially more realized return because they affect more refugees.

These findings have several implications. Theoretically, they extend existing research on refugee return by distinguishing preferences from realized behavior. Prior studies have taught us a great deal about when refugees want to return, particularly how conditions at home and in asylum shape those preferences. But these findings should translate into predictions about realized return only when refugees can meaningfully choose whether to remain abroad. Government coercion or armed conflict can remove that option, producing return among refugees who would otherwise prefer to stay. Realized return is therefore a constrained choice: understanding when refugees actually return requires considering both where they prefer to live and whether they are able to remain in asylum.

This distinction matters for how repatriation is understood within the refugee regime. Voluntary return after peace can provide a durable solution when refugees want to return and can do so safely, but observed return is not necessarily evidence that these conditions have been met. Some of the largest return movements instead occur under coercive pressure and while conflict continues at home. This raises questions about the welfare and reintegration of refugees who return under such conditions (Fransen, Ruiz and Vargas-Silva, 2017), as well as the consequences of mass return for origin communities and post-conflict politics (Schwartz, 2019; Blair, Van Dijke and Wright, 2026). The conditions driving return may therefore shape both returnees' prospects and the consequences of repatriation after they arrive home, and warrant further investigation.

The analysis also underscores the difficulty of measuring refugee return. Existing UNHCR statistics omit many forced and spontaneous returns, including the coercive movements central to the argument here. GRR provides reconstructed measures of refugee stocks and realized return, together with new data on forced-return policies, but it relies on expert reporting and remains an improved yet imperfect measure of movements that are often difficult to observe with precision. Future research should continue to triangulate administrative and archival sources with new forms of measurement, including satellite-based estimates of population movement and border-crossing data.

Several questions follow. First, when and why do governments impose forced-return policies? The Appendix suggests that severe coercion is concentrated in large, unresolved displacement crises. Population size is its most consistent correlate, while crisis duration, origin-country violence, and host characteristics are much less robust predictors. More detailed case analysis could examine the domestic pressures behind these decisions, the role of international aid and diplomacy, and whether host governments face meaningful costs for violating the refugee regime's core protection principle. Future research should also examine the role of origin governments, which may encourage return to signal normalization, but resist it when they lack the capacity to absorb returnees or regard particular groups as political or security threats.

Second, the analysis captures movement from an asylum country to the country of origin but not refugees' broader trajectories. Because it begins after refugees enter

asylum, it excludes maritime pushbacks, border closures, and other measures used to prevent access to protection (FitzGerald, 2019; Goodwin-Gill, 2011; Schwartz, 2025). These measures are especially common in high-income countries, where governments often seek to prevent refugees from reaching their territory rather than remove them after arrival. The data also do not capture temporary return or repeated movement between origin and asylum (Masterson and Vidarte, 2026), trajectories that remain difficult to study cross-nationally because existing data rarely track these movements.

Third, the persistence of displacement after peace requires explanation. Of the 41 wars in the data that ended and remained ended for at least five years, only 12 were followed by the return of even half the refugee population within those five years. This is striking given the temporary logic of the refugee regime, under which protection is expected to end as conditions at home improve. Existing research points to several reasons refugees may remain abroad, including fear of repression, limited housing or livelihoods at home, and stronger economic and social ties in asylum. Yet why some refugee populations remain displaced long after war ends remains poorly understood.

These questions also speak to broader debates about international cooperation, law, and institutions. *Non-refoulement* is the core legal principle of the refugee regime, yet states use policies that induce refugees to return while conflict continues at home. UNHCR sometimes facilitates repatriation under these conditions because refusing to participate may leave refugees to return without assistance or jeopardize the organization's access to refugee populations. This illustrates the constraints international organizations face when implementation depends on state cooperation.

Taken together, the findings shift attention from return alone to the political processes that determine when displacement ends. Repatriation is not simply the endpoint that follows the restoration of peace. It may result from the erosion of protection in asylum, while refugees who are no longer fleeing active war may remain abroad for years or decades. This creates a broader puzzle for the study of international cooperation. Host governments often have the practical ability to push refugees out despite their legal obligations, while providing asylum may require a much longer commitment than the regime's temporary-protection logic suggests. Why, then, does displacement persist after peace, and why do so many states continue to provide asylum rather than force refugees to leave? My book project takes up these questions by examining how refugee preferences and the policies of origin and asylum governments jointly determine when displacement ends, and why states sustain the refugee regime despite weak enforcement and potentially enduring costs.

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Peace or Push? Understanding Global Patterns of Refugee Return

Mae MacDonald

Supplementary Materials

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1 Global Refugee Return (GRR) Dataset

The Global Refugee Return (GRR) dataset is an original global dataset on refugee stocks, returns, and the conditions shaping repatriation. It covers origin–asylum dyad-years from 1980 to 2023. The dataset makes two core contributions: (1) improved stock and repatriation data that corrects systematic errors in UNHCR figures and incorporates forced returns; and (2) new indicators of *de facto* refugee conditions in asylum, including coercive return policies.

Unit of analysis. Each observation records conditions for refugees from a specific country of origin residing in a specific country of asylum in a given year.

Geographic coverage. The dataset includes all **origin countries** with at least 25,000 refugees in exile in any given year, consistent with UNHCR’s threshold for a protracted refugee situation. For each origin country, the dataset includes the principal **asylum countries** that together account for at least 75% of the origin’s total refugee population, calculated using UNHCR refugee stock data. Dyad-years in which no refugee population is present are excluded.

Temporal coverage. The dataset spans 1980–2023. The year 1989 is missing for all variables coded from the World Refugee Survey (WRS), because the 1990 WRS report was not published.

Primary sources. GRR draws on three primary sources: (1) the World Refugee Survey (WRS), 1980–2008, providing annual country reports on refugee stocks and conditions in asylum; (2) the U.S. Department of State Country Reports on Human Rights Practices (USHR), 2009–2023, with data extracted from Section 2.d on freedom of movement; and (3) the UNHCR population statistics database, which supplies the stock and voluntary-return baseline for the years after 2008 and for the high-income asylum countries that enter only in the extended series. Additional variables are drawn from the UCDP Armed Conflict Dataset, the Ethnic Power Relations (EPR) Dataset, DWRAP (Blair et al., 2022), the World Bank World Development Indicators, and the CEPII GeoDist dataset. The UNHCR financing measures are compiled from UNHCR Executive Committee reports and archival documents for 1980–1997 and from UNHCR statistical yearbooks and financial reports for 1998–2023. Neither source contributes to the refugee stock or return series.

Core and extended versions. There are two versions of GRR. The **core** dataset covers Global South asylum countries from 1980 through 2008 and rests on the World Refugee Survey evidence in which I have the greatest confidence: 1,623 dyad-years across 105 dyads, 58 origin populations and 61 asylum countries. The **extended** series adds high-income asylum countries and continues through 2023 using the U.S. Department of State’s annual country reports on human rights practices, giving 7,526 dyad-years across 293 dyads, 87 origin populations and 93 asylum countries. Because the State Department reports do not always provide stock or voluntary-return estimates, the extended series takes UNHCR figures as a baseline and supplements them with additional evidence on forced returns and unregistered populations. The extended series therefore offers broader geographic and temporal coverage, and the core series more consistent measurement.

1.1 Returns Absent from UNHCR’s Series

The dataset’s return series departs from UNHCR’s most sharply in a small number of large episodes.

Table 1: Largest return episodes recorded by GRR and absent from UNHCR.

Dyad	Year	GRR return	UNHCR return	Unrecorded	Of which forced
IRQ-IRN	1991	1,295,000	0	1,295,000	0
AFG-IRN	2020	1,151,789	947	1,150,842	1,150,842
AFG-IRN	2019	668,209	1,952	666,257	666,257
AFG-IRN	1993	471,500	0	471,500	—
AFG-IRN	2008	409,000	3,656	405,344	406,000
AFG-IRN	2007	379,650	7,462	372,188	363,000
AFG-PAK	1993	358,800	0	358,800	0
ETH-SOM	1984	320,554	0	320,554	0
ETH-SOM	1991	320,000	0	320,000	0
AFG-IRN	2017	274,291	1,202	273,089	273,089
AFG-IRN	2018	275,061	1,972	273,089	273,089
AFG-IRN	2001	263,501	18,000	245,501	120,000

Table 1 lists the twelve dyad-years in which GRR records the most return that UNHCR does not, with the forced component shown alongside. It is placed here, with the description of the dataset, because it is the clearest single illustration of what the GRR coding adds: the largest gaps are overwhelmingly forced movements, which is the case for building the series in the first place. Section 1.5 takes the comparison up systematically.

Table 1 lists the largest return episodes recorded by GRR but absent or nearly absent from UNHCR’s series. In several Afghan cases, UNHCR’s figure is close to GRR’s voluntary-return count and excludes most of the forced-return component. UNHCR records no return for Iraqi Kurds from Iran in 1991 or for Ethiopians from Somalia in 1984 and 1991. The gap across these twelve dyad-years equals 18% of all returns recorded by GRR.

1.2 Codebook

Tables 2–6 describe all variables used in the main analysis. Variables with the prefix **a_** refer to asylum-country characteristics, **o_** to origin-country characteristics, and **d_** to dyad-level variables. Variable suffixes follow a consistent convention: **_z** denotes a standardized (z-scored) variable, **_l** a log-transformed variable, and **_zl** a standardized log-transformed variable. In the regression models, several variables are transformed for estimation. Binary high/low indicators (e.g., **a_conflict_high**, **a_conflict_low**) are derived from the ordinal variables by splitting them into separate dummies for the high and low categories, with zero as the reference. Continuous variables used in standardized or logged form in the regressions are noted below; standardization is performed on the estimation sample (dyad-years with more than 1,000 refugees in the previous year).

Table 2: Codebook: Identifiers

Variable	Description
id	Unique dyad identifier, formatted as COO-COA.
year	Year of observation.
coo	Country of origin name.
coo_iso	Country of origin ISO 3166-1 alpha-3 code.
coa	Country of asylum name.
coa_iso	Country of asylum ISO 3166-1 alpha-3 code.

Table 3: Codebook: Stock and Return Variables

Variable	Description	Source
refugees	Stock of forcibly displaced persons in the dyad at year-end, including refugees, asylum-seekers, individuals in refugee-like situations, and unregistered refugees. Primary stock variable.	WRS (1980–2008); UNHCR + USHR unregistered (2009–2023)
refugees_formal	Stock of forcibly displaced persons excluding unregistered refugees.	WRS (1980–2008); UNHCR (2009–2023)
trepat	Total refugees repatriated (voluntary + forced).	Derived: vrepat + frepat
vrepat	Voluntary repatriation count.	WRS (1980–2008); UNHCR (2009–2023)
frepat	Forced repatriation count.	WRS (1980–2008); USHR (2009–2023)

Table 4: Codebook: Asylum-Country Variables (prefix: **a_**)

Variable	Description	Source
a_conflict	Conflict intensity in the asylum country (UCDP): 0 = none, 1 = low (25–999 battle deaths), 2 = high ($\geq 1,000$). In regressions, split into a_conflict_high (=1 if 2) and a_conflict_low (=1 if 1); no conflict is the reference category.	UCDP Armed Conflict Dataset
a_conflict_bd	Battle deaths in the asylum country (UCDP ‘best’ estimate). Available from 1989.	UCDP Battle-Related Deaths Dataset
a_forced	Coercive forced-return policies in the asylum country at the dyad level: 0 = none; 1 = moderate (threats or encouragement to induce return without policy implementation, or implemented policies targeting <10% of the refugee stock); 2 = severe (implemented policies intended or expected to affect $\geq 10\%$ of the refugee stock). Coding reflects the policy itself, not realized return. In regressions, split into a_forced_high (severe) and a_forced_low (moderate); no forced return is the reference category. See Table 14 for example coding.	Hand-coded from WRS and USHR
a_camp	Ordinal share of the refugee population living in camps: 0 = no camps, 1 = <10% in camps, 2 = 10–50%, 3 = 50–95%, 4 = 95–100%.	Hand-coded from WRS and USHR
a_camp_most	Binary: 1 if most refugees live in camps (a_camp $\in$ {3, 4}), 0 otherwise.	Derived from a_camp
a_urban_z	Standardized (z-score) measure of settlement type. Underlying ordinal variable: 3 if no camps or <10% in camps (a_camp $\in$ {0, 1}), 2 if 10–50% in camps (a_camp = 2), 1 if 50–100% in camps (a_camp $\in$ {3, 4}).	Derived from a_camp
a_funding	UNHCR funding allocated to the asylum country (USD, country-year level).	UNHCR ExCom (1980–2009); UNHCR Yearbooks (2010–2023)
a_funding_pp	UNHCR funding per refugee in the asylum country. In regressions, used in standardized log form (a_funding_pp_zl).	Derived: a_funding / asylum-country refugee stock
a_rights	Refugee rights index (inverse-covariance weighted) from DWRAP. In regressions, standardized (a_rights_z).	Blair et al. (2022)

Table 4: Codebook: Asylum-Country Variables (prefix: **a_**) (*continued*)

Variable	Description	Source
a_gdp_1	Log GDP per capita of the asylum country.	World Bank WDI
a_edu	Mean years of schooling for the 20–24 age cohort of the asylum country’s own resident population – its citizens, not the refugees it hosts (a near-completion measure of recent educational attainment), linearly interpolated from 5-year reporting intervals to annual frequency.	Wittgenstein Centre
a_region	World Bank regional classification of the asylum country.	World Bank

Table 5: Codebook: Origin-Country Variables (prefix: **o_**)

Variable	Description	Source
o_conflict	Conflict intensity in the origin country (UCDP): 0 = none, 1 = low (25–999 battle deaths), 2 = high ($\geq 1,000$). Matched to the specific conflict driving displacement. Missing for non-conflict displacement (e.g., Venezuela). In regressions, split into o_conflict_high and o_conflict_low ; no conflict is the reference category.	UCDP Armed Conflict Dataset
o_conflict_bd	Battle deaths in the origin country (UCDP ‘best’ estimate). Available from 1989. Matched to the specific conflict producing refugees.	UCDP Battle-Related Deaths Dataset
o_conflict_years	Years elapsed in the origin-country conflict spell matched to this displacement. The counter increments in any year of armed conflict at either intensity, carries through a single lapse year, and resets to zero after two consecutive years without conflict; it is the age of the current spell rather than a cumulative total. Left-truncated at 1980, so spells beginning earlier are counted from the panel’s first year. Because o_conflict is matched to the conflict driving each displacement, the counter can differ across the hosts of one origin in the same year. In regressions, used in standardized log form (o_conflict_years_zl), captured as $\log(\text{years} + 0.01)$ and standardized; 68 per cent of dyad-years are zero, so the standardized measure separates an active spell from none more sharply than it distinguishes long spells from short ones.	Derived from UCDP Armed Conflict Dataset
o_funding_repat_pp	UNHCR repatriation funding allocated to the origin country, divided by the origin’s global refugee population (USD per refugee). In regressions, used in standardized log form (o_funding_repat_pp_zl).	UNHCR ExCom (1980–1999); UNHCR Yearbooks (2011–2023). 2000–2010 missing.
o_eth_name	Name of the refugee group’s ethnic identity (e.g., “Kurds,” “Tutsi,” “Bosniaks/Muslims”).	EPR Refugee Dataset
o_eth_id	EPR group identifier (gwgroupid) for the refugee group.	EPR Refugee Dataset; EPR Dataset

Table 5: Codebook: Origin-Country Variables (prefix: o_) (*continued*)

Variable	Description	Source
o_eth_status	EPR power-status category of the refugee group in the origin country (categorical: monopoly, dominant, senior partner, junior partner, powerless, discriminated, irrelevant, self-exclusion, state collapse).	EPR Dataset
o_eth_high	Binary: 1 if the refugee group holds high political status ('monopoly,' 'dominant,' 'senior partner,' or 'junior partner'), 0 otherwise.	EPR Dataset; EPR Refugee Dataset
o_eth_low	Binary: 1 if the refugee group holds low political status ('discriminated' or 'powerless'), 0 otherwise.	EPR Dataset; EPR Refugee Dataset
o_eth_power	Ordinal power score (0–6) derived from o_eth_status : 6 = monopoly, 5 = dominant, 4 = senior partner, 3 = junior partner, 2 = state collapse, 1.5 = discriminated, 1 = powerless, 0.5 = irrelevant, 0 = self-exclusion.	Derived from o_eth_status
o_eth_cat	Three-category collapse of o_eth_status : “In Power” (monopoly, dominant, senior/junior partner), “Excluded” (powerless, discriminated), or “Other” (irrelevant, self-exclusion, state collapse).	Derived from o_eth_status
o_eth_inpower	Binary: 1 if the refugee group is “In Power,” 0 otherwise. Equivalent to o_eth_high .	Derived from o_eth_status
o_eth_excluded	Binary: 1 if the refugee group is “Excluded,” 0 if “In Power,” NA for “Other” categories. Differs from o_eth_low in coding “Other” as NA rather than 0.	Derived from o_eth_status
o_eth_claim	Binary indicator for whether the rebel group in the conflict producing refugees has made an exclusive claim to fight on behalf of an ethnic group (ACD2EPR).	ACD2EPR (Wucherpfennig et al.)
o_civlib_z	Freedom House Civil Liberties index for the origin country, standardized.	Freedom House
o_terror_z	Political Terror Scale for the origin country, standardized.	Political Terror Scale (Amnesty International, Human Rights Watch, U.S. State Department)
o_genocide	Binary indicator for genocide or politicide in the origin country.	PITF Genocide/Politicide Dataset
o_pitf_any	Binary indicator for any PITF political violence in the origin country (ethnic war or revolutionary war).	PITF Ethnic/Revolutionary War Datasets
o_pagree	Binary indicator for a peace agreement signed in the origin country in a given year.	UCDP Peace Agreement Dataset
o_gdp_l	Log GDP per capita of the origin country.	World Bank WDI
o_edu	Mean years of schooling for the 20–24 age cohort of the origin country’s own resident population – citizens who did not flee, not the refugee population abroad (a near-completion measure of recent educational attainment), linearly interpolated from 5-year reporting intervals to annual frequency.	Wittgenstein Centre
o_region	World Bank regional classification of the origin country.	World Bank

Table 6: Codebook: Dyad-Level Variables (prefix: **d_**)

Variable	Description	Source
d_distance	Distance between origin and asylum countries (km). In regressions, used in log form (d_distance_1).	CEPII GeoDist
d_contig	Binary: 1 if the countries share a border, 0 otherwise.	CEPII GeoDist

1.3 Summary Statistics

Table 7 reports summary statistics for every variable in the main specification, followed by the between-dyad measures that enter the pooled column only. Statistics are computed on the estimation sample (dyad-years with more than 1,000 refugees in the previous year).

Table 7: Summary statistics

Variable	Obs	Mean	Std. Dev.	Min	Max
Return and stock					
Total Repatriation	5109	6,513.72	48,844.54	0.00	1,700,000.00
Refugee Stock (Lagged)	5131	129,495.83	400,251.60	1,002.00	6,013,249.00
Return Rate (% of stock at risk)	5109	2.99	11.03	0.00	100.00
Forced return					
Forced Return: severe	4415	0.03	0.18	0.00	1.00
Forced Return: moderate	4415	0.07	0.25	0.00	1.00
Conflict in COA: high	5131	0.07	0.26	0.00	1.00
Conflict in COA: low	5131	0.18	0.38	0.00	1.00
Preference-based explanations					
War Ending	5131	0.09	0.28	0.00	1.00
Conflict in COO: high	5131	0.12	0.32	0.00	1.00
Conflict in COO: low	5131	0.22	0.42	0.00	1.00
GDP (COO)	4845	-0.00	1.00	-2.98	2.22
GDP (COA)	5095	-0.00	1.00	-2.23	1.45
Civil Liberties Restrictions (COO)	5086	0.00	1.00	-2.82	1.44
Refugee Rights	4911	-0.00	1.00	-2.16	3.79
Conflict Duration	5131	0.00	1.00	-0.78	1.83
Measures entering the pooled specification only					
Distance (log)	5069	7.49	1.20	2.35	9.69
Contiguous	5069	0.49	0.50	0.00	1.00
Refugee Stock at Risk (log)	5131	9.92	1.86	6.91	15.61
Refugee Convention	5131	0.85	0.36	0.00	1.00
Common spoken language	5097	0.35	0.48	0.00	1.00
Religious similarity (z)	4956	-0.00	1.00	-1.48	1.63
Kin share of origin population (z)	5131	0.00	1.00	-0.52	3.24

Note: Computed on the filtered sample: dyad-years with a lagged refugee stock above 1,000.

Figure 1 shows the distribution of the dependent variable. The left panel displays the full range of total repatriation counts with the y-axis capped to show detail among non-zero values. The right panel plots positive values on a log scale, showing a roughly log-normal distribution among dyad-years with nonzero repatriation. The heavy concentration of zeros, extreme right skew, and count nature of the dependent variable motivate the use of Poisson pseudo-maximum likelihood estimation with the log stock at risk as the offset.

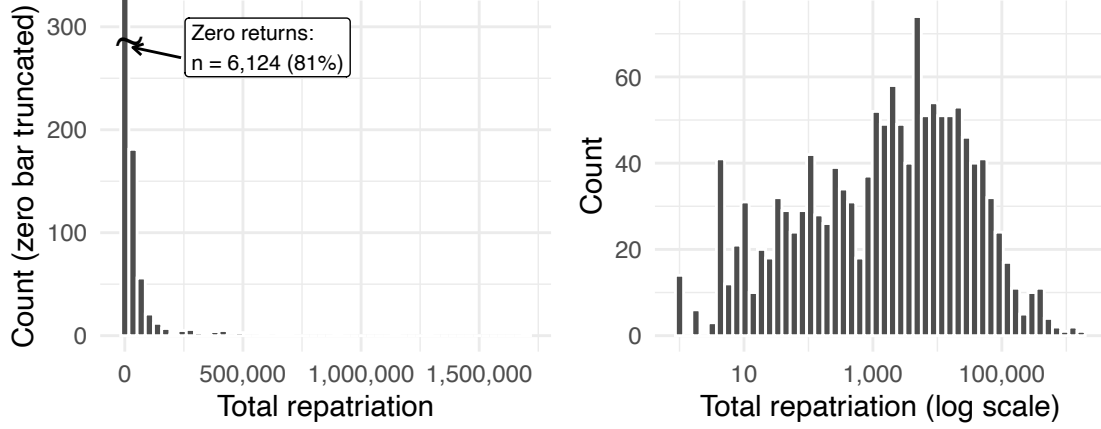


Figure 1: Distribution of the dependent variable (total repatriation). Left panel shows the raw distribution; right panel uses a log scale and excludes zeros.

1.4 Cases

The analysis includes 293 dyads. The tables below list the dyads with the most return, the largest single-year movements, and the largest populations with little or no return. They use the full GRR panel rather than the estimation sample, so they also include dyads omitted from the fixed-effects models.

Table 8: The twenty dyads with the most recorded return, 1980–2023

Dyad	Origin → asylum	Years	Total return	Peak stock	Return/peak (%)	% yrs COO conflict	Push ever
AFG-IRN	Afghanistan → Iran	1980–2023	7,507,327	6,352,317	118	100	Both
AFG-PAK	Afghanistan → Pakistan	1980–2023	7,018,397	3,997,133	176	100	Both
IRQ-IRN-KUR	Iraq (Kurds) → Iran	1980–2023	1,426,658	160,000	892	33	Both
RWA-COD	Rwanda → Congo - Kinshasa	1980–2023	1,190,071	1,000,000	119	49	Both
MOZ-MWI	Mozambique → Malawi	1985–2023	1,126,986	1,085,000	104	44	None
BDI-TZA	Burundi → Tanzania	1980–2023	1,049,147	443,700	236	53	Both
ETH-SOM	Ethiopia → Somalia	1980–2023	1,039,754	705,000	147	52	COA conflict
MMR-THA	Myanmar (Burma) → Thailand	1984–2023	735,024	1,653,500	44	72	Both
KHM-THA	Cambodia → Thailand	1981–2023	631,632	365,000	173	45	Both
SDN-SSD-UGA	Sudan → Uganda	1985–2023	623,004	1,037,898	60	97	COA conflict
SSD-SDN	South Sudan → Sudan	2014–2023	572,695	852,080	67	80	COA conflict
SYR-TUR	Syria → Turkey	2009–2023	559,246	3,737,369	15	87	COA conflict
SOM-ETH	Somalia → Ethiopia	1986–2023	532,300	419,000	127	81	COA conflict
UKR-POL	Ukraine → Poland	2002–2023	468,475	958,519	49	45	None
ETH-ERI-SDN	Ethiopia → Sudan	1980–2023	433,357	730,000	59	26	COA conflict
LBR-GIN	Liberia → Guinea	1988–2023	404,223	420,000	96	14	COA conflict
AGO-COD	Angola → Congo - Kinshasa	1980–2023	383,166	312,700	123	40	Both
UGA-SDN	Uganda → Sudan	1980–2010	382,250	250,000	153	95	COA conflict
MOZ-ZAF	Mozambique → South Africa	1984–2023	359,700	300,000	120	50	Both
CIV-LBR	Côte d'Ivoire → Liberia	2002–2023	328,376	128,067	256	18	COA conflict

Note: Return over peak stock can exceed 100 per cent where a population arrived and left more than once over the period.

Table 9: The forty largest single-year return episodes

Dyad	Year	Origin → asylum	Return	Stock at risk	Rate (%)	COO	COA	Forced
AFG-PAK	2002	Afghanistan → Pakistan	1,700,000	2,100,000	81.0	Low	Low	Moderate
IRQ-IRN-KUR	1991	Iraq (Kurds) → Iran	1,295,000	1,366,400 [†]	94.8	Low	Low	None
AFG-IRN	2020	Afghanistan → Iran	1,151,789	2,701,142	42.6	High	Low	Severe
AFG-PAK	1992	Afghanistan → Pakistan	910,000	3,997,133	22.8	High	Low	None
RWA-COD	1996	Rwanda → Congo - Kinshasa	700,000	900,000	77.8	Low	High	None
AFG-IRN	2019	Afghanistan → Iran	668,209	2,701,142	24.7	High	Low	Severe
MOZ-MWI	1994	Mozambique → Malawi	600,000	700,000	85.7	None	None	None
AFG-IRN	2005	Afghanistan → Iran	489,600	1,702,800	28.8	High	Low	Severe
AFG-PAK	2005	Afghanistan → Pakistan	475,100	967,200	49.1	High	None	Severe
AFG-IRN	1993	Afghanistan → Iran	471,500	2,862,500	16.5	High	Low	Severe

Table 9: The forty largest single-year return episodes (*continued*)

Dyad	Year	Origin → asylum	Return	Stock at risk	Rate (%)	COO	COA	Forced
AFG-IRN	2004	Afghanistan → Iran	419,500	1,600,000	26.2	Low	None	Severe
AFG-PAK	2004	Afghanistan → Pakistan	409,100	1,200,000	34.1	Low	Low	None
AFG-IRN	2008	Afghanistan → Iran	409,000	2,414,700	16.9	High	Low	Severe
UKR-POL	2022	Ukraine → Poland	403,000	1,361,519 [†]	29.6	High	None	None
MOZ-MWI	1993	Mozambique → Malawi	400,000	1,085,000	36.9	None	None	None
SSD-SDN	2023	South Sudan → Sudan	386,793	796,831	48.5	None	High	Moderate
AFG-PAK	2016	Afghanistan → Pakistan	381,275	2,366,950	16.1	High	Low	None
AFG-IRN	2007	Afghanistan → Iran	379,650	2,190,400	17.3	High	Low	Severe
AFG-IRN	2002	Afghanistan → Iran	376,300	2,855,000	13.2	Low	None	Severe
AFG-PAK	1993	Afghanistan → Pakistan	358,800	2,075,000	17.3	High	None	None
AFG-PAK	2003	Afghanistan → Pakistan	343,537	1,500,000	22.9	Low	Low	Moderate
ETH-SOM	1984	Ethiopia → Somalia	320,554	700,000	45.8	None	Low	None
ETH-SOM	1991	Ethiopia → Somalia	320,000	355,000	90.1	None	High	None
AFG-IRN	1992	Afghanistan → Iran	300,000	3,030,555	9.9	High	None	None
AFG-IRN	2003	Afghanistan → Iran	285,400	2,500,000	11.4	Low	None	Severe
AFG-IRN	2018	Afghanistan → Iran	275,061	2,451,142	11.2	High	Low	Severe
AFG-IRN	2017	Afghanistan → Iran	274,291	2,451,149	11.2	High	Low	Severe
AFG-IRN	2006	Afghanistan → Iran	264,350	1,940,000	13.6	High	Low	Severe
AFG-IRN	2001	Afghanistan → Iran	263,501	1,982,000	13.3	High	Low	Severe
AFG-PAK	2008	Afghanistan → Pakistan	261,220	2,018,150	12.9	High	High	Severe
AFG-PAK	2007	Afghanistan → Pakistan	253,590	2,280,000	11.1	High	Low	Severe
KHM-THA	1992	Cambodia → Thailand	249,000	365,000	68.2	Low	None	None
AFG-IRN	1995	Afghanistan → Iran	245,972	1,645,500	14.9	High	None	Severe
AFG-IRN	2016	Afghanistan → Iran	234,298	3,951,175	5.9	High	Low	Severe
AFG-IRN	1994	Afghanistan → Iran	227,387	1,925,000	11.8	High	None	Severe
ETH-SOM	1983	Ethiopia → Somalia	200,000	700,000	28.6	Low	Low	None
RWA-COD	1997	Rwanda → Congo - Kinshasa	189,000	200,000	94.5	High	High	Moderate
AFG-IRN	2000	Afghanistan → Iran	183,600	1,825,000	10.1	High	Low	Severe
SDN-SSD-UGA	2021	South Sudan → Uganda	180,356	889,054	20.3	Low	None	None
SYR-TUR	2018	Syria → Turkey	177,305	3,424,399	5.2	High	Low	Moderate

Note: *dagger* marks dyad-years in which recorded return exceeds the lagged stock, where the denominator is the year-end stock plus return rather than the lagged stock, so the rate is an upper bound. COO and COA give conflict intensity in the origin and asylum country that year; Forced gives the asylum country's forced-return policy.

Table 10: The non-return cases

Dyad	Origin → asylum	Years	Refugee-years (m)	Peak stock	Total return	% yrs COO conflict	Push ever
PSE-JOR	Palestinian Territories → Jordan	1981–2008	44.62	2,439,718	233	89	None
VNM-CHN	Vietnam → China	1980–2021	11.97	321,502	0	15	COA conflict
VEN-USA	Venezuela → Colombia	2009–2023	11.93	2,875,743	0	0	COA conflict
PSE-SYR	Palestinian Territories → Syria	1980–2016	10.62	592,200	233	83	COA conflict
PSE-LBN	Palestinian Territories → Lebanon	1981–2023	9.24	500,000	233	81	COA conflict
VEN-PER	Venezuela → Peru	2003–2023	6.98	1,542,004	0	0	COA conflict
ESH-DZA	Western Sahara → Algeria	1980–2023	5.89	165,000	0	21	COA conflict
COL-ECU	Colombia → Ecuador	1990–2023	4.71	674,500	5,102	85	COA conflict
CHN-IND	China → India	1984–2023	4.01	123,000	0	0	COA conflict
UKR-RUS	Ukraine → Russia	1996–2023	3.56	1,275,925	0	42	COA conflict
PHL-MYS	Philippines → Malaysia	1981–2023	2.70	136,075	0	91	COA conflict
BTN-NPL	Bhutan → Nepal	1991–2023	2.33	133,700	0	0	Both
RUS-USA	Russia → United States	1988–2023	2.22	245,238	0	0	None
ERI-ETH	Eritrea → Ethiopia	1993–2023	2.03	181,271	0	8	COA conflict
TUR-DEU	Turkey → Germany	2000–2023	1.98	181,344	0	96	None
IRN-TUR	Iran → Turkey	1982–2023	1.98	257,000	2,208	53	Both
CHN-USA	China → United States	1988–2023	1.96	173,873	0	0	Moderate only
VEN-BRA	Venezuela → Brazil	2011–2023	1.93	510,499	0	0	None
GTM-USA	Guatemala → United States	1988–2023	1.76	192,596	0	20	Moderate only
ARM-AZE	Armenia → Azerbaijan	1992–2013	1.56	229,000	0	38	COA conflict

Note: The twenty largest dyads whose cumulative return never reaches 1 per cent of peak stock. Ranked by refugee-years, the sum of the annual stock over the dyad's observed history, which measures how much displacement the dyad represents rather than how long it is observed. The three Palestinian dyads each record the same 233 returns in 1986, which is one reported figure carried across all three rather than three independent observations.

1.5 Comparison of GRR and UNHCR

I compare GRR’s refugee-stock and return series with the corresponding UNHCR series. GRR’s stock data include unregistered refugees, while its total-return measure includes both voluntary and forced return. The comparison uses the GRR core dataset and excludes populations that UNHCR does not record, such as Palestinians under UNRWA. Restricting the sample to dyad-years observed in both sources leaves 1,415 dyad-years across 96 dyads.

Table 3 of the paper reports the main comparison. This shows that the two sources are very similar on voluntary return: UNHCR records 19.1 million returns, compared with 19.5 million in GRR. The larger difference is in forced return, much of which does not appear in the UNHCR series. Thirty-one per cent of forced-return episodes recorded by GRR have no corresponding UNHCR return. The gap is largest in the 1980s. In that decade, UNHCR records 71 per cent of GRR’s voluntary-return total, while two-thirds of forced-return episodes recorded by GRR are absent from UNHCR. The correspondence improves considerably after the 1980s, although the two series never fully converge.

This section extends that comparison in several directions. I examine the stock data over the full panel, show the differences in the data through dyadic plots, identify the episodes responsible for the largest differences, check the internal consistency of each series, and test whether replacing GRR with UNHCR changes the paper’s main estimates.

1.5.1 Comparing Refugee Stocks

In this section, I compare GRR and UNHCR on stock data, similar to the analysis of returns in the main paper. This subsection uses the core dataset, covering 1980–2008 and excluding asylum in Europe and North America. Logged formal stocks correlate at 0.86 and are exactly equal in 7% of dyad-years. The two series therefore track each other closely, but they are not identical.

Within the core dataset, the larger difference concerns GRR’s broad stock, which includes unregistered refugees absent from UNHCR’s formal statistics. Table 11 compares registered and broad stocks in the same way that the paper compares voluntary and forced return. For registered refugees, UNHCR records 0.82 of GRR’s total. For the broad stock, the ratio falls to 0.61, a difference of 90.6 million unregistered refugee stock-years. Of the 424 dyad-years in which GRR records an unregistered population, 108, or 25%, have no UNHCR stock at all, while another 62 have an UNHCR figure below one-tenth of GRR’s.

The stock and return comparisons therefore show a similar pattern: populations missing from formal registration are also less likely to appear in recorded returns. The timing differs, however. Return coverage is weakest in the 1980s and improves thereafter, whereas coverage of GRR’s broad stock is weakest in the 2000s, when the ratio falls to 0.44. This is also the decade in which GRR records the largest unregistered refugee population.

Table 11: Registered and unregistered refugee stocks, GRR against UNHCR

Quantity	1980s	1990s	2000s	All
Sample				
Dyad-years compared	371	638	605	1,614
Dyads	55	90	89	104
Aggregate calibration				
GRR registered stock	78,321,850	104,907,809	83,528,312	266,757,971
GRR unregistered stock	14,177,285	27,405,166	49,059,410	90,641,861
UNHCR recorded stock	73,605,396	86,767,560	58,795,551	219,168,507
UNHCR / GRR registered	0.94	0.83	0.70	0.82
UNHCR / GRR total	0.80	0.66	0.44	0.61
Population capture				
Dyad-years with unregistered population	98	162	164	424
invisible in UNHCR	38 (39%)	40 (25%)	30 (18%)	108 (25%)
token in UNHCR	5	26	31	62
Share of unregistered volume				
Unregistered stock-years in invisible dyad-years	35%	45%	10%	25%

Note: Core dataset only: 1980–2008, asylum outside Europe and North America. Stocks are summed over dyad-years, so the unit is a stock-year. A dyad-year is emphinvisible when UNHCR records no stock for it and emptoken when UNHCR’s figure is below a tenth of GRR’s broad stock. Built to the same row blocks as the comparison of voluntary and forced return in the paper.

In the later years of the extended dataset, GRR and UNHCR are more closely aligned. That convergence partly reflects a change in how GRR is constructed. After the *World Refugee Survey* ends, GRR takes UNHCR’s registered stock as its starting point, then adds unregistered refugee populations identified in State Department reports and other archival sources. The two series therefore share the same baseline after 2009, but GRR remains broader where those additional populations can be identified.

During the *World Refugee Survey* period, the two stock series are identical in 55.9% of dyad-years, while GRR exceeds UNHCR in roughly one-quarter. In the State Department period, they are identical in 95.3%, and GRR exceeds UNHCR in fewer than one dyad-year in twenty. Where GRR does exceed UNHCR, the median difference is 76%.

The return data show the same shift. GRR and UNHCR report identical return figures in 9.7% of dyad-years before 2009, compared with 84.0% after 2009. GRR’s independent contribution is therefore concentrated before 2009. In later years, it is better understood as UNHCR’s series supplemented with archival additions and corrections. The comparison between the two sources is therefore most informative for 1980–2008, when they were produced more independently.

1.5.2 Internal Consistency

Table 12: Internal-consistency diagnostics on the stock series

Series	Returns exceed prior stock (%)	Unexplained collapse (%)	V-shaped reversal (%)	Spike then collapse (%)	Unchanged from prior year (%)	Exact multiple of 10,000 (%)	Median abs. log change
GRR stock	0.365	3.28	0.848	0.348	8.37	7.49	0.111
UNHCR stock	0.437	3.54	0.941	0.282	4.15	1.82	0.113

Note: The first four columns are impossibilities or near-impossibilities; the last three are fingerprints of estimation rather than measurement.

The internal-consistency checks are similar across the two sources. Pairing GRR’s returns with its stock puts 0.36% of dyad-years above a rate of one, and the corresponding share for UNHCR is similar. GRR holds a stock unchanged from the previous year more often than UNHCR does, and reports round multiples of ten thousand more often. This follows from the coding rule: when an archival range does not change, GRR

carries the previous estimate forward rather than moving between bounds, and the underlying sources often report rounded figures. The result is a more stable within-dyad series that is also visibly less granular.

1.5.3 Replicating the Main Analysis

Table 13: Alternative return series and offsets

Term	(1) GRR returns GRR stock	(2) UNHCR returns GRR stock	(3) UNHCR returns UNHCR stock
Forced Return: severe	1.02*** (0.21)	0.35 (0.23)	0.15 (0.25)
Forced Return: moderate	0.50* (0.21)	-0.14 (0.12)	-0.24 [†] (0.13)
Conflict in COA: high	0.54 [†] (0.30)	-0.06 (0.30)	-0.18 (0.29)
Conflict in COA: low	0.82*** (0.21)	0.10 (0.19)	0.12 (0.22)
War Ending	0.63** (0.20)	0.95*** (0.27)	0.88*** (0.24)
Conflict in COO: high	0.13 (0.35)	0.73 [†] (0.41)	0.82* (0.41)
Conflict in COO: low	0.02 (0.22)	0.18 (0.24)	0.21 (0.27)
GDP (COA)	-1.15 [†] (0.64)	0.28 (0.71)	-0.71 (0.73)
GDP (COO)	0.38 (0.29)	-0.08 (0.33)	-0.06 (0.34)
Civil Liberties Restrictions (COO)	-0.55*** (0.16)	0.03 (0.17)	0.14 (0.15)
Refugee Rights	0.04 (0.16)	0.05 (0.22)	0.05 (0.20)
Conflict Duration	-0.09 (0.09)	-0.05 (0.11)	-0.08 (0.11)
Dyad-years	2,327	2,297	2,126
Dyads	110	109	107

Note: Every column is the paper’s specification — the right-hand side of the baseline table, dyad and year fixed effects, log stock at risk as offset, dyad-clustered standard errors — re-estimated with the dependent variable and the offset swapped for other measures. Column 1 is the paper’s baseline, reproduced coefficient for coefficient. Samples differ across columns because a rung drops the dyad-years whose own offset is missing or zero. Significance: ^{dagger} 0.10, * 0.05, ** 0.01, *** 0.001.

This section replaces GRR’s return and stock measures with UNHCR’s and examines how the results change. In Table 13, Column 1 reproduces the main specification. Column 2 replaces the outcome with UNHCR’s return series, while Column 3 also replaces the refugee stock, yielding a specification based entirely on UNHCR measures.

The results change substantially. Neither severe forced-return policies nor conflict in the asylum country is associated with higher return in the UNHCR specifications. The coefficient on severe forced return falls from 1.02 in Column 1 to 0.35 when UNHCR’s return series is used, and to 0.15 when UNHCR’s stock measure is used as well. Neither UNHCR estimate is distinguishable from zero. The same is true of conflict in the asylum country, which is coded independently of either return series.

Other coefficients also move in less expected directions. Conflict in the origin is positively associated with return in both UNHCR specifications, even after including the covariates and fixed effects. In the full UNHCR specification, a moderate forced-return policy is associated with *lower* return, although only weakly significantly. At the same time, none of the economic or political conditions in either the origin or asylum country that are commonly thought to shape return is statistically significant in either UNHCR specification.

One possible explanation is that UNHCR’s return series reflects where the organization is able to observe and record return most consistently. If coverage is systematically greater in some settings than others, those coverage patterns could enter the estimates along with the underlying return process. I do not have direct evidence that this is what produces the results, however, so I do not treat this explanation as a finding. More generally, neither GRR nor UNHCR provides a ground-truth measure of return.

1.5.4 Dyadic Return Patterns

Figures 2 and 3 plot refugee stocks and total returns over time for the forty largest dyads in the core dataset: the two largest asylum destinations for each of the twenty largest origin countries by displaced population.

GRR is shown in black and UNHCR in red. Across most dyads, UNHCR records lower stocks and returns than GRR.

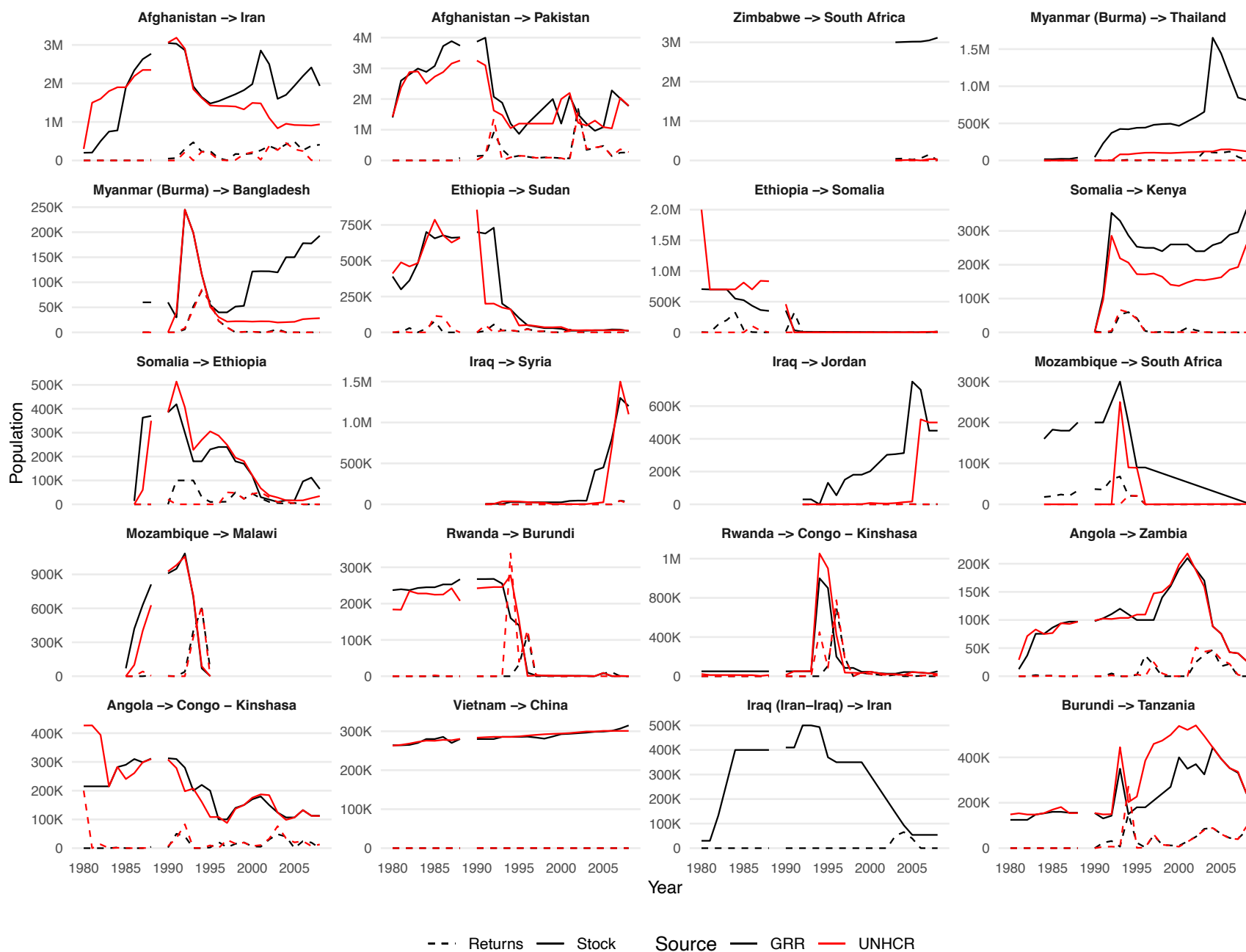


Figure 2: Refugee Stocks and Returns by Dyad (1 of 2)

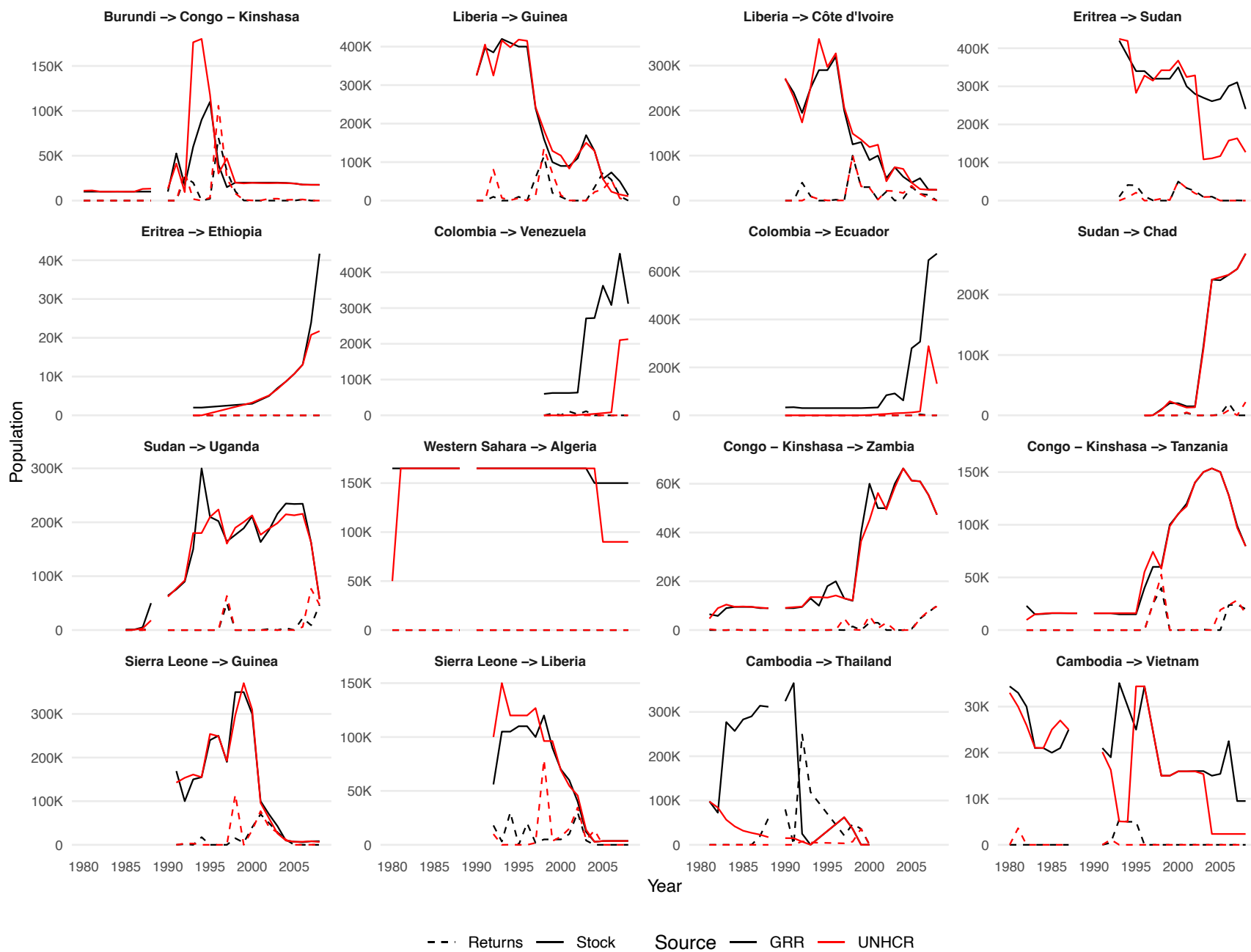


Figure 3: Refugee Stocks and Returns by Dyad (2 of 2)

Table 14: Examples of Forced Return Policies

Moderate forced return policies	Severe forced return policies
“In a move protested by UNHCR, authorities in Congo-Brazzaville forced 19 asylum seekers back to Congo-Kinshasa in April.” (COD-COG, 2002) “The Tibetan refugee leadership encourages most of those on pilgrimages and many who come to study to return to Tibet, both to counteract the overwhelming Chinese presence in Tibet and because the two dozen Tibetan refugee settlements in India cannot absorb so many people.” (CHN-IND, 1999) “There were scattered reports that the government intended to expel Palestinians who had not arrived in 1948, but who had come to Lebanon from other Arab countries, such as Egypt and Syria. If implemented, this could affect up to 50,000.” (PAL-LEB, 1993) “UNHCR expressed concern that some refugees were being manipulated to repatriate prematurely to Congo-Kinshasa based on misinformation about conditions there. Some refugees reportedly repatriated and had to flee again to Rwanda because conditions in Congo-Kinshasa remained dangerous.” (COD-RWA, 2000) “In western Sudan, 45,000 Chadian refugees remain in UNHCR-assisted reception centers despite government and UNHCR attempts to persuade them to return to Chad.”	“South Africa enacted new legislation in 1991, the Aliens Control Act, giving immigration authorities even more power to seek out and deport Mozambicans... South Africa has forcibly repatriated thousands of Mozambican refugees to a land of deprivation and death in recent years. This practice violates international standards and must cease,” USCR stated. (MOZ-ZAF, WRS 1993) “In April, the Government announced that it would expel all Rwandans whose cases it rejected and the number of applicants declined substantially... Deterred by the low acceptance rates, some 2,000 never registered with the Government and more than 10,500 of those who registered dropped their claims and returned.” (RWA-BDI, WRS 2006) “The events in Zaire produced strains in Congo that resulted in the Congolese authorities forcibly deporting some 10,000 Zaireans in November.” (COD-COG, WRS 1992) “Yemeni authorities forcibly returned 438 Somalis, at least 200 of whom were UNHCR-recognized refugees. They had been part of a group of 1,200 persons who had been rounded up in Aden as part of a sweep of the city for illegal immigrants.” (SOM-YEM, 1996) “UNHCR protested when the government’s roundup of 1,000 undocumented immigrants in January resulted in the detention of several refugees who possessed UNHCR permits. Persons judged to be undocumented were deported without UNHCR screening to determine claims to refugee status.” (COD-ZMB, 1993)

1.6 Forced-Return Measurement and Validation

This section sets out how the forced-return measure is coded and discusses validation checks.

1.6.1 Measurement

The *Forced Return* variable is hand-coded from the qualitative country narratives in the *World Refugee Survey* and, after the Survey ceased publication, the U.S. State Department’s *Country Reports on Human Rights Practices*. Non-zero dyad-years are assigned to one of two categories, *moderate* or *severe*, using two criteria: (i) whether forced return is implemented or only threatened, and (ii) the share of the refugee stock the policy is intended or expected to affect.

A dyad-year is coded **severe** when the asylum government implements a forced return policy intended or expected to affect at least 10% of the refugee stock. Severe policies include large-scale arrests and deportations, non-renewal of residence cards or refugee status, closures of camp schools and marketplaces, and cuts to food aid expressly intended to induce return. A dyad-year is coded **moderate** when the asylum government either (a) threatens or encourages forced return without policy implementation (e.g., political rhetoric, parliamentary debate, public ultimatums), or (b) implements a forced return policy targeting fewer than 10% of the refugee stock. The 10% threshold separates broad coercive policies, which we expect to shift return rates, from narrower actions targeting specific individuals or groups but not intended to bring about widespread return.

Coding depends only on the policy itself, not on the return outcome. A policy that cuts food aid across all camps to induce return is coded as severe regardless of whether refugees subsequently repatriate. This rules out the most direct form of measurement endogeneity: the variable captures the policy environment refugees face rather than the return rate it is intended to predict.

Table 14 provides illustrative examples of moderate and severe coding decisions. The GRR codebook records a verbatim source quote for every non-zero observation, allowing external verification of all coding decisions.

1.6.2 Validation

In core GRR, forced-return policies and forced-return counts are coded from the same archival sources, raising the possibility of a mechanical association. I address this in three ways: by asking whether policies are coded in years with no return, whether the result persists when forced returns are removed from the outcome, and whether the measure agrees with an independently coded benchmark. The third check compares GRR’s record of observed practice with DWRAP’s coding of statutory protection against *refoulement*.

Table 15: Return in the years a policy is coded

Policy	Dyad-years	Zero return	Return below 1% of stock	Median return rate
None	3,988	76.3%	84.8%	0.00%
Moderate	290	35.9%	58.3%	0.35%
Severe	150	8.0%	22.0%	10.64%

Note: If the measure were read off the return, the zero-return column would be near empty.

Moderate policy is coded in 290 dyad-years, and 36% record no return at all. Severe policy is coded in 150 dyad-years, of which 8% record no return. This is possible because the policy measure captures threats, ultimatums, camp closures, deadlines, and other government actions, rather than return itself. Moreover, most returns are not coded from forced-return passages: 81% of all returns in GRR are voluntary, and forced returns occur in only 4.1% of dyad-years.

A remaining concern is that the association could still depend on the coding of forced returns. Table 18 below replicates the main analysis using only voluntary returns as the dependent variable. The severe-policy coefficient falls from 1.02 to 0.42 when forced returns are removed, or 41 per cent of the baseline, but remains statistically significant.

This pattern is reassuring for two reasons. First, if the baseline result were driven mainly by mechanical overlap in the coding of forced-return policy and forced-return counts, the association should largely disappear once forced returns are removed from the outcome. Second, its persistence for voluntary return suggests that coercive policies may also increase returns classified as voluntary, but undertaken under substantial pressure. The smaller coefficient is therefore consistent with severe policies producing both directly forced return and ostensibly voluntary return under coercive conditions.

1.6.3 De Jure Law and De Facto Practice

DWRAP provides an independent benchmark for the GRR forced-return measure. It codes statutory refugee policy for 90 host countries, covering 95% of the panel. Its non-*refoulement* measure distinguishes statutes that permit *refoulement*, are silent or ambiguous, or explicitly protect against it. DWRAP measures national law, while GRR measures observed practice toward a particular refugee population.

Table 16: Forced-return practice by statutory position on *refoulement*

Statutory position	Dyad-years	Hosts	Any policy	Severe policy
Explicit provision permitting <i>refoulement</i>	509	34	15.3%	3.9%
Silent or ambiguous	698	29	21.3%	7.0%
Explicit right to non- <i>refoulement</i>	3,044	66	6.8%	2.5%

Note: Restricted to dyad-years in which both measures are coded. Forced-return practice is hand-coded from the archival sources; the statutory position is DWRAP’s independent coding of national law.

Table 17: Forced return and the statutory right to non-refoulement

Specification	Explicit right to non-refoulement	Dyad-years
Any, host + year FE	-1.15*** (0.34)	3,321
Any, dyad + year FE	-1.09* (0.48)	2,128
Severe, host + year FE	-1.29* (0.61)	2,268
Severe, dyad + year FE	-1.66* (0.83)	970

Note: Logit of hand-coded forced return on the statutory right to non-refoulement, compared with the two remaining DWRAP positions combined. Standard errors clustered by dyad. The dyad-fixed-effects columns are identified from statutory changes within a host over time.

The main contrast is in the expected direction. Forced return occurs in 6.8% of dyad-years in which the host grants an explicit right to non-refoulement, compared with 21.3% where the statute is silent or ambiguous. The statutory right is negatively associated with coercion in each specification in Table 17, including models with dyad and year fixed effects. This correspondence between an independently coded legal measure and observed practice provides evidence for the validity of the GRR measure.

The three statutory categories are not monotonic: hosts whose laws permit refoulement record less coercion than those whose laws are silent or ambiguous. Because the permitting category is small and concentrated among relatively few hosts, Table 17 compares an explicit right to non-refoulement with the other two positions combined.

Formal protection does not eliminate coercion. Among dyad-years in which the host grants an explicit right to non-refoulement, 206 record a forced-return policy and 77 record a severe policy, across 30 hosts. Statutory protection is associated with less coercion, but it remains distinct from state practice.

As a further check, I am constructing an independently coded measure of forced-return policy from Amnesty International reports. The coding and analysis is ongoing.

2 Robustness Checks

Unless otherwise noted, each robustness model changes one feature of column 1 in the main table and holds the remainder of the specification fixed.

The prose below describes what each check does and reports the estimates only where they change a conclusion or add something the table does not already show. Where nothing is said about a coefficient, it keeps the sign, significance and approximate magnitude it has in the main table.

2.1 Alternative Measures of Return

Because GRR return counts are hand-coded from archival sources, some annual variation may reflect reporting error. I therefore vary the dependent variable in several ways.

Model 1 of Table 18 reproduces the baseline, Model 2 uses voluntary return only, and Model 3 uses a centered three-year moving average of total return.

Two estimates move. Removing forced movements from the outcome reduces the severe-policy coefficient from 1.02 to 0.42 and turns the moderate-policy coefficient negative and insignificant (-0.09), consistent with the outcome reflecting both forced movement and voluntary return under coercive conditions. The post-war coefficient falls from 0.63 to 0.39 under the moving average; unlike severe policy it rests on only 452 dyad-years and is sensitive to smoothing the outcome.

Table 18: Alternative Measures of the Dependent Variable

Dep. Var.:	Total Repatriation Total Returns	Voluntary Repatriation Voluntary Returns	Total Repat. (3-yr MA) 3-Year MA
Model:	(1)	(2)	(3)
<i>Variables</i>			
<i>Forced return</i>			
Forced Return: severe	1.02*** (0.21)	0.42* (0.17)	0.91*** (0.23)
Forced Return: moderate	0.50* (0.21)	-0.09 (0.12)	0.49* (0.20)
Conflict in COA: high	0.54† (0.30)	0.29 (0.30)	0.39† (0.23)
Conflict in COA: low	0.82*** (0.21)	0.23† (0.13)	0.70*** (0.16)
<i>Preference-based explanations</i>			
War Ending	0.63** (0.20)	0.73*** (0.19)	0.39* (0.19)
Conflict in COO: high	0.13 (0.35)	0.10 (0.36)	0.33 (0.34)
Conflict in COO: low	0.02 (0.22)	0.12 (0.23)	0.12 (0.18)
GDP (COO)	0.38 (0.29)	0.18 (0.31)	0.16 (0.26)
GDP (COA)	-1.15† (0.64)	-0.28 (0.62)	-1.00 (0.62)
Civil Liberties Restrictions (COO)	-0.55*** (0.16)	-0.51** (0.17)	-0.51*** (0.14)
Refugee Rights	0.04 (0.16)	0.14 (0.15)	0.00 (0.16)
Conflict Duration	-0.09 (0.09)	-0.11 (0.09)	-0.14 (0.09)
<i>Fixed-effects</i>			
Dyad	Yes	Yes	Yes
Year	Yes	Yes	Yes
<i>Fit statistics</i>			
Observations	2,327	2,281	2,207
Squared Correlation	0.75	0.84	0.80
Pseudo R ²	0.79	0.78	0.82

Clustered (Dyad) standard-errors in parentheses

*Signif. Codes: ***, 0.001, **, 0.01, *, 0.05, †: 0.1*

Standard errors clustered by dyad in parentheses. All models: Poisson PML with log stock-at-risk offset and dyad and year fixed effects. Model 3 uses a centered 3-year moving average of total returns as the dependent variable.

Table 19: Discrete return episodes: seven codings of the dependent variable

Episode rule: Model:	Returns (count) (1) Poisson	Rate > 0.05 (2) Logit	Rate > 0.10 (3) Logit	Rate > 0.25 (4) Logit	> 25,000 (5) Logit	> 50,000 (6) Logit	Mean + 1 SD (7) Logit	3× median (8) Logit
<i>Variables</i>								
<i>Forced return</i>								
Forced Return: severe	1.02*** (0.21)	2.84*** (0.34)	2.70*** (0.33)	2.10*** (0.53)	3.31*** (0.41)	4.24*** (0.67)	2.04*** (0.37)	2.04*** (0.49)
Forced Return: moderate	0.50* (0.21)	0.69** (0.26)	0.46 (0.30)	0.53 (0.43)	0.67 (0.41)	0.45 (0.59)	0.74* (0.31)	0.57† (0.32)
Conflict in COA: high	0.54† (0.30)	0.23 (0.40)	-0.01 (0.41)	0.02 (0.44)	1.08 (0.70)	1.25 (1.12)	0.30 (0.41)	0.34 (0.45)
Conflict in COA: low	0.82*** (0.21)	0.26 (0.31)	-0.15 (0.30)	-0.04 (0.36)	0.27 (0.57)	1.29 (0.91)	-0.10 (0.33)	0.00 (0.35)
<i>Preference-based explanations</i>								
War Ending	0.63** (0.20)	0.69** (0.26)	0.74* (0.32)	0.77* (0.31)	1.29* (0.50)	1.40 (0.88)	0.85** (0.28)	0.68* (0.34)
Conflict in COO: high	0.13 (0.35)	-0.25 (0.35)	-0.38 (0.39)	-0.20 (0.54)	-0.03 (0.57)	0.97 (1.16)	-0.01 (0.36)	0.15 (0.43)
Conflict in COO: low	0.02 (0.22)	-0.23 (0.25)	-0.40 (0.27)	-0.52 (0.36)	-0.56 (0.49)	-0.04 (0.45)	-0.58* (0.27)	-0.33 (0.32)
GDP (COO)	0.38 (0.29)	0.19 (0.34)	0.15 (0.32)	0.50 (0.47)	0.44 (0.70)	0.59 (1.34)	0.78† (0.41)	0.35 (0.43)
GDP (COA)	-1.15† (0.64)	-0.18 (0.62)	-0.67 (0.70)	-0.29 (0.95)	-0.45 (0.81)	-1.29 (1.71)	-0.47 (0.67)	-0.21 (0.66)
Civil Liberties Restrictions (COO)	-0.55*** (0.16)	-0.67** (0.22)	-0.75** (0.28)	-1.06*** (0.29)	-0.58† (0.31)	-0.97* (0.43)	-0.48* (0.21)	-0.43† (0.22)
Refugee Rights	0.04 (0.16)	0.35* (0.16)	0.40** (0.15)	0.28 (0.18)	0.51* (0.25)	-0.08 (0.58)	0.12 (0.15)	0.37* (0.16)
Conflict Duration	-0.09 (0.09)	0.29* (0.13)	0.33* (0.17)	0.25 (0.21)	0.64** (0.23)	0.37 (0.41)	0.35* (0.16)	0.17 (0.20)
Rule type	Count	Level	Level	Level	Level	Level	Change	Change
Dyads	110	93	84	68	55	32	107	72
Episode-years in sample	—	496	367	187	195	98	274	255
<i>Fixed-effects</i>								
Dyad	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>Fit statistics</i>								
Observations	2,327	2,023	1,834	1,530	1,240	732	2,300	1,743
Pseudo R ²	0.79	0.25	0.25	0.25	0.37	0.50	0.17	0.17

Clustered (Dyad) standard-errors in parentheses

Signif. Codes: ***, 0.001, **, 0.01, *, 0.05, †, 0.1

Column 1 is the paper's specification: Poisson PML on the return count with the log stock at risk as offset. Columns 2 to 8 are conditional logits on the episode codings named in the column headings. All models carry dyad and year fixed effects and cluster standard errors by dyad. Logit coefficients are log odds ratios and are not on the scale of column 1. The conditional logit drops dyads and years without variation in the outcome, which is why the observation and dyad counts fall below column 1.

Table 20: Discrete return episodes: estimators

Dep. Var.:	Logit (1)	Rate > 0.10 LPM (2)	LPM, logit rows (3)	Onset logit (4)	Episode onset Onset hazard (5)	Onset LPM (6)	Returns (count) PPML, logit rows (7)
Model:	Logit	OLS	OLS	Logit	Logit	OLS	Poisson
<i>Variables</i>							
<i>Forced return</i>							
Forced Return: severe	2.70*** (0.33)	0.35*** (0.05)	0.43*** (0.05)	2.58*** (0.50)	2.10*** (0.32)	0.24*** (0.05)	1.06*** (0.22)
Forced Return: moderate	0.46 (0.30)	0.04† (0.03)	0.04 (0.04)	0.78* (0.33)	0.65* (0.28)	0.05* (0.02)	0.54* (0.23)
Conflict in COA: high	-0.01 (0.41)	0.01 (0.04)	-0.01 (0.05)	0.51 (0.56)	-0.03 (0.30)	0.04 (0.03)	0.51 (0.33)
Conflict in COA: low	-0.15 (0.30)	-0.01 (0.03)	-0.02 (0.04)	0.12 (0.37)	-0.21 (0.25)	0.01 (0.02)	0.85*** (0.21)
<i>Preference-based explanations</i>							
War Ending	0.74* (0.32)	0.08** (0.03)	0.11* (0.05)	0.60 (0.45)	0.42 (0.32)	0.03 (0.02)	0.60** (0.21)
Conflict in COO: high	-0.38 (0.39)	-0.04 (0.03)	-0.06 (0.05)	0.65 (0.59)	0.23 (0.39)	0.03 (0.03)	0.05 (0.37)
Conflict in COO: low	-0.40 (0.27)	-0.04 (0.03)	-0.05 (0.04)	0.00 (0.45)	-0.06 (0.35)	0.00 (0.03)	0.01 (0.22)
GDP (COO)	0.15 (0.32)	0.03 (0.02)	0.04 (0.04)	0.37 (0.40)	-0.23 (0.18)	0.03† (0.02)	0.34 (0.30)
GDP (COA)	-0.67 (0.70)	-0.12* (0.05)	-0.09 (0.08)	-1.32† (0.72)	-1.19*** (0.19)	-0.10** (0.04)	-1.09 (0.68)
Civil Liberties Restrictions (COO)	-0.75** (0.28)	-0.05** (0.02)	-0.10** (0.03)	-0.42 (0.28)	-0.25* (0.12)	-0.02 (0.01)	-0.60*** (0.17)
Refugee Rights	0.40** (0.15)	0.02 (0.01)	0.04* (0.02)	0.47** (0.17)	0.28*** (0.08)	0.02 (0.01)	0.06 (0.16)
Conflict Duration	0.33* (0.17)	0.03* (0.01)	0.04† (0.02)	0.16 (0.27)	0.18 (0.18)	0.01 (0.01)	-0.10 (0.09)
Years Since Last Episode (log)					-0.31* (0.14)		
Sample Dyads	Panel 84	Panel 245	Logit rows 84	At risk 74	At risk 241	At risk 234	Logit rows 84
Fixed-effects							
Dyad	Yes	Yes	Yes	Yes		Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fit statistics							
Observations	1,834	3,913	1,834	1,266	3,221	3,214	1,834

Clustered (Dyad) standard-errors in parentheses
*Signif. Codes: ***, 0.001, **, 0.01, *, 0.05, †: 0.1*

The dependent variable is the ten per cent rate rule throughout: an episode is a dyad-year in which return exceeds a tenth of the stock at risk. Columns 4 to 6 use episode onset, one in the first year of a run, estimated on the dyad-years at risk of starting an episode; interior years of a run are removed from the risk set rather than coded zero. Column 5 is the discrete-time hazard: year fixed effects and log years since the last episode, without dyad fixed effects, because duration is collinear with dyad and year effects together; its dyad count is the number of clusters rather than the number of absorbed effects. Column 7 refits the paper's count model, outcome and offset unchanged, on the rows the conditional logit retains. Standard errors clustered by dyad throughout. Logit, linear-probability and PPML coefficients are on three different scales and should not be compared across columns of different estimators.

A further way of varying the outcome is to discard the exact number returned. In Tables 19 and 20 I recode it as an indicator for whether a dyad experiences a substantial return episode, which tests whether the results hold for movements too large to be explained by minor coding differences. There is no unique threshold, so I use seven alternatives. The *level* rules apply common thresholds across dyads: return above five, ten, or twenty-five percent of the population at risk, or more than twenty-five or fifty thousand people. The *change* rules define an episode relative to the dyad’s history: a return rate more than one within-dyad standard deviation above its mean, or at least three times its median positive return rate. Table 19 applies each definition to the main specification, with the count model shown for comparison; Table 20 holds the threshold at ten percent and varies the estimator. I also re-estimated at nine rate thresholds from one to thirty-three percent, not reported.

Two features of the recoding are worth recording. How much of the panel counts as an episode depends heavily on which rule is used, from 129 dyad-years under the fifty-thousand threshold to 559 above the five percent rate, or 2.5 to 10.9 percent of the panel. And the choice determines how much of the panel a conditional logit can use, since only dyads whose coded outcome varies contribute: from 38 dyads under the fifty-thousand rule to 122 under the within-dyad standard deviation, so the change rules rest on more dyads than their episode counts alone suggest. A substantial episode occurs in 51 percent of severe-policy dyad-years against 8 percent otherwise. Across the nine unreported thresholds the severe-policy coefficient stays between 2.01 and 3.07 and the war-ending coefficient between 0.67 and 0.89, both significant at every one. The only estimate that weakens is the post-war window when the outcome is episode onset rather than the presence of an episode.

2.2 Alternative Measures of Conflict and War Endings

2.2.1 Alternative Definitions of a War Ending

The post-war window is based on UCDP data. An ending is the last year of a UCDP high-intensity spell at the origin, and a dyad-year counts as post-war only if the origin does not have a high-intensity war in it; a low-intensity war may be running. There may be concern about this measure, that it does not signify the end of war because a low-intensity war continues.

I therefore code several other ways of measuring war endings. The first is Fearon’s (2017) civil-war list, which dates wars by definitive beginnings and endings rather than by an annual battle-death threshold, merged with interstate wars coded to a refugee origin. Fearon’s original coding runs through 2014. An extended version carries thirty-one of those cases forward with revised end dates and adds seventeen cases.

Table 21 reports these definitions separately. Column 1 is the UCDP list the paper uses. Column 2 uses the extended Fearon list merged with the inter-state endings and Column 3 keeps only the cases Fearon coded and this project did not revise, so the pair shows whether the result depends on the author-coded extension or on the use of definitive conflict dates. Column 4 removes the requirement that an origin have at least 25,000 refugees in the ending year, the one filter that selects endings on a quantity related to the outcome. Column 5 adds the ending year itself, so the window is identified from years one to three rather than zero to three. Column 6 turns to a formal rather than a battlefield definition of an ending, adding an indicator for a peace agreement at the origin in the previous year, drawn from the UCDP Peace Agreements Dataset and merged to the origin’s active conflict via UCDP `conflict_id`.

Table 21: Robustness: Alternative Definitions of a War Ending

Model:	Main (1)	Fearon + inter-state (2)	Total Repatriation Fearon original (3)	No size floor (4)	+ Ending yr (5)	+ Peace ag. (6)
<i>Variables</i>						
<i>Forced return</i>						
Forced Return: severe	1.02*** (0.21)	1.13*** (0.20)	1.14*** (0.20)	1.13*** (0.20)	1.02*** (0.21)	1.01*** (0.20)
Forced Return: moderate	0.50* (0.21)	0.49* (0.21)	0.50* (0.21)	0.49* (0.21)	0.50* (0.21)	0.53** (0.20)
Conflict in COA: high	0.54† (0.30)	0.65* (0.32)	0.64* (0.33)	0.65* (0.32)	0.54† (0.30)	0.66* (0.30)
Conflict in COA: low	0.82*** (0.21)	0.85*** (0.23)	0.85*** (0.23)	0.85*** (0.23)	0.83*** (0.21)	0.87*** (0.20)
<i>Preference-based explanations</i>						
War Ending	0.63** (0.20)	0.98*** (0.18)	1.02*** (0.20)	0.97*** (0.18)	0.63** (0.20)	0.62*** (0.19)
War Ending (Year Of)					0.08 (0.31)	
Conflict in COO: high	0.13 (0.35)	0.28 (0.27)	0.28 (0.27)	0.29 (0.27)	0.10 (0.38)	0.10 (0.34)
Conflict in COO: low	0.02 (0.22)	0.18 (0.21)	0.16 (0.21)	0.18 (0.21)	0.02 (0.22)	-0.03 (0.21)
GDP (COO)	0.38 (0.29)	0.35 (0.28)	0.32 (0.28)	0.36 (0.28)	0.39 (0.29)	0.42 (0.29)
GDP (COA)	-1.15† (0.64)	-1.15* (0.58)	-1.12† (0.58)	-1.19* (0.56)	-1.15† (0.64)	-1.23* (0.59)
Civil Liberties Restrictions (COO)	-0.55*** (0.16)	-0.56*** (0.16)	-0.56*** (0.17)	-0.55*** (0.16)	-0.56*** (0.16)	-0.56*** (0.16)
Refugee Rights	0.04 (0.16)	-0.01 (0.14)	-0.02 (0.15)	-0.01 (0.15)	0.04 (0.16)	0.08 (0.16)
Conflict Duration	-0.09 (0.09)	-0.05 (0.08)	-0.05 (0.08)	-0.05 (0.08)	-0.08 (0.09)	-0.05 (0.08)
Peace Agreement (COO, t-1)						-0.40* (0.17)
All controls	Yes	Yes	Yes	Yes	Yes	Yes
Ending list	UCDP spells	Extended + inter-state	Fearon original only	No 25,000 floor	UCDP spells	UCDP spells
Endings	76	41	33	75	76	76
<i>Fixed-effects</i>						
Dyad	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes
<i>Fit statistics</i>						
Observations	2,327	2,327	2,327	2,327	2,327	2,214
Pseudo R ²	0.79	0.80	0.80	0.80	0.79	0.80

Clustered (Dyad) standard-errors in parentheses

*Signif. Codes: ***: 0.001, **: 0.01, *: 0.05, †: 0.1*

All columns are the specification of column 1 of the main table with the post-war window rebuilt; nothing else changes. Every column holds the at-peace test fixed on UCDP conflict spans and varies the list of war endings. Column 1 is the list the paper uses: an ending is the last year of a UCDP high-intensity spell, so it is a year in which battle deaths fall below the threshold rather than the end of a named war. Column 2 uses an extended Fearon civil-war list merged with the inter-state endings. Column 3 keeps only the cases Fearon coded and this project did not revise, so it excludes both the post-2014 extension and the revised end dates. Column 4 removes the requirement that the origin have at least 25,000 refugees in the ending year, which is the one filter that selects endings on a quantity related to the outcome. Column 5 adds the ending year itself, so the window is identified from years one to three rather than zero to three. Column 6 keeps the UCDP list and adds an indicator for a peace agreement at the origin in the previous year, a formal rather than a battlefield definition of an ending. PPML with the log stock at risk as offset, dyad and year fixed effects, SEs clustered by dyad.

The post-war coefficient is positive under every definition, and the UCDP list the paper uses gives the smallest estimate: 0.63, against 0.98 on the extended Fearon list merged with inter-state endings and 1.02 on Fearon’s original cases alone. UCDP records an ending whenever annual battle deaths fall below the high-intensity threshold, including interruptions in conflicts that later resume, whereas the Fearon lists use a single definitive ending for each named war. The estimate is largest for Fearon’s original cases, so the result does not depend on this project’s extension of that list. The ending year itself is not distinguishable from zero (0.08), consistent with the timing reported in the main text.

The size floor makes little difference. Because it applies to the definitive-dates lists rather than to the UCDP measure, the relevant comparison is Column 4 against Column 2: removing it increases the list from 41 endings to 75 and leaves the window at 0.97 against 0.98, and raising it to 100,000 moves the coefficient to 1.08 on 31 endings. The UCDP window used in the paper has no size threshold.

The peace agreement in Column 6 enters negative and significant (-0.40). This is surprising as it does not align with the war ending result.

2.2.2 Alternative Measures of Origin-Country Conflict and Repression

In this section I vary the way that conflict in both the origin and the asylum country is coded. Model 1 reproduces the baseline. Model 2 replaces the ordinal conflict indicators with logged battle deaths, a continuous measure of intensity available from 1989. Model 3 dates origin-country conflict at $t - 2$ to allow for a delayed response. Model 4 replaces the origin conflict levels with a de-escalation indicator equal to one when conflict falls from high-intensity to low or none, capturing transitions rather than levels. Model 5 adds the PITF measure of warfare and violence, combining its Ethnic War and Revolutionary War datasets, which ends in 2017.

I then also vary the way that political repression is coded. In the main analysis this is the Freedom House civil liberties measure. Model 6 adds two further measures alongside it. The Political Terror Scale (PTS) measures state-perpetrated physical integrity violations, including torture, political imprisonment, extra-judicial killings, and disappearances, on a 1–5 ordinal scale; I use the mean of three sources (Amnesty International, Human Rights Watch, and the U.S. State Department), standardized across the estimation sample (`o_terror_z`). The second is a binary indicator equal to one in any country-year with an ongoing episode of genocide or politicide, drawn from the Political Instability Task Force (PITF) Geno-Politicide event dataset (`o_genocide`). The PITF episode list ends in 2017, so observations after 2017 are coded as missing and Model 6 runs on a shorter panel.

The Political Terror Scale is indistinguishable from zero once civil liberties are controlled for (-0.17), so the repression result operates through the Freedom House measure rather than through physical-integrity violations specifically. Surprisingly, the genocide/politicide indicator enters positive and significant (1.07), opposite to expectation.

2.2.3 Ethnic Group Political Power

In the main table, I examine correlates of shared ethnic and social ties between refugees and hosts. In this section, I examine factors related to the political status of the refugees’ ethnic group in both the origin and asylum country, where there are shared ethnic ties. A group that gains power at home might be more likely to go back, and one whose kin hold power in the host might be more likely to stay. I test both with the EPR data in Models 1 and 2 of Table 23. I find that refugees from politically dominant groups return at higher rates than those from excluded ones. The standing of co-ethnics in the host has no bearing on return: neither EPR-TEK indicator is distinguishable from zero (-0.24 for kin in power, 0.28 for kin excluded).

The ordinal EPR status variable (`o_eth_status`) takes nine values from ‘self-exclusion’ to ‘monopoly’. Refugee groups are matched to EPR group identifiers via the EPR Refugee Dataset, which links populations of at least 2,000 refugees to EPR groups in neighboring or proximate asylum countries (within 950 km of the origin border). For dyads outside this coverage I code the refugee group manually from World

Table 22: Robustness: Alternative Measures of Conflict and Repression

Model:	Main (1)	Battle deaths (2)	COO t-2 (3)	Total Repatriation De-escalation (4)	PITF war (5)	+ PTS/Genocide (6)
<i>Variables</i>						
<i>Forced return</i>						
Forced Return: severe	1.02*** (0.21)	0.83*** (0.20)	1.03*** (0.21)	1.06*** (0.21)	1.04*** (0.22)	0.99*** (0.22)
Forced Return: moderate	0.50* (0.21)	0.45* (0.22)	0.56** (0.21)	0.54* (0.21)	0.51* (0.21)	0.43† (0.23)
Conflict in COA: high	0.54† (0.30)		0.73* (0.31)	0.49† (0.29)	0.53† (0.31)	0.58† (0.32)
Conflict in COA: low	0.82*** (0.21)		0.89*** (0.20)	0.82*** (0.21)	0.84*** (0.21)	0.81*** (0.21)
<i>Preference-based explanations</i>						
War Ending	0.63** (0.20)	0.61*** (0.15)	0.56*** (0.15)	0.71*** (0.16)	0.56** (0.20)	0.69*** (0.20)
Conflict in COO: high	0.13 (0.35)				0.07 (0.36)	0.22 (0.36)
Conflict in COO: low	0.02 (0.22)				0.06 (0.22)	0.13 (0.20)
GDP (COO)	0.38 (0.29)	0.33 (0.29)	0.45 (0.29)	0.34 (0.29)	0.41 (0.29)	0.20 (0.31)
GDP (COA)	-1.15† (0.64)	-1.14* (0.47)	-1.23† (0.65)	-1.18† (0.65)	-1.07† (0.63)	-0.73 (0.63)
Civil Liberties Restrictions (COO)	-0.55*** (0.16)	-0.53*** (0.15)	-0.54*** (0.15)	-0.57*** (0.15)	-0.59*** (0.16)	-0.56** (0.19)
Refugee Rights	0.04 (0.16)	0.22 (0.16)	0.10 (0.16)	0.04 (0.15)	0.02 (0.17)	0.06 (0.18)
Conflict Duration	-0.09 (0.09)	-0.05 (0.10)	-0.03 (0.08)	-0.04 (0.08)	-0.15 (0.10)	-0.11 (0.09)
<i>Alternative conflict and repression measures</i>						
Battle Deaths (COA, log)		0.14*** (0.04)				
Battle Deaths (COO, log)		-0.02 (0.04)				
Conflict in COO: high (t-2)			-0.21 (0.19)			
Conflict in COO: low (t-2)			-0.08 (0.13)			
De-escalation (COO)				-0.41† (0.24)		
PITF War/Violence (COO)					1.39* (0.60)	
Political Terror Scale (COO)						-0.17 (0.14)
Genocide/Politicide (COO)						1.07*** (0.36)
All controls	Yes	Yes	Yes	Yes	Yes	Yes
Years	1980-2022	1990-2022	1982-2022	1980-2022	1980-2022	1980-2022
Fixed-effects						
Dyad	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes
Fit statistics						
Observations	2,327	2,134	2,128	2,327	2,238	1,994
Pseudo R ²	0.79	0.81	0.81	0.80	0.79	0.80

Clustered (Dyad) standard-errors in parentheses

Signif. Codes: ***: 0.001, **: 0.01, *: 0.05, †: 0.1

All models carry the full specification of the main table, including the post-war window and the Freedom House civil liberties measure, and differ only in how conflict and repression are measured. Columns 2 to 4 replace the ordinal conflict levels; columns 5 and 6 add measures alongside them. Column 6 adds the Political Terror Scale (the mean of the Amnesty International, Human Rights Watch and U.S. State Department ratings, standardised) and a binary indicator for an ongoing episode of genocide or politicide (PITF), whose episode list ends in 2017. PPML with the log stock at risk as offset, dyad and year fixed effects, SEs clustered by dyad.

Refugee Survey reports.¹ EPR coverage is unavailable for some refugee groups, reducing the estimation sample.

2.3 Alternative Explanatory Variable Measures

Table 24 tests sensitivity to alternative operationalizations and additional covariates. Model 1 reproduces the baseline. Model 2 replaces continuous origin and asylum GDP per capita with quartile indicators (Q1 reference) to relax linearity assumptions. Model 3 lags all independent variables by one year, addressing the possibility that contemporaneous asylum-country conflict partly follows refugee return or refugee mobilization.

Model 4 adds UNHCR funding per refugee in the asylum country. This measure is difficult to interpret, because it is a proxy for two things at once: what UNHCR spends on organising repatriation, and what it spends on supporting refugees where they are, including camps and health care. The first would be expected to raise return and the second to reduce the pressure to leave, and the funding series does not separate them. I therefore include it as a control rather than as a test of a mechanism, and the positive coefficient (0.95) should not be interpreted causally.

Models 5 and 6 add conditions in the asylum country. Model 5 includes educational attainment among the resident citizen populations of the origin and asylum countries, which is a national characteristic of each country rather than a property of the refugees themselves; no comparable schooling measure exists for refugee populations at this scale. Model 6 adds the asylum country’s urban share and an indicator that most refugees live in camps.

Lagging every independent variable by a year raises severe forced return (1.18 against 1.02) and lowers the post-war window (0.52 against 0.63).

2.4 Alternative Data Subsets

Table 25 tests whether the results hold across different subsets of the data. Model 1 reproduces the baseline on the extended GRR estimation panel. Model 2 restricts to the State Department-report period beginning in 2009.

Model 3 excludes dyads associated in the literature with the “refugee warrior” hypothesis, under which refugee populations, camps, or exile communities may become militarized and contribute to conflict spillover in host states (Zolberg 1989; Salehyan and Gleditsch 2006).² Excluding these cases assesses whether the main results are driven by settings in which refugee presence may be more plausibly endogenous to asylum-country conflict. Model 4 excludes asylum countries in Europe and North America. Model 5 restricts to dyads where refugees originally fled armed conflict rather than non-war forms of persecution. Model 6 restricts to the top three asylum countries per origin by refugee stock, addressing concerns that refugees from some countries are more dispersed across multiple asylum destinations than others; because origins vary in the number of asylum countries they appear in, highly dispersed origins contribute more dyads and receive greater implicit weight when each dyad-year is weighted equally.³

Severe forced return survives every subset, but the post-war window does not, and the source split is where it fails. Read Model 2 against column 2 of the main table and the two halves of the panel do not agree. Severe forced return is 0.62 in the World Refugee Survey years and 1.60 in the State Department years, significant

¹The same ethnic group is carried forward as long as the refugee population is continuous; the assignment is set to missing when a discontinuity in the population could plausibly indicate a change in ethnic composition. Two case-specific adjustments warrant note: EPR classifies Somalis as ‘irrelevant’ within Somalia on grounds of ethnic homogeneity, and I reclassify this group as high-status; Peuhl/Goula refugees from the Central African Republic appear in the EPR Refugee Dataset but not in the EPR Power Dataset, and are coded as ‘powerless’ based on desk research. The 2021 EPR Refugee values are carried forward from 2020.

²The excluded dyads are: RWA–COD, RWA–TZA, AFG–PAK, KHM–THA, PSE–LBN, PSE–JOR, and NIC–HND.

³Origins vary in the number of asylum destinations they appear in because the sample rule adds asylum countries for each origin until the cumulative displaced population reaches 75 per cent of the origin’s total. The median origin appears in 3 asylum countries and the most dispersed in 10, and 33 of 84 origins appear in more than three. The restriction therefore leaves most origins untouched and retains 86 per cent of the estimating sample.

Table 23: Ethnic political status and refugee return

Model: <i>Variables</i>	Total Repatriation	
	Origin status (EPR) (1)	Kin in host (EPR-TEK) (2)
<i>Forced return</i>		
Forced Return: severe	1.27*** (0.24)	1.04*** (0.22)
Forced Return: moderate	0.53* (0.21)	0.55* (0.23)
Conflict in COA: high	0.43 (0.36)	0.58* (0.29)
Conflict in COA: low	1.03*** (0.20)	0.86*** (0.21)
<i>Preference-based explanations</i>		
War Ending	0.54** (0.17)	0.64** (0.20)
Conflict in COO: high	0.38 (0.31)	0.12 (0.35)
Conflict in COO: low	0.42* (0.21)	0.01 (0.22)
GDP (COO)	0.13 (0.24)	0.38 (0.29)
GDP (COA)	-2.25*** (0.52)	-1.16† (0.63)
Civil Liberties Restrictions (COO)	-0.49*** (0.14)	-0.56*** (0.16)
Refugee Rights	0.11 (0.18)	0.05 (0.16)
Conflict Duration	-0.26** (0.10)	-0.08 (0.09)
<i>Ethnic political status</i>		
Ethnic Status: Monopoly (COO)	-1.72** (0.55)	
Ethnic Status: Senior Partner (COO)	-0.80* (0.39)	
Ethnic Status: Junior Partner (COO)	-0.83† (0.43)	
Ethnic Status: Powerless (COO)	-0.78† (0.46)	
Ethnic Status: Discriminated (COO)	-1.08* (0.44)	
Ethnic Status: Self-Excluded (COO)	-2.54*** (0.63)	
Ethnic Status: State Collapse (COO)	-1.83*** (0.47)	
Co-ethnic Kin in Power (COA)		-0.24 (0.46)
Co-ethnic Kin Excluded (COA)		0.28 (0.38)
All controls	Yes	Yes
<i>Fixed-effects</i>		
Dyad	Yes	Yes
Year	Yes	Yes
<i>Fit statistics</i>		
Observations	1,718	2,327
Pseudo R ²	0.81	0.79

Clustered (Dyad) standard-errors in parentheses

*Signif. Codes: ***: 0.001, **: 0.01, *: 0.05, †: 0.1*

Column 1 adds the ordinal EPR status of the displaced group in the origin, with Dominant omitted as the reference; EPR coverage is incomplete, which is why the sample is smaller. Column 2 adds the EPR-TEK indicators for whether the group's co-ethnics in the asylum country hold power or are excluded. Both carry the full specification of column 1 of the main table. PPML with the log stock at risk as offset, dyad and year fixed effects, SEs clustered by dyad.

Table 24: Robustness: Alternative Independent Variables

	Total Repatriation					
Model:	Main (1)	GDP Quartiles (2)	Lagged IVs (3)	+ Funding (4)	+ Education (5)	+ Urban/Camp (6)
<i>Variables</i>						
<i>Forced return</i>						
Forced Return: severe	1.02*** (0.21)	1.04*** (0.21)	1.18*** (0.30)	1.21*** (0.24)	1.05*** (0.21)	1.02*** (0.22)
Forced Return: moderate	0.50* (0.21)	0.52** (0.17)	0.25 (0.24)	0.56* (0.22)	0.52* (0.21)	0.51* (0.21)
Conflict in COA: high	0.54† (0.30)	0.80** (0.27)	0.49* (0.23)	-0.09 (0.35)	0.47 (0.32)	0.54† (0.27)
Conflict in COA: low	0.82*** (0.21)	0.69*** (0.20)	0.56** (0.18)	0.92*** (0.26)	0.87*** (0.20)	0.82*** (0.20)
<i>Preference-based explanations</i>						
War Ending	0.63** (0.20)	0.54* (0.22)	0.52* (0.25)	0.34 (0.26)	0.55** (0.21)	0.58** (0.21)
Conflict in COO: high	0.13 (0.35)	0.03 (0.36)	-0.26 (0.32)	-0.30 (0.39)	0.13 (0.35)	0.08 (0.36)
Conflict in COO: low	0.02 (0.22)	0.01 (0.21)	-0.29 (0.20)	-0.13 (0.23)	0.07 (0.22)	0.03 (0.21)
GDP (COO)	0.38 (0.29)		0.14 (0.30)	0.58† (0.31)	0.44 (0.29)	0.36 (0.29)
GDP (COA)	-1.15† (0.64)		-0.26 (0.70)	-1.15 (0.74)	-1.30* (0.65)	-1.10† (0.64)
Civil Liberties Restrictions (COO)	-0.55*** (0.16)	-0.59*** (0.18)	-0.27† (0.16)	-0.44* (0.17)	-0.58*** (0.17)	-0.55*** (0.15)
Refugee Rights	0.04 (0.16)	0.10 (0.16)	0.13 (0.16)	0.04 (0.15)	0.02 (0.17)	0.06 (0.16)
Conflict Duration	-0.09 (0.09)	-0.07 (0.09)	0.07 (0.08)	-0.07 (0.10)	-0.12 (0.10)	-0.10 (0.10)
<i>Alternative and additional measures</i>						
GDP (COA): Q2		-0.69* (0.27)				
GDP (COA): Q3		-0.54 (0.47)				
GDP (COA): Q4		-1.02 (1.29)				
GDP (COO): Q2		-0.77** (0.30)				
GDP (COO): Q3		-0.46 (0.43)				
GDP (COO): Q4		-0.18 (0.45)				
UNHCR Funding p.p. (COA)				0.95** (0.30)		
Education (COA)					-0.50 (1.15)	
Education (COO)					-0.25 (0.81)	
Urban						-0.29 (0.25)
Majority in Camps						-0.29 (0.55)
<i>Fixed-effects</i>						
Dyad	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes
<i>Fit statistics</i>						
Observations	2,327	2,327	2,164	1,928	2,202	2,325
Pseudo R ²	0.79	0.80	0.80	0.82	0.80	0.79

Clustered (Dyad) standard-errors in parentheses

Signif. Codes: ***: 0.001, **: 0.01, *: 0.05, †: 0.1

Poisson PML, log stock-at-risk offset, dyad + year FEs, SEs clustered by dyad. M1: column 1 of the main table. M2: GDP (COO/COA) quartiles (Q1 ref.). M3: all independent variables at t-1, including the war-ending window. M4: + UNHCR funding p.p. (z-log). M5: + educational attainment in the resident citizen populations of both countries. M6: + urban share and an indicator for the majority of the refugee population living in camps; the urban and camp series are missing for roughly 650 dyad-years, so M6 runs on a smaller sample.

in both, so the coercion result holds on either source and is markedly stronger on the later one. The window is 0.56 in the Survey years and -0.17, indistinguishable from zero, in the State Department years.

Two things are confounded in that contrast and this design cannot separate them. The later period has fewer war endings with large populations still in exile, so there may simply be less of the phenomenon to detect; the State Department subsample is 901 dyad-years. The sources also differ in what they record: the Survey reports on return movements directly, whereas the State Department reports are organised around human rights conditions and give return figures less consistently, which is why GRR uses UNHCR as a baseline for those years. The war-ending estimate therefore rests mainly on the pre-2009 evidence. Separating period from source effects would require coding both sources for their overlapping years.

Table 25: Robustness: Alternative samples and periods

Dep. Var.:	Total Repatriation					
Model:	Main	USHR	Excl.Warrior	Excl.ENA	War.Orig	Top.3.COA
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Variables</i>						
<i>Forced return</i>						
Forced Return: severe	1.02*** (0.21)	1.60*** (0.27)	1.59*** (0.30)	0.98*** (0.21)	1.00*** (0.21)	0.97*** (0.21)
Forced Return: moderate	0.50* (0.21)	0.25 (0.32)	0.33 (0.22)	0.51* (0.21)	0.50* (0.21)	0.51* (0.21)
Conflict in COA: high	0.54† (0.30)	-0.95* (0.46)	0.68** (0.23)	0.60* (0.29)	0.49 (0.31)	0.60* (0.30)
Conflict in COA: low	0.82*** (0.21)	0.63* (0.30)	0.51* (0.22)	0.82*** (0.21)	0.86*** (0.20)	0.84*** (0.22)
<i>Preference-based explanations</i>						
War Ending	0.63** (0.20)	-0.17 (0.36)	0.49* (0.20)	0.58** (0.22)	0.63** (0.20)	0.61** (0.21)
Conflict in COO: high	0.13 (0.35)	-0.20 (0.50)	-0.05 (0.35)	0.08 (0.36)	0.17 (0.34)	0.07 (0.37)
Conflict in COO: low	0.02 (0.22)	-0.29 (0.29)	-0.27 (0.19)	0.03 (0.22)	0.04 (0.21)	-0.02 (0.21)
GDP (COO)	0.38 (0.29)	2.54*** (0.53)	0.48† (0.27)	0.29 (0.30)	0.41 (0.29)	0.36 (0.30)
GDP (COA)	-1.15† (0.64)	-5.09** (1.76)	-1.72*** (0.47)	-1.20† (0.65)	-1.16† (0.64)	-1.30* (0.65)
Civil Liberties Restrictions (COO)	-0.55*** (0.16)	-0.32 (0.37)	-0.51** (0.16)	-0.61*** (0.17)	-0.58*** (0.16)	-0.55*** (0.16)
Refugee Rights	0.04 (0.16)	0.01 (0.27)	0.07 (0.14)	0.07 (0.15)	0.04 (0.18)	0.07 (0.17)
Conflict Duration	-0.09 (0.09)	0.23 (0.16)	0.01 (0.09)	-0.10 (0.09)	-0.08 (0.09)	-0.07 (0.09)
All controls	Yes	Yes	Yes	Yes	Yes	Yes
Source	Both	USHR	Both	Both	Both	Both
Years	1980-2022	2009-2022	1980-2022	1980-2022	1980-2022	1980-2022
<i>Fixed-effects</i>						
Dyad	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes
<i>Fit statistics</i>						
Observations	2,327	901	2,242	1,877	2,099	2,007
Pseudo R ²	0.79	0.82	0.79	0.79	0.80	0.79

Clustered (Dyad) standard-errors in parentheses

*Signif. Codes: ***: 0.001, **: 0.01, *: 0.05, †: 0.1*

All models: Poisson PML with log stock-at-risk offset, dyad and year fixed effects. Standard errors clustered by dyad. Model 1 is column 1 of the main table. Model 2 restricts to the years coded from State Department reports; the World Refugee Survey years are not a column here because they are column 2 of the main table. Model 3 excludes dyads linked in the literature to the refugee warrior hypothesis. Model 4 excludes dyads with asylum countries in Europe and North America. Model 5 restricts to dyads where refugees originally fled armed conflict (`o_pers_nonwar == 0`). Model 6 restricts to the top 3 asylum countries per origin-year by refugee stock. The checks that substitute UNHCR returns or the registered stock are reported in Table 13.

2.5 Alternative Specifications

Table 26 varies the model specification. Model 1 reproduces the baseline Poisson model. Models 2 and 3 are OLS on a transformed outcome, $\log(y + 1)$ and $\text{asinh}(y)$, the two transforms usually reached for when a count has many zeros. Model 4 estimates a negative binomial model, which relaxes the Poisson equidispersion

assumption. Models 5 and 6 add two-way fixed effects on each side of the dyad in turn. Model 5 uses origin-by-year fixed effects, which absorb all time-varying origin-country confounders, including origin-country conflict, ethnic status, conflict years and the post-war window itself, and identify the asylum-side variables from variation across the hosts of one origin in one year. Model 6 uses asylum-by-year fixed effects, which do the same on the other side: they absorb everything common to a host in a given year, including its own conflict, its income and its statutory refugee-rights position, and identify the coefficients from variation across the origin groups that host receives in that year. The forced-return measures are coded by dyad rather than by host, so they survive this second design.

Coefficient magnitudes are not comparable across estimators, and in Models 2 and 3 they are not interpretable on their own terms either. With 71 per cent of the dyad-years they use recording no return, neither $\log(y+1)$ nor $\text{asinh}(y)$ is invariant to the units in which return is measured: counting returnees in thousands rather than in people moves the severe-policy estimate by about two-thirds in both columns, while leaving the Poisson estimate in column 1 exactly where it is. The two OLS columns establish that the association survives a different functional form; the proportional reading of the coercion result belongs to Model 1, where 1.02 is a 177 per cent increase in the return count.

Under origin-by-year fixed effects the severe-policy coefficient is larger than in the baseline (1.34 against 1.02), though the high-intensity asylum-conflict estimate is imprecise there. Under asylum-by-year fixed effects it is 1.60. The coercion result therefore holds both across the hosts of one origin in one year and across the origin groups inside one host in one year.

2.5.1 Alternative Clustering of the Standard Errors

The baseline clusters by dyad because the shocks used in the model — a host government’s policy and a war ending at the origin — can persist within a dyad across years. Alternative groupings account for dependence among dyads that share an origin, a host, or a calendar year.

Table 27 reports the standard errors and p -values under each scheme; the point estimates are common to all of them. Clustering by year, or two-way by dyad and year, gives the largest standard errors, and even there the severe-policy estimate is significant at the 0.001 level and the war-ending estimate at the 0.01 level.

Table 27: Alternative clustering of the standard errors

Clustered by	Groups	Forced Return: severe		War Ending	
		SE	p	SE	p
Dyad (baseline)	110	0.21***	<0.001	0.20**	0.002
Origin country	51	0.19***	<0.001	0.18***	<0.001
Asylum country	65	0.20***	<0.001	0.20**	0.002
Origin + asylum, two-way	116	0.17***	<0.001	0.19**	0.001
Year	41	0.27***	<0.001	0.18***	<0.001
Dyad + year, two-way	151	0.25***	<0.001	0.22**	0.003
Heteroskedastic, no clustering	–	0.23***	<0.001	0.16***	<0.001

Note: The model is column 1 of the main table throughout, so the coefficients do not vary: severe forced return is 1.02 and the war-ending window 0.63 in every row. Stars are on the coefficient’s significance under that variance estimator. ‘Groups’ is the number of clusters, summed across dimensions for the two-way estimators. The dyad-plus-year row should be read with care: with 41 year groups the variance matrix is not positive semi-definite and fixest repairs it; the year-clustered row is the cleaner conservative benchmark.

2.6 Event Study around Onsets of Severe Forced Return

The main text reports a stacked event study around the onset of severe forced-return policies, which addresses the concern that host governments adopt coercive policies after return has already begun to rise. It finds no

Table 26: Robustness: Alternative Specifications

Dep. Var.:	Total Repatriation	log(trepat+1)	asinh(trepat)	Total Repatriation		
Model:	Main (1) Poisson	OLS log (2) OLS	OLS asinh (3) OLS	Neg.Bin. (4) Neg. Bin.	Origin-Yr (5) Poisson	Asylum-Yr (6) Poisson
<i>Variables</i>						
<i>Forced return</i>						
Forced Return: severe	1.02*** (0.21)	4.47*** (0.41)	4.76*** (0.44)	2.65*** (0.51)	1.34*** (0.34)	1.60* (0.80)
Forced Return: moderate	0.50* (0.21)	1.65*** (0.29)	1.81*** (0.32)	1.23* (0.57)	1.36*** (0.16)	0.82† (0.44)
Conflict in COA: high	0.54† (0.30)	0.57 (0.46)	0.59 (0.49)	0.77 (0.50)	0.67 (0.51)	
Conflict in COA: low	0.82*** (0.21)	0.39 (0.30)	0.42 (0.32)	0.22 (0.39)	0.97*** (0.18)	
<i>Preference-based explanations</i>						
War Ending	0.63** (0.20)	0.85*** (0.24)	0.92*** (0.25)	0.78** (0.27)		0.12 (0.23)
Conflict in COO: high	0.13 (0.35)	-0.47 (0.43)	-0.50 (0.46)	0.81 (0.56)	-0.58 (0.78)	0.39 (0.52)
Conflict in COO: low	0.02 (0.22)	-0.34 (0.26)	-0.37 (0.28)	-0.25 (0.28)	0.05 (0.41)	0.27 (0.30)
GDP (COO)	0.38 (0.29)	0.28 (0.26)	0.29 (0.28)	0.33 (0.53)		-0.02 (0.56)
GDP (COA)	-1.15† (0.64)	-0.92 (0.57)	-0.99 (0.62)	-0.76 (1.04)	-2.14† (1.11)	
Civil Liberties Restrictions (COO)	-0.55*** (0.16)	-0.51** (0.17)	-0.55** (0.19)	-0.48 (0.32)		-0.65† (0.33)
Refugee Rights	0.04 (0.16)	0.23 (0.20)	0.25 (0.21)	0.33 (0.29)	0.10 (0.16)	
Conflict Duration	-0.09 (0.09)	0.21 (0.14)	0.22 (0.15)	0.37† (0.21)	0.09 (0.18)	-0.07 (0.18)
Log Stock at Risk		0.78*** (0.10)	0.84*** (0.11)			
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Estimator	Poisson	OLS	OLS	NegBin	Poisson	Poisson
DV	trepat	log(y+1)	asinh(y)	trepat	trepat	trepat
<i>Fixed-effects</i>						
Dyad	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes		
coo_iso-Year					Yes	
coa_iso-Year						Yes
<i>Fit statistics</i>						
Observations	2,327	3,913	3,913	2,327	1,397	1,013
Pseudo R ²	0.79	0.15	0.15	0.04	0.93	0.93

Clustered (Dyad) standard-errors in parentheses

Signif. Codes: ***: 0.001, **: 0.01, *: 0.05, †: 0.1

All models include dyad fixed effects and cluster standard errors by dyad; alternative clustering schemes are reported separately. Origin-Yr and Asylum-Yr are origin-by-year and asylum-by-year fixed effects, and y is the recorded return count. Models 2 and 3 are OLS on a transformed outcome and enter the log stock at risk directly in place of the offset, which is why they run on more dyad-years than the Poisson columns. Neither transform is invariant to the units of the outcome once the outcome has zeros, and 71 per cent of the dyad-years they use record no return: measuring return in thousands rather than in people moves the severe-policy estimate from 4.47 to 1.52 in Model 2 and from 4.76 to 1.80 in Model 3, on identical data, while leaving Model 1 unchanged. Columns 2 and 3 therefore test whether the sign and significance survive a different functional form, and their magnitudes should not be read as effect sizes or compared with column 1. Model 4 uses negative binomial. Model 5 adds origin-by-year and Model 6 asylum-by-year fixed effects; variables the fixed effects absorb are omitted, which is why the origin-side rows are empty in Model 5 and the asylum-country conflict, income and refugee-rights rows are empty in Model 6. The forced-return measures survive Model 6 because they are coded by dyad and so vary across the origin groups one host receives in one year.

pre-trend two or three years before onset, a sharp rise in the onset year and a peak one year later, and notes that the design rests on few onsets and cannot rule out anticipation in the omitted year immediately before implementation.

This section reports the underlying table, for two- and three-year windows, and the linear version of the estimator. Severe forced return begins 49 times and ends 51 times in the panel, so treatment is not absorbing; each onset therefore defines its own cohort with a window of $\pm K$ years, and controls are dyads observed throughout the same period whose host records no severe policy in any year of that window. The two-year window contains 34 cohorts across 24 dyads and the three-year window 23 cohorts across 17 dyads. Alongside PPML on the return count I estimate a linear model on return as a share of the stock at risk, which retains dyads with no within-dyad variation in return and so uses about 13,000 stacked dyad-years against about 5,300 under PPML.

Table 28: Event study around onsets of severe forced return

Event year	PPML, $K = 2$	Rate, $K = 2$	PPML, $K = 3$	Rate, $K = 3$
$t - 3$	—	—	0.23 (0.21)	-0.03 [†] (0.02)
$t - 2$	-0.31 (0.43)	-0.00 (0.02)	0.21 (0.32)	-0.00 (0.02)
$t - 1$	(ref.)	(ref.)	(ref.)	(ref.)
$t + 0$	0.71* (0.28)	0.11** (0.04)	0.96*** (0.24)	0.06* (0.03)
$t + 1$	0.83 [†] (0.45)	0.07 [†] (0.04)	1.53*** (0.29)	0.10* (0.05)
$t + 2$	0.34 (0.34)	0.03 (0.04)	0.88** (0.30)	0.06 (0.05)
$t + 3$	—	—	0.42 (0.31)	0.02 (0.02)

The two estimators and the two windows give the same picture as the main text. The lead coefficients are small and not distinguishable from zero in any of the four: joint tests of the leads yield $p = 0.47$ for PPML and $p = 0.88$ for the return rate when $K = 2$, and $p = 0.55$ and $p = 0.11$ when $K = 3$.

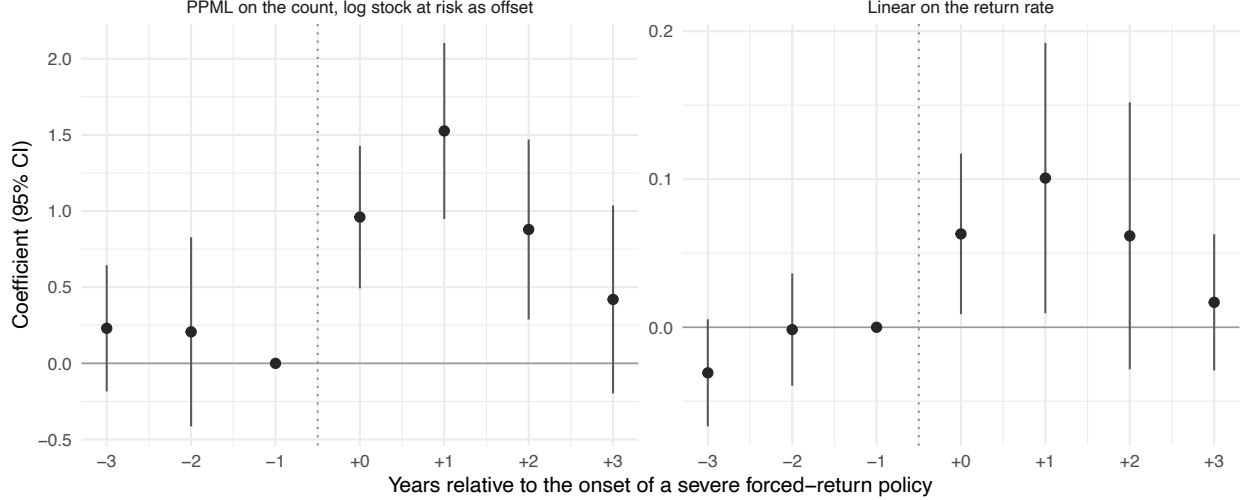


Figure 4: Event study around onsets of severe forced return, three-year window, both estimators

Figure 4 plots the three-year window under both estimators. The vertical scales are not comparable, since the PPML coefficients are on the log scale and the linear-model coefficients are changes in the return rate. One further limitation is worth adding to those noted in the main text: the controls are dyads without severe coercion during the relevant window, not a randomly assigned comparison group, so the estimates address whether return was already rising before recorded policy onset rather than the full endogeneity of policy adoption.

2.7 Influential Cases

2.7.1 Dropping the Largest Cases

The top five percent of dyad-years account for 88.5 percent of recorded return. Table 29 tests whether this concentration drives the estimates by dropping the largest dyads or dyad-years and, separately, restricting the sample to the ten largest dyads.

The severe-policy coefficient is larger after each exclusion, not smaller: 1.41 after dropping the five largest dyads, 1.50 after dropping the largest one percent of dyad-years and 1.46 after dropping the largest five percent, against 1.02 in the baseline, and significant at the 0.001 level in each. Among the ten largest dyads alone it is smaller (0.79) and marginally significant on 275 dyad-years.

The post-war estimate is less stable, and the exclusions move it downward. Dropping the five largest dyads lowers it from 0.63 to 0.42 and dropping the largest one percent leaves it at 0.52; dropping the largest five percent reduces it to 0.19 and among the ten largest dyads it is -0.11, neither significant. The window contains only 452 dyad-years, so trimming the largest five percent removes much of its support. Low-intensity asylum-country conflict also becomes imprecise after excluding the largest cases, but is strongly associated with return among the ten largest dyads (1.26). Severe policy is therefore robust across the distribution, while the post-war window and low-intensity asylum conflict depend more on which large episodes are included.

2.7.2 Leave-One-Out Analyses

Figure 5 re-estimates the main model after excluding each origin country, asylum country, origin region, and asylum region in turn. Table 30 reports exclusions that change a coefficient's sign or move it by more than one.

No single origin, host, region or dyad accounts for the severe-policy result. The largest change at the dyad level is 0.25: excluding Afghanistan–Pakistan raises the coefficient from 1.02 to 1.27 and excluding Rwanda–DR Congo raises it to 1.25, and every estimate stays significant at the 0.001 level.

The larger shifts occur for coefficients that are weakly estimated in the baseline. High-intensity asylum-country conflict, significant only at the ten percent level (0.54), becomes negative when Sub-Saharan Africa is excluded as either an origin or an asylum region, although the resulting estimates are not significant, and it is most sensitive to Rwanda–DR Congo and the two Afghan dyads. The moderate-policy coefficient also changes when Asia and MENA are excluded, and origin-country conflict changes sign under several exclusions but is never distinguishable from zero in the baseline. The asylum-conflict coefficient is therefore the least stable of the main asylum-side results.

Table 29: Robustness: Concentration of Returns

Dependent Variable: Model:	Total Repatriation				
	Full (1)	Drop top 5 dyads (2)	Drop top 1% yrs (3)	Drop top 5% yrs (4)	Top 10 dyads (5)
<i>Forced return</i>					
Forced Return: severe	1.02*** (0.21)	1.41*** (0.24)	1.50*** (0.19)	1.46*** (0.34)	0.79† (0.45)
Forced Return: moderate	0.50* (0.21)	0.11 (0.19)	0.40* (0.18)	0.56* (0.22)	0.67 (0.44)
Conflict in COA: high	0.54† (0.30)	0.52* (0.25)	0.53* (0.23)	0.36 (0.24)	0.33 (0.66)
Conflict in COA: low	0.82*** (0.21)	0.32 (0.26)	0.33 (0.25)	0.03 (0.18)	1.26** (0.38)
<i>Preference-based explanations</i>					
War Ending	0.63** (0.20)	0.42 (0.26)	0.52* (0.22)	0.19 (0.19)	-0.11 (0.64)
Conflict in COO: high	0.13 (0.35)	0.07 (0.34)	0.13 (0.29)	-0.53† (0.29)	-1.23 (0.79)
Conflict in COO: low	0.02 (0.22)	-0.12 (0.20)	0.02 (0.18)	-0.33* (0.17)	-0.64 (0.45)
GDP (COO)	0.38 (0.29)	0.62* (0.30)	0.36 (0.27)	0.59** (0.23)	1.25 (0.80)
GDP (COA)	-1.15† (0.64)	-0.47 (0.60)	-0.53 (0.48)	0.21 (0.56)	-4.04** (1.41)
Civil Liberties Restrictions (COO)	-0.55*** (0.16)	-0.45* (0.19)	-0.55** (0.18)	-0.33† (0.18)	-0.85** (0.29)
Refugee Rights	0.04 (0.16)	0.07 (0.14)	0.12 (0.13)	0.03 (0.13)	0.36** (0.14)
Conflict Duration	-0.09 (0.09)	0.05 (0.10)	-0.02 (0.09)	0.06 (0.10)	0.34 (0.31)
Fixed-effects					
Dyad	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes
Fit statistics					
Observations	2,327	2,199	2,289	2,081	275
Squared Correlation	0.75	0.61	0.57	0.33	0.88
Pseudo R ²	0.79	0.67	0.64	0.44	0.87

Clustered (Dyad) standard-errors in parentheses

Signif. Codes: ***: 0.001, **: 0.01, *: 0.05, †: 0.1

Table 30: Influential regions

Variable	Region Dropped	Drop Type	Main Est.	New Est.	Change
Conflict in COA: high	Sub-Saharan Africa	Origin Region	0.54+	-0.33	0.86
Conflict in COA: high	Sub-Saharan Africa	Asylum Region	0.54+	-0.24	0.77
Forced Return: moderate	Asia & MENA	Origin Region	0.50*	-0.04	0.54
Forced Return: moderate	Asia & MENA	Asylum Region	0.50*	-0.04	0.54
Conflict in COO: low	Sub-Saharan Africa	Asylum Region	0.02	-0.39	0.41
Conflict in COO: low	Sub-Saharan Africa	Origin Region	0.02	-0.33	0.35
Conflict in COO: low	Latin America & Caribbean	Origin Region	0.02	-0.01	0.03
Conflict in COO: low	Latin America & Caribbean	Asylum Region	0.02	-0.01	0.03

Note: Regions whose exclusion causes a sign change or an absolute coefficient shift above 1. Stars indicate significance of each coefficient: + p<0.10, * p<0.05, ** p<0.01, *** p<0.001. Bold rows indicate sign reversals where the new coefficient remains statistically significant at p<0.05; sign flips on coefficients that fall back to non-significance after exclusion are shown but not bolded.

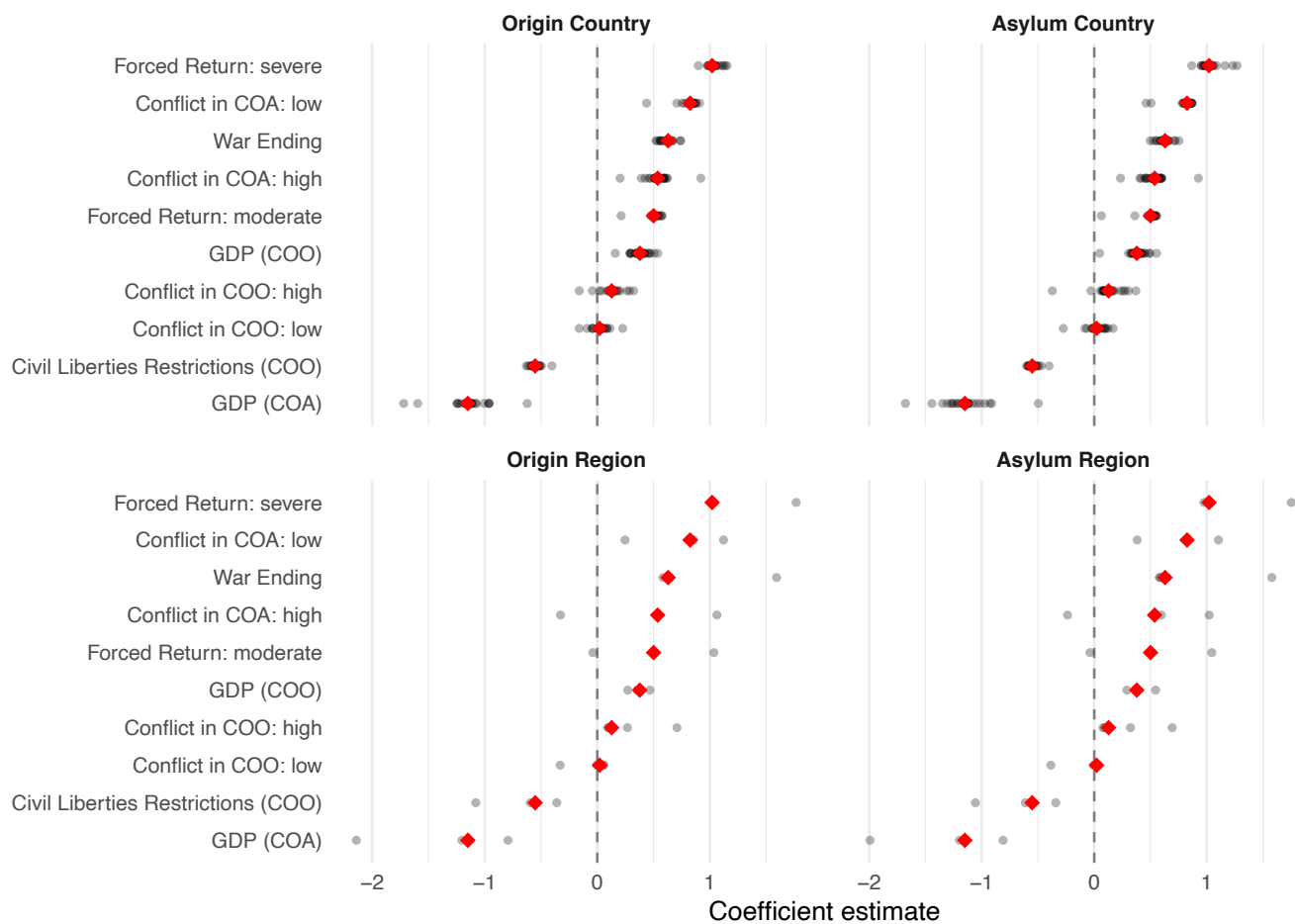


Figure 5: Coefficient stability across leave-one-out analyses

Table 31: Most influential dyads

Variable	Dyad	Main Est.	New Est.	Change
Conflict in COA: high	AFG-PAK	0.54+	0.93***	0.39
Conflict in COA: high	RWA-COD	0.54+	0.25	0.28
Conflict in COA: high	AFG-IRN	0.54+	0.34	0.20
Conflict in COA: high	SOM-ETH	0.54+	0.46	0.08
Conflict in COA: high	MOZ-MWI	0.54+	0.47	0.07
Conflict in COA: low	AFG-IRN	0.82***	0.42+	0.40
Conflict in COA: low	AFG-PAK	0.82***	0.51*	0.32
Conflict in COA: low	MMR-THA	0.82***	0.71***	0.11
Conflict in COA: low	KHM-THA	0.82***	0.91***	0.08
Conflict in COA: low	RWA-COD	0.82***	0.77***	0.06
Forced Return: severe	AFG-PAK	1.02***	1.27***	0.25
Forced Return: severe	RWA-COD	1.02***	1.25***	0.23
Forced Return: severe	AFG-IRN	1.02***	0.86***	0.16
Forced Return: severe	BDI-TZA	1.02***	1.15***	0.13
Forced Return: severe	MMR-THA	1.02***	0.95***	0.07
Forced Return: moderate	AFG-IRN	0.50*	0.10	0.41
Forced Return: moderate	AFG-PAK	0.50*	0.36*	0.14
Forced Return: moderate	ETH-SOM	0.50*	0.56**	0.06
Forced Return: moderate	BDI-TZA	0.50*	0.55*	0.05
Forced Return: moderate	RWA-COD	0.50*	0.55*	0.05
GDP (COA)	AFG-IRN	-1.15+	-0.45	0.70
GDP (COA)	KHM-THA	-1.15+	-1.73***	0.58
GDP (COA)	UGA-COD	-1.15+	-1.42*	0.27
GDP (COA)	UGA-SDN	-1.15+	-1.34*	0.19
GDP (COA)	BDI-TZA	-1.15+	-0.96	0.19
War Ending	ETH-SOM	0.63**	0.75***	0.12
War Ending	AFG-PAK	0.63**	0.52**	0.11
War Ending	MMR-THA	0.63**	0.72***	0.09
War Ending	LBR-GIN	0.63**	0.54**	0.09
War Ending	ETH-ERI-SDN	0.63**	0.55**	0.08
Civil Liberties Restrictions (COO)	AFG-IRN	-0.55***	-0.46**	0.09
Civil Liberties Restrictions (COO)	ETH-ERI-SDN	-0.55***	-0.63***	0.08
Civil Liberties Restrictions (COO)	AFG-PAK	-0.55***	-0.47**	0.08
Civil Liberties Restrictions (COO)	IRQ-IRN-IRN	-0.55***	-0.50***	0.06
Civil Liberties Restrictions (COO)	KHM-THA	-0.55***	-0.60***	0.05
Conflict in COO: high	RWA-COD	0.13	-0.19	0.32
Conflict in COO: high	ETH-SOM	0.13	0.37	0.24
Conflict in COO: high	AFG-PAK	0.13	0.27	0.15
Conflict in COO: high	MOZ-MWI	0.13	0.26	0.13
Conflict in COO: high	AFG-IRN	0.13	0.00	0.13
Conflict in COO: low	RWA-COD	0.02	-0.20	0.22
Conflict in COO: low	MOZ-MWI	0.02	0.17	0.15
Conflict in COO: low	AFG-PAK	0.02	-0.08	0.10
Conflict in COO: low	AFG-IRN	0.02	0.12	0.10
Conflict in COO: low	ETH-SOM	0.02	0.12	0.10
GDP (COO)	KHM-THA	0.38	0.16	0.22
GDP (COO)	AFG-PAK	0.38	0.55+	0.18
GDP (COO)	MMR-BGD	0.38	0.50+	0.12
GDP (COO)	MMR-THA	0.38	0.27	0.11
GDP (COO)	SOM-ETH	0.38	0.49+	0.11

Note: For each key variable, the five dyads whose exclusion produces the largest absolute change in the coefficient. Stars indicate significance of each coefficient: + $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Bold rows indicate sign reversals where the new coefficient remains statistically significant at $p < 0.05$.

2.8 Aggregate Distribution and Model-Adjusted Contrasts

The main text evaluates H2 with a mutually exclusive comparison of four contexts: neither an asylum-side shock nor a post-war period, a post-war period only, an asylum-side shock only, and both conditions. That comparison identifies where observed return is concentrated without assigning a counterfactual motive to individual movements. This section first disaggregates the descriptive patterns by policy level and conflict intensity. It then reports a pooled model-adjusted contrast as a robustness check on whether observed compositional differences account for the descriptive ordering. The model-adjusted quantities are not a second test of H2 and are not interpreted as causal effects.

2.8.1 Detailed Descriptive Comparison

Table 32 splits the asylum-shock indicator into its parts, so that the contribution of each policy level and each conflict intensity can be read separately. The categories within each panel are mutually exclusive, but the policy and conflict panels should not be added to one another, because the same dyad-year can appear in both.

The policy disaggregation shows a marked difference between moderate and severe measures. Severe policy occupies 2.9 per cent of dyad-years and 17.6 per cent of the stock-years at risk, but 35.3 per cent of all recorded return, an aggregate rate of 10.05 per cent against 5.00 per cent across the whole sample. Moderate-policy dyad-years have a return rate of 5.11 per cent, close to the sample-wide rate, and account for 14.4 per cent of recorded return. The aggregate category for any policy, which accounts for 49.7 per cent of recorded return, therefore combines substantively different policy levels.

The conflict disaggregation does not show a monotone intensity gradient. Low-intensity conflict dyad-years have a return rate of 6.64 per cent, slightly faster than high-intensity conflict at 6.12 per cent, and carries 49.8 per cent of recorded return against 13.6 per cent, because it covers 37.5 per cent of the stock-years at risk against 11.1 per cent. Much of the aggregate association between asylum-country conflict and return therefore reflects the number of refugees exposed to conflict, while conflict intensity itself does not order the return rates.

Table 32: Recorded return by policy level and asylum-conflict intensity

Context	Dyad-years	Stock-years at risk (m)	Return (m)	Share of return	Return rate
Forced-return policy in the asylum country					
No policy recorded	4,669 (91.4%)	454.79 (68.3%)	16.73	50.3%	3.68%
Moderate policy	290 (5.7%)	93.80 (14.1%)	4.79	14.4%	5.11%
Severe policy	150 (2.9%)	116.96 (17.6%)	11.75	35.3%	10.05%
Conflict in the asylum country					
No conflict in asylum	3,852 (75.4%)	341.92 (51.4%)	12.17	36.6%	3.56%
Low intensity	896 (17.5%)	249.70 (37.5%)	16.58	49.8%	6.64%
High intensity	361 (7.1%)	73.94 (11.1%)	4.53	13.6%	6.12%
<i>All dyad-years</i>	5,109 (100.0%)	665.55 (100.0%)	33.28	100.0%	5.00%

Note: Each block partitions the same 5,109 dyad-years, so the total row is the benchmark for both. Dyad-years and stock-years carry their share of the sample total in parentheses. The return rate is total recorded return over total stock-years at risk within the block, not the mean of dyad-year ratios, which would weight a small refugee population like a large one.

Two qualifications apply. First, the forced-return measure is uncoded in 681 of the 5,109 dyad-years, and those rows are counted as no policy recorded rather than as confirmed policy absences. The exposure they represent is small – 1.6 per cent of stock-years and 0.1 per cent of recorded return – because coverage is worst for high-income asylum states, whose sources discuss refugee treatment in aggregate rather than by origin group (Section 3.2). Treating these observations as no recorded policy can understate the prevalence of forced-return measures, although their small share of stock-years and returns limits their influence on the aggregate comparison. Second, these descriptive comparisons do not adjust for differences in composition. The severe-policy category, in particular, consists of a small number of very large hosting relationships: its average dyad-year holds approximately 780,000 refugees at risk, against 205,000 in the average high-intensity

asylum-conflict dyad-year. The next subsection assesses whether a pooled covariate adjustment changes the ordering of the three focal conditions.

2.8.2 Model-Adjusted Aggregate Contrasts

Table 33 supplements the mutually exclusive descriptive comparison with contrasts from Model 4 of the main paper. For each condition, the model is used to predict aggregate return under the observed data and after setting that condition to its reference value while holding the remaining covariates fixed. The table also reports the frequency of each condition and the refugee stock-years exposed to it. These rows overlap and therefore cannot be summed.

Table 33: Exposure and model-adjusted aggregate contrasts

Condition	Dyad-years present	Stock-years exposed	Reduction in fitted return
Severe forced-return policy	132 (3.5%)	99.9m	25.0%
Post-war window (1–3 years)	390 (10.2%)	75.7m	16.9%
High-intensity asylum conflict	280 (7.3%)	55.0m	4.4%

Note: The final column uses the pooled PPML specification in Model 4 of the main paper and reports the percentage reduction in aggregate fitted return after setting the listed condition to its reference value, holding the remaining covariates fixed. The model includes 3,810 dyad-years in 249 dyads and retains 66.5 per cent of stock-years in the estimation frame. Conditions can overlap, so the rows are not additive. The final column is a model-implied descriptive contrast, not an identified causal effect.

Setting severe forced-return policy to zero reduces fitted aggregate return by 25.0 per cent, compared with 16.9 per cent for the post-war window and 4.4 per cent for high-intensity asylum-country conflict. This ordering parallels the narrow descriptive comparison. Severe policy occurs in only 132 dyad-years, compared with 390 post-war dyad-years, but covers 99.9 million refugee stock-years against 75.7 million, and the average severe-policy dyad-year holds 756,928 refugees at risk against 193,991 in the average post-war dyad-year. In Model 4, the severe-policy coefficient is also larger than the post-war coefficient (1.03 versus 0.72), so the adjusted ordering reflects both exposure and the fitted proportional association.

The conditions remain nonexclusive: 69 dyad-years contain two of the three factors, and setting all three to their reference values jointly reduces fitted return by 42.2 per cent rather than the 46.4 per cent obtained by adding the separate contrasts. More importantly, changing a covariate on a fitted surface does not identify how many refugees were caused to return by that condition. The model-adjusted comparison is therefore used only to show that the ordering between severe policy and the post-war window persists after adjustment for the observed covariates in Model 4.

3 Extensions and Supporting Analyses

3.1 Correlates of Global Refugee Return

Figure 6 compares the share of recorded return that occurs under each condition with the share of the population at risk exposed to it. The conditions are not mutually exclusive. These are descriptive comparisons without controls or fixed effects.

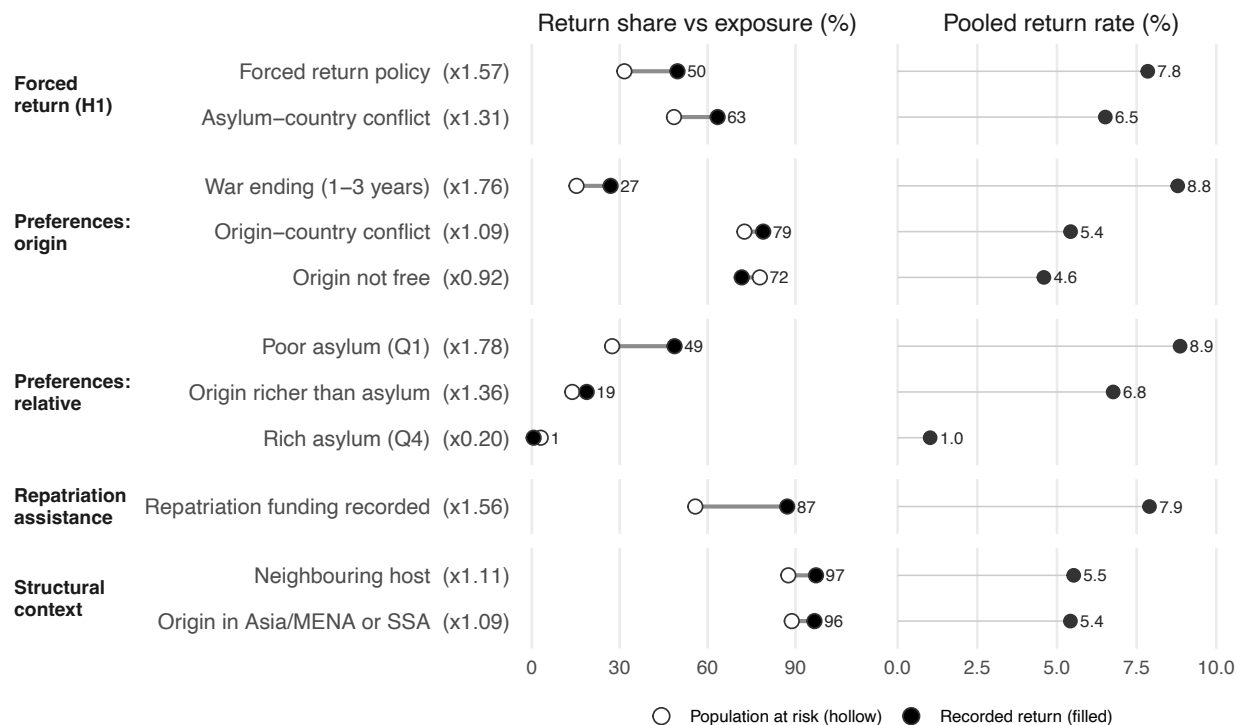


Figure 6: Correlates of global refugee return

3.2 The Implementation of Forced Return Policies

The main analysis treats forced-return policy as a constraint on refugees' ability to remain in asylum. This section asks which hosts and refugee populations are most likely to face that constraint, and when governments adopt it. Possible explanations concern the host government, characteristics of the refugee population, and the trajectory of the displacement crisis. Coercion may be more likely under authoritarian governments or weak legal protections, against large, socially distant, or militarized populations, or as conflict and displacement persist.

The analysis uses the full GRR policy panel, retaining dyad-years with a coded forced-return measure and more than 1,000 refugees in the preceding year. Severe policy appears in 150 dyad-years (3.4%), while either moderate or severe policy appears in 441 (9.9%). These are descriptive rather than causal estimates, but they show which associations persist across different comparisons.

Five tables follow. Table 34 enters each candidate correlate on its own, with no controls, to show what the intuitive explanations look like before composition is addressed. Table 35 then holds host-year, origin-year and dyad characteristics fixed in turn, so each column poses the question to a different comparison set: different refugee populations in the same host and year, different hosts of the same origin in the same year, and the same dyad over time. Table 36 repeats that ladder for social and ethnic ties between origin and host, and Table 38 for origin battle deaths and reported militarization, the two security explanations that

most plausibly compete with population size. Table 37 turns from which populations to when: it reports the probability that a severe policy begins, by how long the dyad has been in the panel and by whether the population is above or below 100,000.

The clearest result, in Tables 34 and 35, is the importance of the size of the refugee population targeted by the policy. In the fully adjusted pooled model, a ten-fold increase in lagged dyad-specific stock is associated with 8.10 times the odds of severe policy. The relationship remains positive when comparing different refugee populations within the same host-year and when comparing the same dyad over time. It is also the dyad-specific population, rather than the host country's overall refugee burden, that matters: once both are included, the association with total host-country refugee population largely disappears.

Other proposed explanations are much less consistent. Democracy and refugee-rights protections are associated with less coercion in simple comparisons, but these relationships change sign or disappear after adjustment. Origin-country violence and reported refugee militarization are also associated with coercion in the bivariate data, but neither consistently predicts severe policy once refugee stock and other differences are taken into account. Nor is there consistent evidence that social proximity between refugees and hosts protects against coercion.

The timing of severe policy, in Table 37, reinforces the importance of population size but provides little evidence for a general "host fatigue" process. Severe policy often occurs during ongoing conflict in the origin, but the association with conflict weakens substantially once refugee stock is included. Similarly, the risk of severe-policy onset does not rise steadily as a crisis ages, nor is there a common population threshold at which coercion begins. Large refugee populations are instead exposed to severe policy at different points in the life of a crisis.

Taken together, the results suggest that severe forced-return policy is most consistently associated with the size of the particular refugee population a government is hosting. Large populations often arise in protracted, unresolved conflicts, but battlefield violence, crisis duration, host political institutions, militarization, and social ties do not independently predict coercion with the same consistency. The evidence therefore points less to a general threshold of "host fatigue" than to the pressures associated with accommodating large refugee populations during unresolved displacement crises.

Table 34: Correlates of forced-return policy, entered one at a time

Correlate	Severe policy	Any policy	Dyad-years
Scale of the refugee population			
Refugee stock in this dyad (log, lagged)	1.35*** (0.10)	0.91*** (0.05)	4,448
Total refugees hosted by the asylum country (log)	0.83*** (0.10)	0.53*** (0.06)	4,448
Camp residence (ordinal)	0.48*** (0.08)	0.39*** (0.05)	4,425
Refugees reported militarized	1.53*** (0.30)	1.60*** (0.21)	4,360
Host government and host pressure			
GDP p.c. in asylum (log)	-0.53*** (0.09)	-0.47*** (0.05)	4,412
V-Dem polyarchy (host)	-0.41*** (0.09)	-0.43*** (0.05)	4,448
Freedom House political rights (host)	0.36*** (0.09)	0.42*** (0.05)	4,111
DWRAP refugee-rights index (host)	-0.12 (0.08)	-0.24*** (0.05)	4,285
Conflict in asylum: high	-0.42 (0.37)	0.45** (0.16)	4,448
Conflict in asylum: low	1.03*** (0.17)	0.70*** (0.11)	4,448
Origin-country conditions			
Conflict in origin: high	1.33*** (0.18)	1.02*** (0.12)	4,448
Conflict in origin: low	0.32 [†] (0.18)	0.28* (0.11)	4,448
Battle deaths in origin (log)	0.63*** (0.08)	0.43*** (0.05)	4,113
GDP p.c. in origin (log)	-1.05*** (0.10)	-0.59*** (0.06)	4,179
Civil-liberties restrictions in origin	0.72*** (0.11)	0.56*** (0.06)	4,403
Cumulative years of origin conflict (log)	0.67*** (0.08)	0.45*** (0.05)	4,448
Dyadic relationship and duration			
Distance origin-asylum (log)	-0.36*** (0.07)	-0.42*** (0.05)	4,399
Origin and asylum contiguous	1.67*** (0.24)	1.47*** (0.13)	4,399
Composite cultural-tie index	0.39*** (0.08)	0.44*** (0.05)	4,274
Years since the dyad entered the panel	-0.12 (0.09)	-0.13* (0.05)	4,448

Note: Each cell is a separate logit of the policy indicator on that correlate alone, with no controls and no fixed effects; standard errors in parentheses. Continuous measures are standardised on the policy sample, so coefficients are per standard deviation. These are descriptive associations and are not to be read as effects.

Table 35: Predictors of forced-return policy, holding host-year, origin-year and dyad fixed

Variable	(1) Pooled, severe	(2) Pooled, any	(3) Same host-year, severe	(4) Same origin-year, severe	(5) Same dyad, severe	(6) Same dyad, any
Scale of the refugee population						
Refugee stock in this dyad (log, lagged)	1.71*** (0.32)	0.71*** (0.17)	4.23*** (1.01)	2.20 [†] (1.32)	1.19** (0.40)	0.21 (0.22)
Total refugees hosted by the asylum country (log)	-0.05 (0.24)	0.15 (0.21)	—	-0.74 (1.19)	-0.29 (0.46)	0.38* (0.19)
Camp residence (ordinal)	0.34 (0.24)	0.19 (0.18)	0.63 (1.39)	-0.86 (0.68)	0.67 (0.69)	0.20 (0.28)
Host government and host pressure						
GDP p.c. in asylum (log)	0.18 (0.34)	0.63* (0.28)	—	3.52* (1.42)	-1.26 (1.63)	-1.80* (0.78)
V-Dem polyarchy (host)	0.94** (0.36)	0.46* (0.21)	—	-1.03 (1.26)	0.55 (0.77)	0.23 (0.55)
DWRAP refugee-rights index (host)	0.40* (0.19)	-0.02 (0.16)	—	1.04* (0.48)	0.64* (0.32)	0.14 (0.31)
Conflict in asylum: high	-0.84 (0.56)	0.23 (0.47)	—	-1.39 (1.94)	-0.18 (0.83)	0.00 (0.45)
Conflict in asylum: low	0.48 (0.35)	0.32 (0.36)	—	-0.57 (1.08)	0.53 (0.51)	-0.01 (0.36)
Origin-country conditions						
Conflict in origin: high	0.04 (0.45)	0.11 (0.31)	4.00 (2.46)	—	0.03 (0.86)	-0.04 (0.39)
Conflict in origin: low	0.03 (0.48)	0.09 (0.30)	0.91 (1.90)	—	0.30 (0.46)	0.24 (0.35)
GDP p.c. in origin (log)	-0.85** (0.30)	-0.22 (0.18)	-2.12* (0.90)	—	-0.91 (0.99)	-0.21 (0.50)
Civil-liberties restrictions in origin	-0.17 (0.24)	0.11 (0.22)	-4.70*** (1.34)	—	0.56 (0.39)	0.52* (0.21)
Cumulative years of origin conflict (log)	0.10 (0.24)	0.14 (0.13)	1.19 (1.23)	—	0.10 (0.37)	0.15 (0.21)
Dyadic relationship and duration						
Distance origin-asylum (log)	-0.09 (0.26)	-0.26 (0.18)	-7.16** (2.68)	-0.43 (0.57)	—	—
Origin and asylum contiguous	-0.01 (0.58)	1.21 [†] (0.73)	-4.57 [†] (2.48)	6.94* (2.88)	—	—
Composite cultural-tie index	0.19 (0.21)	0.20 (0.15)	-0.78 (0.95)	-1.84 (1.36)	—	—
Years since the dyad entered the panel	0.15 (0.36)	0.13 (0.23)	-1.84 [†] (1.07)	-3.71* (1.58)	—	—
Specification						
Fixed effects	Year	Year	Asylum × year	Origin × year	Dyad, year	Dyad, year
Dyad-years	3,622	3,828	225	285	917	1,977

Note: Logit coefficients, standard errors clustered by dyad in parentheses. Column 1 and 2 are pooled with year effects. Column 3 compares different origin groups facing the same asylum government in the same year, so every host-level characteristic is absorbed by construction. Column 4 compares hosts of the same origin in the same year, so every origin-level characteristic is absorbed. Columns 5 and 6 compare a dyad to itself over time, so time-invariant dyadic measures drop mechanically. A dash means the design absorbs the variable; textitsep. means the coefficient is a report of a cell with no counter-examples rather than an estimate. All models converged.

Table 36: Social and ethnic ties as predictors of forced-return policy

Tie measure	(1) Year FE	(2) + region	(3) + geography	(4) + observables	(5) Origin $\times$ year	(6) Asylum $\times$ year
Severe forced-return policy						
Common official language	0.49 (0.54)	0.25 (0.58)	0.30 (0.64)	0.35 (0.39)	1.15 (1.60)	5.25 (4.93)
Common spoken language	0.22 (0.56)	0.07 (0.57)	0.10 (0.66)	0.19 (0.38)	0.49 (1.41)	-0.27 (1.26)
Religious similarity, broad	0.36 (0.30)	0.19 (0.25)	0.14 (0.28)	0.27 (0.22)	-2.10 (1.30)	-1.45 (1.07)
tradition						
Religious similarity, denominational	-0.06 (0.12)	-0.46* (0.22)	-0.65** (0.22)	-0.38 (0.28)	-2.10 (1.40)	-3.41 [†] (1.77)
Any transnational ethnic kin	1.24* (0.61)	0.70 (0.69)	0.49 (0.72)	0.49 (0.66)	-1.54 (2.19)	sep.
Kin share of origin population	0.21* (0.10)	0.09 (0.14)	0.02 (0.15)	0.02 (0.14)	-0.33 (0.55)	-1.08 (0.75)
Colonial or common-state history	0.65 (0.54)	0.69 (0.52)	0.53 (0.58)	1.03* (0.45)	2.07 (1.77)	-4.41 (3.35)
Summary index of all ties	0.35 (0.26)	0.19 (0.28)	0.11 (0.35)	0.23 (0.22)	-1.45 (1.26)	-0.53 (1.00)
Any forced-return policy						
Common official language	0.40 (0.32)	0.20 (0.32)	0.17 (0.34)	0.20 (0.30)	0.56 (0.81)	3.15*** (0.79)
Common spoken language	0.07 (0.31)	-0.11 (0.31)	-0.20 (0.34)	-0.12 (0.29)	-0.23 (0.81)	0.97 (0.99)
Religious similarity, broad	0.42* (0.18)	0.27 (0.19)	0.15 (0.18)	0.33 [†] (0.18)	0.76* (0.35)	0.07 (0.54)
tradition						
Religious similarity, denominational	0.13 (0.13)	-0.16 (0.18)	-0.39 [†] (0.20)	-0.31 (0.24)	-0.47 (0.45)	-1.29 (0.79)
Any transnational ethnic kin	1.33*** (0.35)	1.08* (0.45)	0.71 [†] (0.41)	0.94* (0.37)	0.76 (0.74)	2.00** (0.66)
Kin share of origin population	0.32** (0.11)	0.25* (0.12)	0.10 (0.16)	0.13 (0.15)	0.09 (0.42)	1.31*** (0.34)
Colonial or common-state history	0.25 (0.34)	0.21 (0.33)	-0.15 (0.42)	-0.28 (0.41)	-0.31 (0.84)	-0.24 (1.26)
Summary index of all ties	0.42** (0.14)	0.30* (0.14)	0.10 (0.16)	0.22 (0.16)	0.38 (0.58)	0.95 [†] (0.53)

Note: The outcome here is forced-return policy, not return. Each cell is a separate logit with one tie measure entered on its own alongside the stated conditioning set, standard errors clustered by dyad. The measures are almost time-invariant, so there are no dyad fixed effects; column 5 identifies the tie from variation across the hosts of one origin in one year and column 6 from variation across the origin groups in one host in one year. Column 3 adds distance and contiguity, column 4 adds the full pooled observable set. textitsep. marks a cell with no counter-examples rather than an estimate. Conclusions from Section refapp:ties, which asks which hosts refugees return from, do not transfer here.

Table 37: Onset risk of severe policy by displacement age and population size

Years since the dyad entered the panel	All sizes	Under 100,000 refugees	100,000 refugees or more
0–1	0.00% (0/20)	0.00% (0/11)	0.00% (0/9)
2–3	3.98% (9/226)	3.64% (6/165)	4.92% (3/61)
4–5	1.53% (4/262)	1.04% (2/193)	2.90% (2/69)
6–10	1.36% (9/663)	0.83% (4/481)	2.75% (5/182)
11–15	1.16% (8/687)	0.60% (3/497)	2.63% (5/190)
16–20	1.16% (7/604)	0.00% (0/464)	5.00% (7/140)
21+	0.79% (11/1400)	0.18% (2/1106)	3.06% (9/294)

Note: Each cell is the probability that a new severe forced-return policy begins, with onsets over dyad-years at risk in parentheses. The risk set is dyad-years whose previous CALENDAR year is observed in the sample and carried no severe policy, so 1988 and 1990 are never treated as consecutive. Age is measured from the dyad's first year of observed refugee presence in the full panel.

Table 38: Origin battle deaths and reported militarization as predictors of forced-return policy

Specification	Severe policy	Any policy	N (severe)	N (any)
Origin battle deaths (log, per SD)				
Raw, year FE	0.69** (0.25)	0.47*** (0.13)	3,985	4,113
+ log refugee stock	0.21 (0.19)	0.16 (0.10)	3,985	4,113
+ stock and origin GDP p.c.	0.10 (0.18)	0.11 (0.10)	3,792	3,916
+ distance and contiguity	0.11 (0.18)	0.10 (0.11)	3,745	3,868
Full pooled covariates	0.45 (0.51)	-0.36 (0.42)	3,493	3,613
Same host-year	1.72* (0.73)	0.24 (0.24)	221	733
Host and year FE	0.30 (0.25)	0.09 (0.13)	2,000	2,946
Dyad and year FE	0.42 (0.33)	0.11 (0.15)	893	1,959
Violence in the host's other neighbours, host and year FE	-0.06 (0.23)	0.05 (0.14)	2,055	3,006
Refugees reported militarized				
Raw, year FE	1.41* (0.55)	1.51*** (0.40)	4,121	4,345
Full pooled covariates	0.81 [†] (0.47)	1.14** (0.40)	3,576	3,778
Same host-year	sep.	5.34*** (1.58)	229	751
Host and year FE	0.72 (0.47)	1.74*** (0.48)	2,136	3,145
Same origin-year	0.58 (1.22)	1.61 [†] (0.86)	287	802
Dyad and year FE, contemporaneous	-0.05 (0.50)	1.27* (0.53)	1,022	2,198
Dyad and year FE, one-year lag	-0.72 (0.45)	0.25 (0.41)	946	2,062

Note: Each column is a progressively more demanding comparison. Logit coefficients on the row variable, standard errors clustered by dyad. Battle deaths are UCDP counts and begin in 1989, so those models run on the smaller sample the N columns report. The militarization flag is spliced from two non-overlapping World Refugee Survey coding series and is sometimes read from the same archival narrative as the forced-return code itself, which is why the contemporaneous and lagged within-dyad estimates are shown side by side. textitsep. marks a cell with no counter-examples rather than an estimate.

3.3 Conditional Effects of the Post-War Window

A more demanding version of the preference account implies that the association between peace at home and return should itself depend on how attractive remaining in asylum is. The main text reports the substance of that test; Table 39 reports the models in full, including the association of each condition with return outside the post-war window, the number of post-war dyad-years on each side of the split, and the fit statistics.

Each column interacts the post-war window with a binary condition divided at the sample median: asylum-country income, origin-country income, and civil-liberties restrictions in the origin. The war-ending row gives the coefficient when the condition is absent, the interaction shows how it changes when the condition is present, and their sum gives the corresponding conditional coefficient. All columns include the full main specification and the continuous measure from which the binary condition is constructed.

Table 39: Interactions between the post-war window and permissive conditions

Term	(1) Poor asylum	(2) Rich origin	(3) Free origin
Window and condition			
Condition absent means	Rich asylum	Poor origin	Repressive origin
War ending, condition absent	0.02 (0.24)	0.62** (0.23)	0.74** (0.24)
Condition alone, no war ending	0.19 (0.28)	0.52 (0.32)	-0.46 (0.35)
War ending x condition	0.82*** (0.24)	-0.07 (0.30)	-0.57 (0.44)
War ending, condition present	0.84*** (0.21)	0.55 [†] (0.29)	0.17 (0.45)
Support and fit			
Post-war dyad-years, condition absent	147	182	250
Post-war dyad-years, condition present	169	134	66
All main-specification variables	Yes	Yes	Yes
Dyad and year fixed effects	Yes	Yes	Yes
Dyad-years	2,327	2,327	2,327
Pseudo R2	0.798	0.795	0.797

Asylum-country income is the only moderator with an interaction distinguishable from zero (0.82). In richer asylum countries the post-war coefficient is 0.02 and is not statistically significant; in poorer ones it is 0.84. Neither of the other two interactions is distinguishable from zero: the origin-income interaction is -0.07 and the civil-liberties interaction is -0.57. Outside the post-war window, the civil-liberties term is also not distinguishable from zero (-0.46). These main effects are distinct from whether the conditions moderate the post-war association.

The full sample contains 452 post-war dyad-years. The splits are 147 and 169 for asylum income, 182 and 134 for origin income, and 250 and 66 for civil liberties. The civil-liberties column is the least well supported of the three, with 66 post-war dyad-years in freer origins.

4 Literature Review

Figure 1 is based on a systematic search of nineteen journals from 2006 through 2026:

- **Political science** (twelve): *American Political Science Review*, *American Journal of Political Science*, *Journal of Politics*, *Comparative Political Studies*, *British Journal of Political Science*, *World Politics*, *International Organization*, *International Studies Quarterly*, *Journal of Peace Research*, *Journal of Conflict Resolution*, *Political Science Research and Methods*, and *PNAS*.
- **Migration and refugee studies** (five): *Journal of Refugee Studies*, *International Migration Review*, *Journal of Ethnic and Migration Studies*, *International Migration*, and *Comparative Migration Studies*.
- **Development economics** (two): *World Development* and the *Journal of Development Economics*.

I supplemented the journal search by tracing citations to and from core studies, reviewing the 2024 World Bank–UNHCR Joint Data Center review of refugee return, and searching for combinations of *refugee return*, *repatriation*, *return intentions*, *IDP return*, and *internally displaced return*.

I conducted an additional search focused on qualitative research. It combined return and repatriation terms with *interview*, *ethnography*, *focus group*, *life history*, *decision-making*, *home*, *belonging*, *waiting*, *refusal*, and *circular*, together with publisher-site searches and citation tracing from key qualitative studies. The qualitative search was targeted rather than a second journal-by-journal census.

A study is included if it (a) reports original empirical evidence, (b) examines refugees or internally displaced people displaced by conflict, and (c) analyzes qualitative return decisions, stated return intentions or preferences, or realized return. I exclude reviews, conceptual and normative work, studies of state policy or pressure to leave, and studies of post-return reintegration unless return itself is an outcome. Working papers and online-first articles are included.

I code three things. *Outcome* distinguishes qualitative evidence on return decisions, stated intentions or preferences, and realized return. *Scope* distinguishes studies of one displacement context from multi-context or cross-national studies. The first category includes studies of several origin populations in one host and of one origin population across several hosts, provided that the study concerns a single displacement context. Only studies spanning several distinct displacement contexts are coded as cross-national. *Population* is shown by marker shape: internally displaced people, Ukrainian refugees, Syrian refugees, and other populations.

Table 40: Studies plotted in Figure 1 of the paper

Outlet	Study	Year	Title	Population	Scope	Outcome	Marker
Journal of Peace Research 61(6)	Costalli, Stefano (2024)	2024	Do UN Peace Operations Help Forcibly Displaced People?	Multiple	Multi-context	Realized return	Other refugees
J Refugee Studies	Zakirova, Komila (2021)	2021	The Road Back Home is Never Long: Refugee Return Migration	Multiple	Multi-context	Realized return	Other refugees
Peace Economics, Peace Science and Public Policy Development in Practice 30	Arias, María Alejandra (2014)	2014	The Desire to Return during Civil War: Evidence for Internally Displaced Populations in Colombia	Colombian	One context	Intentions	IDPs
	Collado, Zenaida C. (2020)	2020	Living in Displacement Contexts: Return Intentions among Internally Displaced Persons in Marawi, Philippines	Marawi IDPs	One context	Intentions	IDPs
International Migration 64(3)	Malashkhia, Mariam (2026)	2026	Integration and Return Intentions among Displaced Individuals	Georgian	One context	Intentions	IDPs

Table 40: Studies plotted in Figure 1 of the paper (*continued*)

Outlet	Study	Year	Title	Population	Scope	Outcome	Marker
European Journal of Psychotraumatology 14(2)	Thompson, Peter Onah (2023)	2023	Posttraumatic Stress Moderates Return Intentions: A Factorial Survey Experiment with Internally Displaced Persons in Nigeria	Conflict-displaced Nigerians	One context	Intentions	IDPs
JEMS	Di Saint Pierre, Francesca (2015)	2015	Return Wishes of Refugees in the Netherlands: The Role of Integration, Host National Identification and Perceived Discrimination	Multiple (Afghan, Iranian, Iraqi, Somali)	One context	Intentions	Other refugees
Comparative Migration Studies	Tefera, Tadele Muche (2026)	2026	Conflict, Drought, and Displacement: Determinants of Return Intentions among South Sudanese Refugees	South Sudanese	One context	Intentions	Other refugees
Population, Space and Place 18(1)	Zimmermann (2012)	2012	Understanding Repatriation: Refugee Perspectives on the Importance of Safety, Reintegration, and Hope	Somali and Afghan	One context	Intentions	Other refugees
International Migration	Özer, Yeşim (2024)	2024	Back for Good? Return Aspirations of Immigrants in the Netherlands	Multiple immigrant groups	One context	Intentions	Other refugees
IMR	Al Husein, Nawras (2023)	2023	Determinants of Intended Return Migration among Refugees: A Comparison of Syrian Refugees in Germany and Turkey	Syrian	One context	Intentions	Syrian
BJPS	Alrababah, Ala (2023)	2023	The Dynamics of Refugee Return: Syrian Refugees and Their Migration Intentions	Syrian	One context	Intentions	Syrian
Political Studies	Gerver, Mollie (2025)	2025	Refugee Resettlement and Preferences	Syrian	One context	Intentions	Syrian
APSR	Ghosn, Faten (2021)	2021	The Journey Home: Violence, Anchoring, and Refugee Decisions to Return	Syrian	One context	Intentions	Syrian
Journal of Population Economics 36	Hannafi, Cyrine (2023)	2023	Social Integration of Syrian Refugees and Their Intention to Stay in Germany	Syrian	One context	Intentions	Syrian
Journal of Immigrant and Refugee Studies	Kaya, Serdar (2020)	2020	Prospects of Return: The Case of Syrian Refugees in Germany	Syrian	One context	Intentions	Syrian
Journal of Immigrant and Refugee Studies	Kayaoglu, Ayselin (2021)	2021	Return Aspirations of Syrian Refugees in Turkey	Syrian	One context	Intentions	Syrian

Table 40: Studies plotted in Figure 1 of the paper (*continued*)

Outlet	Study	Year	Title	Population	Scope	Outcome	Marker
Journal of International Migration and Integration 24(4)	Özkan, Zafer (2023)	2023	Intentions to Return and Migrate to Third Countries: A Socio-Demographic Investigation among Syrians in Turkey	Syrian	One context	Intentions	Syrian
CESifo Working Paper 12118	Adema, Joop (2025)	2025	What Drives Refugees' Return after Conflict?	Ukrainian	One context	Intentions	Ukrainian
Journal of Ethnic and Migration Studies	Hernes, Vilde (2024)	2024	Where Does the Future Lie? Initial Aspirations for Return among Newly Arrived Ukrainian Refugees in Norway	Ukrainian	One context	Intentions	Ukrainian
International Migration 64(3)	Kochaniak, Katarzyna (2026)	2026	Return Intentions under Temporary Protection: Theoretical Perspectives on Migrant Households	Ukrainian households	One context	Intentions	Ukrainian
PLOS ONE 18	Kohlenberger, Judith (2023)	2023	High Self-Selection of Ukrainian Refugees into Europe: Evidence from Krakow and Vienna	Ukrainian	One context	Intentions	Ukrainian
IBS Working Paper 01/2025	Lewandowski, Piotr (2025)	2025	The Role of Job Task Routinization in Shaping Return Intentions: Evidence from Ukrainian War Refugees in Poland	Ukrainian	One context	Intentions	Ukrainian
Population, Space and Place	Novotný, Josef (2026)	2026	Drivers of Return Plans among Ukrainian War Refugees in Czechia: Challenging Common Assumptions and Refining the Theory	Ukrainian	One context	Intentions	Ukrainian
Journal of Ethnic and Migration Studies, online first	Olivo Rumpf, Karelis (2026)	2026	The Interplay between Origin- and Receiving-Country Conditions: Experimental Evidence on Ukrainian Refugees' Return Intentions	Ukrainian	One context	Intentions	Ukrainian
International Migration Review, online first	Perelli-Harris, Brienna (2025)	2025	The Uncertainty of Forced Displacement: How Language and Violence Shaped Displacement Trajectories during Russia's Invasion of Ukraine	Ukrainian	One context	Intentions	Ukrainian

Table 40: Studies plotted in Figure 1 of the paper (*continued*)

Outlet	Study	Year	Title	Population	Scope	Outcome	Marker
International Migration 63(5)	Saijo, Harunobu (2025)	2025	Which Policy Attributes Affect Assisted Voluntary Return and Reintegration Uptake among Ukrainian Evacuees in Japan?	Ukrainian	One context	Intentions	Ukrainian
Journal of Behavioral and Experimental Economics	Vakhitov, Volodymyr (2025)	2025	Return Intentions of Ukrainian Refugees: The Role of National Identity and Pride	Ukrainian	One context	Intentions	Ukrainian
International Migration 62	van Tubergen, Frank (2024)	2024	Return Intentions among Ukrainian Refugees in Europe: A Cross-National Study	Ukrainian women	One context	Intentions	Ukrainian
J Refugee Studies	Žila, Ondřej (2026)	2026	'Partial Returns': Displacement, Mobility, and Translocal Connections in Post-war Bosnia and Herzegovina	Bosnian	One context	Qualitative	IDPs
Journal of Refugee Studies 34(1)	Antwi-Boateng (2021)	2021	The Dilemma of Ghana-Based Liberian Refugees and the Challenges of Repatriation and Integration	Liberian	One context	Qualitative	Other refugees
Journal of Ethnic and Migration Studies	Durodola (2026)	2026	Return to What? Reimagining Home and Belonging after Conflict and Refugee Status Cessation among Liberians in Nigeria	Liberian	One context	Qualitative	Other refugees
International Migration	Eastmond, Marita (2006)	2006	Transnational Returns and Reconstruction in Post-war Bosnia and Herzegovina	Bosnian	One context	Qualitative	Other refugees
J Refugee Studies	Hardgrove, Abby (2009)	2009	Liberian Refugee Families in Ghana: The Implications of Family Demands and Capabilities for Return to Liberia	Liberian	One context	Qualitative	Other refugees
International Migration 52(6)	Harpviken, Kristian Berg (2014)	2014	Split Return: Transnational Household Strategies in Afghan Repatriation	Afghan	One context	Qualitative	Other refugees
Journal of Eastern African Studies 4(1)	Kaiser (2010)	2010	Dispersal, Division and Diversification: Durable Solutions and Sudanese Refugees in Uganda	Sudanese	One context	Qualitative	Other refugees
African Human Mobility Review 6(2)	Khan (2020)	2020	Is Voluntary Repatriation the Preferred Durable Solution? The View of Refugees in South Africa	Multiple refugee origins	One context	Qualitative	Other refugees

Table 40: Studies plotted in Figure 1 of the paper (*continued*)

Outlet	Study	Year	Title	Population	Scope	Outcome	Marker
Journal of Refugee Studies	Krupalija, Mustafa (2026)	2026	Rebuilding Home on Hot Ground: Forced Migration and Return to Srebrenica	Bosnian	One context	Qualitative	Other refugees
Journal of Refugee Studies 19(4)	Muggeridge (2006)	2006	Back Home? Refugees' Experiences of their First Visit back to their Country of Origin	Multiple refugee origins	One context	Qualitative	Other refugees
J Refugee Studies	Nandi, Sarah (2025)	2025	"We were together and we had our own family in each other": Refusing Repatriation and Forging Gendered Belonging as Hijra Refugees in Kolkata	Bangladeshi (Hijra)	One context	Qualitative	Other refugees
Transnational Social Review 6(1-2)	Omata (2016)	2016	Home-making during Protracted Exile: Diverse Responses of Refugee Families in the Face of Remigration	Liberian	One context	Qualitative	Other refugees
JEMS	Omata, Naohiko (2013)	2013	The Complexity of Refugees' Return Decision-Making in a Protracted Exile: Beyond the Home-Coming Model and Durable Solutions	Liberian	One context	Qualitative	Other refugees
International Migration 55(5)	Porobić, Selma (2017)	2017	Daring Life-Return Projects to Post-Dayton Bosnia and Herzegovina	Bosnian returnees	One context	Qualitative	Other refugees
International Migration	Stefansson, Anders H. (2006)	2006	Homes in the Making: Property Restitution, Refugee Return, and Senses of Belonging in a Post-war Bosnian Town	Bosnian	One context	Qualitative	Other refugees
Geoforum 43(1)	Tete (2012)	2012	'Any Place Could Be Home': Embedding Refugees' Voices into Displacement Resolution and State Refugee Policy	Liberian	One context	Qualitative	Other refugees
Comparative Migration Studies	Van Houte, Marieke (2016)	2016	Deconstructing the Meanings of and Motivations for Return: An Afghan Case Study	Afghan	One context	Qualitative	Other refugees
Refugee Survey Quarterly 45(1)	Diab (2026)	2026	Prolonging the Temporary: Female-Headed Syrian Refugee Households on Displacement, Return and Waiting post-Assad	Syrian women heading households	One context	Qualitative	Syrian

Table 40: Studies plotted in Figure 1 of the paper (*continued*)

Outlet	Study	Year	Title	Population	Scope	Outcome	Marker
International Migration 64(3)	Dinçsahin (2026)	2026	From Protracted Liminality to Fractured Homecoming: Intrafamilial Bargaining and the Limits of Voluntary Repatriation for Syrians in Turkey	Syrian	One context	Qualitative	Syrian
Journal of Refugee Studies 39(1)	Enna (2026)	2026	'The Threat of Forced Return Is the Government's Last Resort': Structural, Cultural, and Direct Violence towards Syrian Refugees in Lebanon	Syrian	One context	Qualitative	Syrian
International Migration Review, Online-First	Masterson (2026)	2026	Circular Refugee Migration: Understanding Protracted Displacement Beyond a Refugee-Returnee Binary	Syrian	One context	Qualitative	Syrian
International Migration Review 57(4)	Müller-Funk, Lea (2023)	2023	I Will Return Strong: The Role of Life Aspirations in Refugees' Return Aspirations	Syrian	One context	Qualitative	Syrian
Comparative Migration Studies 14:32	Kokonowskyj, Larissa (2026)	2026	The Role of Family Dynamics in (Return) Migration Aspirations among Ukrainian Protection Holders in Germany	Ukrainian	One context	Qualitative	Ukrainian
Comparative Migration Studies 13:66	Ash, Konstantin (2025)	2025	Who Leaves and Who Returns? IDPs and Returnees after the Russian Invasion of Ukraine	Ukrainian	One context	Realized return	IDPs
JCR	Bove, Vincenzo (2025)	2025	What it Takes to Return: UN Peacekeeping and the Safe Return of Displaced People	South Sudanese	One context	Realized return	IDPs
AJPS	Camarena, Kara Ross (2020)	2020	When Do Displaced Persons Return? Postwar Migration among Christians in Mount Lebanon	Lebanese (Christians)	One context	Realized return	IDPs
International Migration 55(5)	Joireman, Sandra F. (2017)	2017	Ethnic Violence, Local Security and Return Migration: Enclave Communities in Kosovo	Kosovar displaced	One context	Realized return	IDPs
J Refugee Studies	Stefanovic, Djordje (2015)	2015	Home is Where the Heart Is? Forced Migration and Voluntary Return in Turkey's Kurdish Regions	Kurdish	One context	Realized return	IDPs

Table 40: Studies plotted in Figure 1 of the paper (*continued*)

Outlet	Study	Year	Title	Population	Scope	Outcome	Marker
Journal of Conflict Resolution 69(1)	Tellez, Juan F. (2025)	2025	Social Cohesion, Economic Security, and Forced Displacement in the Long Run: Evidence from Rural Colombia	Colombian IDPs	One context	Realized return	IDPs
Working Paper	Weber, Sigrid (2022)	2022	Property Rights and Post-conflict Recovery: Theory and Evidence from IDP Return Movements in Iraq	Iraqi (Yazidi and Sunni Muslim)	One context	Realized return	IDPs
Journal of Refugee Studies 23(1)	Huttunen (2010)	2010	Sedentary Policies and Transnational Relations: A 'Non-sustainable' Case of Return to Bosnia	Bosnian	One context	Realized return	Other refugees
J Refugee Studies	Klinthäll, Martin (2007)	2007	Refugee Return Migration: Return Migration from Sweden to Chile, Iran and Poland 1973-1996	Chilean, Iranian, Polish	One context	Realized return	Other refugees
International Migration 55(5)	Kutlay, Mustafa (2017)	2017	The Turks of Bulgaria: An Outlier Case of Forced Migration and Voluntary Return	Turks of Bulgaria	One context	Realized return	Other refugees
Journal of Ethnic and Migration Studies 51(1)	Omata (2025)	2025	Returning to Fund Refugeehood: Dispersal and Survival between Uganda and South Sudan	South Sudanese	One context	Realized return	Other refugees
International Migration 55(5)	Stefanovic, Djordje (2017)	2017	Peaceful Returns: Reversing Ethnic Cleansing after the Bosnian War	Bosnian	One context	Realized return	Other refugees
Middle Black Sea Journal of Communication Studies 11	Altınışık (2026)	2026	From Forced Migration to Voluntary Return: Psychosocial Factors Influencing Syrian Refugees' Decisions to Return to Their Homeland After the War	Syrian	One context	Realized return	Syrian
J Development Economics	Beaman, Lori (2022)	2022	When Do Refugees Return Home? Evidence from Syrian Displacement in Mashreq	Syrian	One context	Realized return	Syrian
International Review of Sociology	Bimay (2026)	2026	Between Hope and Hesitation: Perceptions and Experiences of Refugee Women Returning from Turkey to Syria	Syrian women	One context	Realized return	Syrian
Working Paper (EBRD)	Adema, Joop (2026)	2026	Refugee Return	Ukrainian	One context	Realized return	Ukrainian

Note: Drawn from the literature audit of 23 August 2026. Scope and Outcome are the two axes of that figure; Marker is the population class the figure shows by shape. Inclusion follows the Figure decision and Strict figure columns of the audit's two inventory tabs; two studies logged there as out of scope are excluded.